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Recent progress of nanomaterials for microwave absorption

Michael Green, Xiaobo Chen*

Department of Chemistry, University of Missouri–Kansas City, Kansas City, MO, 64110, USA

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ABSTRACT

Microwave absorbing materials have received considerable interest over the years for their applications in stealth, communications, and information processing technologies. These materials often require functionalization at the nanoscale so to achieve desirable dielectric and magnetic properties which induce interaction with incident electromagnetic radiation. This article presents a comprehensive review on the recent research progress of nanomaterials for microwave absorption, including the basic mechanism of microwave absorption (e.g., dielectric loss, magnetic loss, dielectric/magnetic loss coupling), measurement principle (e.g., fundamentals of analysis, performance evaluation, common interaction pathways: Debye relaxation, Eddy current loss, natural resonance, size and shape factors), and the advances and performance review in microwave absorption (e.g., absorption bandwidth, reflection loss values, absorption peak position) using various nanomaterials, such as carbon nanotubes, carbon fibers, graphenes, oxides, sulfides, phosphides, carbides, polymers and metal organic frameworks. Overall, this article not only provides an introduction on the fundamentals of microwave absorption research, but also presents a timely update on the research progress of the microwave absorption performance of various nanomaterials.

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* Corresponding author.
E-mail address: chenxiaobo@umkc.edu (X. Chen).
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1. Introduction

Over the past decades the scientific community has witnessed an eruption in the development of nanomaterials, with research exploring their synthesis, fabrication, properties, and applications [1–20]. Nanomaterials have generated tremendous amounts of research interest due to the intrinsic properties, which are causally related to their scale, and are furthermore not observed in macroscale materials [21–31]. These unique nanomaterial properties have revolutionized materials performance in various application domains, and have led to the discovery of new domains of research. For example, due to the small size of nanomaterials, large surface areas can be obtained so to enable many unexpected material properties. In such regard, the negligible surface effects in the bulk materials are magnified over several magnitudes to be significant, which are referred to in the literature as “magnifying effects”. Atomic entities on the surface are normally unsaturated and have many sites to which nanomaterials are rendered chemically active, and both structurally and thermally unstable; such allows for much higher chemical activities, lower melting points, and structural and electronic deficiencies which induce unusual optical and electronic properties [25–33]. Furthermore, the small size of nanomaterials confines the charge excitations and movements within the materials, making continuous electronic structures discrete and gradually changing optical spectra abrupt; such phenomena are known as quantum confinement effects, which occurs when the size of the nanomaterials is on the level of the Bohr radius for semiconductor nanoparticles, or as the surface plasmon resonance effect, when the size echoes the level of optically-driven charge displacements for metal nanoparticles [34–46]. Numerous breakthroughs have been reported in optics, quantum devices, photocatalysis, photovoltaics, et al., due to the exploration of the various exciting properties of nanomaterials [47–50].

Microwave absorption has recently emerged as an innovative application area where nanomaterials and their intrinsic dielectric and magnetic properties can be utilized. Understanding the fundamental properties of these material entities and how their intrinsic characteristics dictate interactions with gigahertz-range electromagnetic radiation is imperative to the continued advancement of 21st century technology [51]. Many civilian and military technological systems require the integration of materials which interact with microwave radiation in specific desirable fashions, such as inhibiting information leakage from computational and communications systems, and material coatings in stealth technology [52,53]. These system applications are oftentimes constrained by functional limitations, such as space, operational frequency, and material/system compatibility [54]. Due to these constraints, it's logical for the scientific community to develop a wide range of materials which are available for application in civilian and military purposes, so that the needs of these unique and specific application domains can converge with the performance of select materials. As the pantheon of scientific literature continues to grow from to such pursuits, as demonstrated in Fig. 1, we must see to it to occasionally review, summarize, and tabulate the results put forth in the literature, so that the work done by those around the globe can be easily observed, compared, and applied.

Most materials which have been developed for the purposes of microwave absorption are of nanoscale morphology. Reports from the literature specifically investigating the propensity for dielectric and magnetic interaction of materials as a function of particle size have reported that, as the size of particulate matter is constrained, electromagnetic interaction tends to increase [55–60]. This understanding is somewhat intuitive – as the particle size decreases for a given material, parameters such as the material surface area are enhanced and thus the effects which lead to dielectric or magnetic interaction, such as phase-dielectrics, [61] unpaired spin-states, [62] eddy-current losses, [51,63] resonance effects, [57] et al., can be enhanced through magnifying effects. Many of the manuscripts which study the intrinsic effects of dielectric and magnetic nanomaterials for microwave absorption don't explicitly investigate their materials as a function of particle size, but nevertheless, the particle size parameter does appear to constitute an important factor when it comes to utilizing these nanomaterials for electromagnetic interference shielding and microwave absorption.

The literature review herein is designed to try and cover not only the broadest of reported responses for the proficiency parameters typically described and reported in the literature, across

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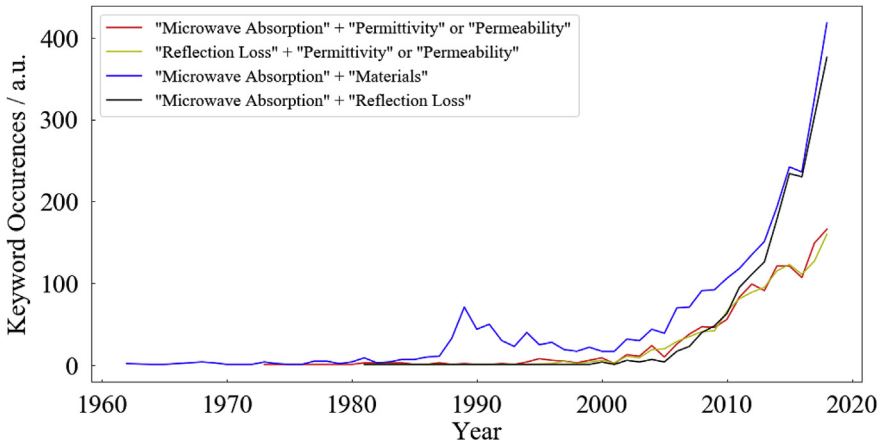


Fig. 1. Keyword occurrences as documented by Scopus for the given phrases by year, 1961–2018.

the broadest domain of nanomaterial types, but also the fundamentals of nanomaterials analysis, so to give the current plethora of reports further context. Additionally, this manuscript includes discussion and summarization of the major nanomaterial classes, and an exhaustive review regarding how these individual classes are examined and applied. Finally, a comprehensive tabulation of nanomaterial responses in the literature to date is supplied herein.

2. Mechanisms of microwave absorption

As an incident microwave is a coupling of an oscillatory electric field and magnetic field, the materials which induce microwave absorption do so by interacting with either one of these two fields, or both, so to drive light/matter interaction at the gigahertz region of the electromagnetic (EM) spectrum. This action is in accordance with Maxwell's equations, where the perturbation of one of the electromagnetic fields by interaction with a material medium will induce a response change of the other, resulting in the dissipation of the entire electromagnetic wave [64]. Dielectric loss is the characteristic electronic interaction between the electric field of the incident electromagnetic radiation and the nanomaterial that results in reflection loss; [65] subsequently, magnetic loss is the characteristic magnetic interaction between the nanomaterial and the electromagnetic wave [52]. Of the multitudes of specific mechanisms of action presented in the literature, each can ultimately be described as a function of predominately electronic interactions, [66] predominately magnetic interactions, [67] or a combination of the two, [68] which, in tandem with the intrinsic properties of the nanomaterial, induce the loss of microwave radiation as it interacts with a material medium. From Maxwell's equations and the associated constitutive relations resulting from such equations, it can be shown that the responses by a nanomaterial to an incident electromagnetic wave are ultimately determined by the materials' bulk permittivity and permeability, [52,69] where permittivity and permeability are defined as complex values so to encapsulate energy storage and dissipation [52].

2.1. Dielectric loss

The energy pathway of dielectric loss is best described as the route which transitions energy from the form of a propagating electric field to that of the thermal output from an interacting nanomaterial. As the electric field of an EM wave propagates in a forward direction, thus changing the polarity of the field sinusoidally and orthogonal to the direction of propagation, charged entities within the vicinity of the wave spatially interact with the electric field so to be drawn in similar motion to the oscillations of the field wave [65]. As the nature of the electric field in the context of electromagnetic radiation is oscillatory in its polarity, the motion initially induces misalignment between the electric field of the EM wave and the charge distribution of a given particle; in instances where this induces charge displacement via the force from the applied electric field, potential is generated within the medium as the nanomaterial is displaced from its original state via motion of the EM wave. As a complex parameter, the real portion of permittivity ε' quantifies this lossless interaction between the wave and the nanomaterial; [52] as the applied force of the EM wave acts on the nanomaterial so to induce polarization in the medium, without a release mechanism so to thermally dissipate the built potential within the medium, the entirety of the electromagnetic wave passes, and the nanomaterial returns to its original state without a net influx of energy. However, if there is a dissipation pathway within the nanomaterial as described by the complex portion of permittivity ε'', which can be thought of as a sort of friction to the displacement of the subatomic particles, then the passing EM wave attenuates, manifesting as the subsequent generation of heat [52]. The point of maximal lossy interaction as described by ε'' is referred to as a relaxation peak, and is associated with the dielectric resonance of the nanomaterial, where the energy dissipation rate from stored energy to thermal energy is maximized [71]. As such, permittivity, expressed as the value equal to the charge required to generate a volt of potential across a meter of substance, though commonly delineated as a relative term ε_r which is the intrinsic permittivity of the nanomaterial ε with respect to the permittivity of free space ε_0 as shown in Equation (1), is the defining characteristic of materials that induce microwave interaction via dielectric loss [72].

$$\varepsilon_r = \varepsilon/\varepsilon_0 = (\varepsilon' - i\cdot\varepsilon'') \quad (1)$$

2.2. Magnetic loss

The magnetic counterpart of permittivity is that of permeability, as shown in Equation (2). However, since there is no elementary particle from which a monopole of magnetic flux can be fielded, [73] permeability is not simply the potential gradient across a distance, but the ratio of the magnetic flux density to that of the incident magnetic field, or the measure of a nanomaterial's capacity to simply sustain a magnetic field within its medium [69]. As with

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the case of permittivity, the relative permeability value, which is the ratio of permeability with respect to that of the permeability of free space, is the typically reported value quantifying magnetic interaction, and is the parameter utilized in this text. Permeability is achieved through a multitude of fashions, which include; eddy-current losses, [51,63] where the incident magnetic portion of the EM field induces an electrical current; and resonance effects, such as natural domain resonance, [57] ferromagnetic resonance, [74] and generally unpaired spin-states, [62] where the electrons of a nanomaterial of interest generate non-cancelling magnetic moments from their spin orientation within a given bulk material domain [74,75]. Supplied with an external magnetic field, the intrinsic and extrinsic magnetic fields of the nanomaterial align with the incident electromagnetic field so to generate a potential associated with alignment, much like in the case of the charged particles with permittivity [69]. The propensity of a nanomaterial to undergo magnetic alignment is quantified by the real part of permeability μ'. As similar to permittivity, without a mechanism for energy dissipation, the interaction between the magnetic field of the incident EM wave and the nanomaterial is lossless and thus does not result in the net transfer of energy [52]. If however there is such mechanism for dissipation, represented by the complex portion of permeability μ", the passing magnetic field will transfer energy, and in the case of gigahertz-range electromagnetic energy, attenuate [52]. The point of maximal lossy interaction as described by μ" can similarly be referred to as a relaxation peak, and is associated with the magnetic resonance of the nanomaterial, [76] where the energy dissipation rate from stored energy to thermal heat is maximized. As with permittivity, the permeability of a given nanomaterial is typically represented by the relative permeability μ_r which is the intrinsic permeability of the nanomaterial μ with respect to the permeability of free space μ_0.

$$\mu_r = \mu/\mu_0 = (\mu' - i\cdot\mu'') \quad (2)$$

2.3. Dielectric/magnetic loss coupling

Maxwell's four equations of electricity and magnetism relate the two fields as an entangled entity that is electromagnetic radiation. As these two fields propagate in a forward direction both orthogonal to each other and the direction of propagation, they, according to the conclusions of such derivations, have wavefunction amplitudes which are equivalent in magnitude over the distance of propagation. As such, the attenuation of one of the fields has an associated effect on the other. However, as both fields have associated potential from which energy can be extracted, the attenuation of all such electromagnetic energy can be done either through interaction of the electric part, the magnetic part, or a combination of both, so to induce the overall effect of absorption [64,67].

3. Measurement principles

Experimentally determining the nanomaterial response to the electric and magnetic portions of incident electromagnetic radiation allows for a suite of calculations so to quantify the physical and experimental parameters of synthesized nanomaterials. These measurement techniques from first-principles have been covered extensively in various National Institute of Standards and Technology (NIST) tech notes [77–79] and textbook publications [52,53] elsewhere, and as such these fundamental considerations shall be discussed only briefly herein, as our interest is ultimately in the nanomaterials which these techniques quantify. Furthermore, focus will also be constrained to non-resonant methods of analysis, as such methodologies allow for the determination of electric and magnetic response properties over a range of frequencies, which is the focus for the majority of the nanomaterials work reviewed herein.

3.1. Fundamentals of analysis

Non-resonant methods of materials analysis can be subcategorized into, firstly, methods which focus on exclusively the reflection of electromagnetic radiation, and secondly, methods which probe both the transmission of an electromagnetic wave, as well as the reflection. Both methodologies have specific techniques which generate data frames that sufficiently characterize the material response, so to derive permittivity and permeability values. From such values, the quantification of parameters such as reflection loss can be derived. Response to incident electromagnetic radiation is quantified as a scattering parameter. Materials are typically dispersed into a low-loss, non-magnetic solidifying medium such as paraffin wax, polyvinyl chloride (PVC), varnish, et al. so to form a bulk material medium (simply referred to as the 'material' herein for distinction) of which the solidifying materials are unresponsive to incident electromagnetic radiation. Such allows for the quantification of the scattering response for the medium strictly as a function of the dielectric or magnetic nanomaterial which is being investigated [52]. For reflection methods, there is only a single scattering parameter to be experimentally determined – that which is the result of a signal sent and received through the incident port, or port 1. This S_{11} parameter is the measured scattering response of an electromagnetic signal directed back towards the radiation source via the signal port. Reflection methods require a pair of S_{11} values for a given material, so that the system of equations which quantify complex permittivity and permeability as a function of these scattering parameters can be subsequently resolved. There are multiple ways of which to accomplish such; measuring varying thicknesses of the same material, probing the material at multiple positions within a transmission line, backing the material with varying metal plates or free space, varying the probe frequencies, and using time-domain techniques – these are all options for parameter quantification [52].

For transmission/reflection lines, the utilization of a two-port device allows for the quantification of electromagnetic response in the forward as well as the reflected directions, thus expanding the system of scattering parameters available for analysis to a 2x2 matrix. A representative sketch of a transmission/reflection line is shown in Fig. 2A, where the S-parameters for the 2x2 matrix are calculated from Maxwell's equations applied for the electromagnetic field response at the material domain boundaries [77,79].

In the 2x2 scattering matrix, the S_{11} parameter is still representative of the response to incident radiation from port 1, along with any subsequent signals emitted via internal reflections caused by the impedance match of the material with respect to free space [70]. S_{12} is now both the signal response of electromagnetic radiation from port 1 traversing through the material medium and continuing in propagation towards the port 2 detector, as well as any signal generated from subsequent internal reflections [77,79]. This can be examined in both directions, generating the subsequent S_{21} and S_{22} parameters. The full pathway for such signals recorded via transmission/reflection lines is demonstrated in Fig. 2B. The basic premise of material testing is equivalent between the two physical devices, coaxial lines and waveguides, which can apply this setup. Coaxial lines typically utilize a circular cross-section along a cylindrical line, with a center conductor passed through the origin, thus allowing the magnetic and electric fields to propagate between the center conducting line and the outer conductor [77]. Waveguides on the other hand are typically rectangular and

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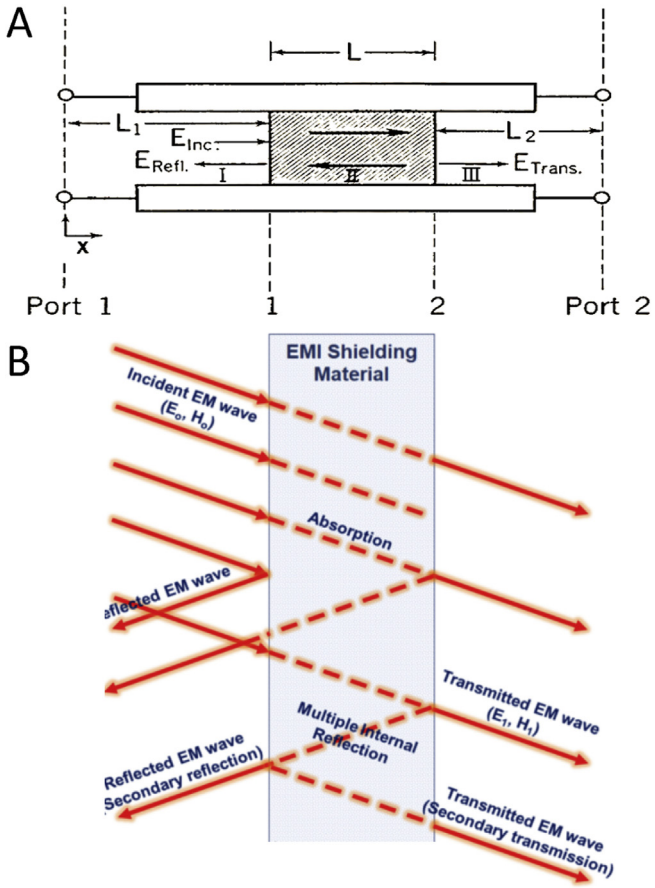


Fig. 2. A) Schematic of a coaxial transmission/reflection line used to quantify scattering parameters. Reproduced with permission from ref. 18. Copyright 1990 the National Institute of Standards and Technology. B) Schematic representation of electromagnetic interaction with a given material medium. Reproduced with permission from ref. 70. Copyright 2018 Elsevier.

utilize no center conductor to propagate electromagnetic radiation, though they can be operated at higher power levels [79]. Regardless of which methodology is chosen so to quantify material response, the scattering parameters extracted can ultimately be used to determine complex permittivity and permeability. There are multiple sets of algorithms available so to convert S-parameters to the desired permittivity and permeability values. From such permittivity and permeability values, further material traits can be described.

The ratio of the lossy action with respect to the lossless is referred to as tangential loss. Analogous terms are available for electric and magnetic interactions, thus yielding parameters for the dielectric loss tangent as well as the magnetic loss tangent, shown in Equation (3). Therein, ε' and ε'' are representative of the relative complex permittivity values for lossless and lossy action, respectively. Furthermore μ' and μ'' are representative of the relative complex permeability values for lossless and lossy action, respectively. Finally, δ is the loss angle between the two components, and $\tan\delta_\varepsilon$ and $\tan\delta_\mu$ are the dielectric and magnetic loss tangents, respectively, or on occasion represented as $tg\delta_\varepsilon$ and $tg\delta_\mu$ [80]. These loss tangents are representations of the loss-rate of energy for a mode of oscillation given a system with dissipating energy. The ratios can be thought of as a sort of net efficiency parameter for the overall energy transfer process; for example, as the product of lossless interaction and the loss tangent describes the power dissipation per unit volume at a given frequency for dielectric interaction [71]. The dielectric loss tangent, being associated with particles which can sustain individual monopoles, is furthermore the summated representation of the various electric interactions which generate electric loss, such as through conduction and polarization [80,81]. Materials which are good dielectrics, where conductivity is considerably small compared to the product of permittivity and angular frequency, have small loss tangents with $\tan\delta_\varepsilon \ll 1$. Conversely, good conductors have large loss tangents with $\tan\delta_\varepsilon >> 1$. The Magnetic loss tangent is more constrained given the absence of the magnetic monopole, though the same understanding of dissipating energy still applies [80,81]. Loss tangents typically peak at frequencies higher than that of the relaxation peaks, as the net dissipation efficiency over the entire transfer process and the energy transfer rate from lossless interaction to lossy are both dependent on the magnitude of the lossless interaction, which varies as a function of frequency and in accordance to the Kramers-Kronig dispersion relations, which describes the casual nature of dissipation requiring storage [71,82].

The inverse of the loss tangent parameters defines the quality factor, which is another term often used to describe material response, shown in Equation (4). Therein, the quality factors for dielectric and magnetic response are Q_ε and Q_μ respectively. The total quality factor Q therefore can be calculated via the sum of the inverse values of the dielectric and magnetic quality factors, as shown in Equation (5). Generally, the lower the Q value, the greater the energy dissipation [53].

$$\tan\delta_\varepsilon = \frac{\varepsilon''}{\varepsilon'};\ \tan\delta_\mu = \frac{\mu''}{\mu'} \tag{3}$$

$$Q_\varepsilon = \frac{1}{\tan\delta_\varepsilon} = \frac{\varepsilon'}{\varepsilon''};\ Q_\mu = \frac{1}{\tan\delta_\mu} = \frac{\mu'}{\mu''} \tag{4}$$

$$\frac{1}{Q} = \frac{1}{Q_\varepsilon} + \frac{1}{Q_\mu} \tag{5}$$

The complex permittivity and permeability parameters derived from experimentation can be utilized so to calculate the parameters for refraction and propagation of the electromagnetic field as it engages with a material interface and traverses through a material medium [78,83]. The instantaneous vector wave equations derived from Maxwell's equations [52] produce solutions defined by the propagation constant for a general lossy medium, shown in Equation (6), which in turn is a complex function of the attenuation and phase constants. As the propagation constant is furthermore a function of the complex refractive index, shown in Equation (7), which in turn is a function of complex permittivity and permeability, parameters such as the refractive index, the extinction coefficient, and finally the attenuation and phase parameters can be calculated directly, as shown in Equation (8) and Equation (9) [52,53,78,83]. Therein, the complex index of refraction $\tilde{n}$, the propagation constant γ, and subsequently the refractive index n, the extinction coefficient k, the attenuation constant α, and the phase constant β, are all directly a function on complex permittivity and permeability, with the propagation factors being furthermore a function of incident electromagnetic frequency, and the speed of light *c*, and *i*, which is the complex unit of $\sqrt{-1}$ [82]. The attenuation constant is in units of nepers/meter, where conversion to the typical reported decibel per meter value is a result of multiplication by the conversion factor of 8.68588 [53]. The phase constant is given in radians/meter, and from such the group velocity of the electromagnetic wave as it propagates through the material medium can be extracted by dividing the angular frequency by the phase constant, shown in Equation (10) [81].

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$$\gamma = i \cdot \frac{2\pi f}{c} \cdot \tilde{n} = (\alpha + i \cdot \beta) \tag{6}$$

$$\tilde{n} = [(\mu' - i \cdot \mu'')(\varepsilon' - i \cdot \varepsilon'')]^{\frac{1}{2}} = (n + i \cdot k) \tag{7}$$

$$n = Re(\tilde{n}); k = Im(\tilde{n}) \tag{8}$$

$$\alpha = Re(\gamma); \beta = Im(\gamma) \tag{9}$$

$$v = \frac{2\pi f}{\beta} \tag{10}$$

As demonstrated in Fig. 3, the derived propagation values can be simultaneously mapped to the response of a wavefunction as it propagates through a material medium. Here, the product of the cosine wavefunction as defined by the phase constant, and the exponential decay function as defined by the attenuation constant, demonstrates the resultant of a model cosine transverse wave as it propagates through space in the x direction [84].

Inverse to the resistivity [85] of a material is a material's propensity to conduct charge across its interface. This conductivity sums over all dissipative effects [86,87] that a material might exhibit, from migrating charge to energy lost from the dispersion of stored energy as defined by ε' [52]. Materials which are said to behave as good conductors exhibit conductivity values on the order of 10^4–$10^8\ S\,m^{-1}$. Materials which are classified as semiconductors manifest conductivity values on the order of 10^4 to $10^{-7}\ S\,m^{-1}$. Finally, materials which are considered insulators exhibit conductivity values below 10^{-12}, as far down as $10^{-20}\ S\,m^{-1}$. A perfect dielectric would have $0\ S\,m^{-1}$ conductivity, whereby achieving such through ε" = 0 [52]. The conductivity of a material can be derived from complex permittivity, as shown in Equation (11), via the product of angular frequency and lossy non-relative dielectric action, resulting as conductivity in units of Siemens per meter, or sometimes referenced as inverse ohm·meter.

For materials which are conductive, it's typical to calculate the distance over which a traversing electromagnetic wave attenuates by a factor of e^{-1}. This value is referred to as skin depth, shown in Equation 12, and is quantified as the inverse of the np/m attenuation constant [53,84]. The functionalization of attenuation as it traverses a material medium can also be understood from the perspective of where the radiation intensity of an incident electromagnetic wave undergoes exponential decay by $I(d) = I_0 \cdot e^{-\alpha \cdot d}$, with I_0 being the initial radiation intensity, d being the distance of traversal in meters, and α being the attenuation constant [84]. Since the skin-depth is defined as the distance over which the electromagnetic field intensity attenuates by a factor of e^{-1} as it traverses a given material medium, setting $I(d) = e^{-1}$ in the decay function yields the distance variable 'd' required to satisfy the equality as equivalent to the inverse of the attenuation constant [53,84]. As the attenuation constant α varies as a function of electromagnetic frequency due to changing permittivity and permeability, so too does the skin depth vary by the changing of the frequency parameter.

The impedance factor, also known as the intrinsic impedance, is the ratio of the transverse component of the electric field within a material medium with respect to that of the transverse component of the magnetic field, as shown in Equation (13). As an electromagnetic wave propagates through a given medium, the strength of the resistance to the manifested electric and magnetic fields within the material is represented by the intrinsic impedance of the material, which can be reduced to terms of permittivity and permeability as a consequence of the constitutive relations which fall from Maxwell's equations [80]. The nature of the impedance factor η in ohms is complex, with the real portion representing the resistance along the real axis and subsequently the action of capacitor and inductor reactance is mapped onto the complex axis, in the positive and negative domains respectively [52].

$$\sigma = 2\pi f \cdot (\varepsilon_0 \cdot \varepsilon'') \tag{11}$$

$$\delta = \alpha^{-1} = \frac{1}{Re(\gamma)} \tag{12}$$

$$\eta = \frac{\boldsymbol{E}}{\boldsymbol{H}} = Z_0 \left[\frac{\mu' - i \cdot \mu''}{\varepsilon' - i \cdot \varepsilon''}\right]^{\frac{1}{2}} \tag{13}$$

3.2. Performance evaluation

In the current pantheon of scientific literature, one of the apex parameters that materials scientists seek to optimize for applications such as radar absorption in stealth technology, [88] electromagnetic interference in signal processing, [89] and information

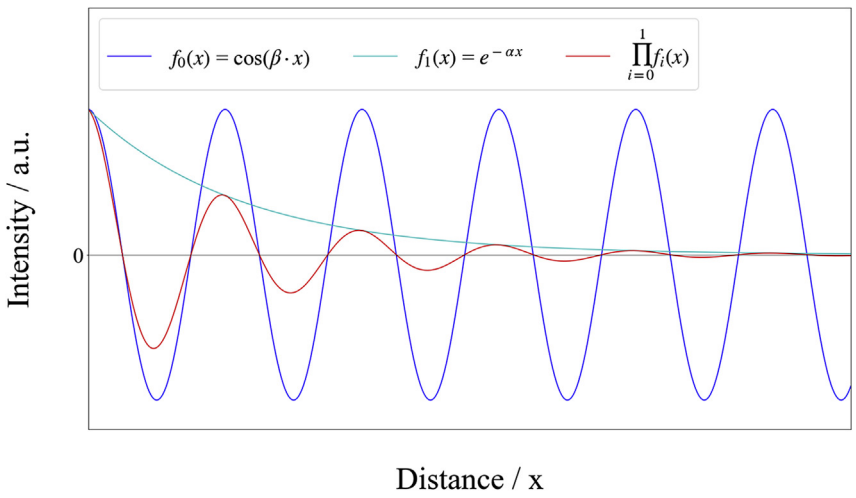


Fig. 3. The attenuation (cyan) and phase (blue) of a model cosine wavefunction (red), defined by predetermined α and β parameters, normalized to an initial amplitude of 1.

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leakage in signal transport, [51] is that of reflection loss (RL). RL, also known as return loss in the fields of telecommunications and signal processing, is a log ratio of incident power with respect to the reflected power scaled by a factor of 10. In the early days of transmission line theory, RL, as derived from materials testing systems [90–92] which were single-port devices backed by a perfect conductor, could be directly expressed via measurement of the S_{11} parameter from experiment, where reflection loss as a unit of decibel is the logarithm of the S_{11} parameter scaled by a factor of 20, as shown in Equation (14) [52,53].

For the development of single-layer plane wave absorbing materials, Naito and Suetake [68,92–95] proposed a mathematical model for relating the reflection coefficient Γ to the input parameters of permittivity and permeability by means of the input impedance, as shown in Equation (15). As mentioned, the historical context of this model pertained to microwave absorbing materials backed by a perfect conductor, analogous to material coatings for an aircraft chassis; [96,97] even though technically this may or may not strictly be an absorbing phenomenon, [62] the term "microwave absorption" was applied and has since remained all-encompassing for systems which may use absorption or cancellation [98,99]. From this analytical model, materials analysis across a three-dimensional domain of RL as a function of thickness and frequency can be examined, since permittivity and permeability themselves are both a function of frequency. Therein, the reflection coefficient, gamma, is a function of impedance Z, where Z_{in} is the input impedance, as shown in Equation (16) and Equation (17), and Z_0 is the characteristic impedance of free space (376.73 Ω, though this term can be normalized out of the expression for the reflection coefficient). Furthermore, ε′ and μ' are the relative permittivity and permeability parameters for lossless interaction, respectively, and ε" and μ" are the relative permittivity and permeability parameters for the lossy interaction of the incident electromagnetic wave. Finally, *f* is the frequency of the incident electromagnetic wave, *d* is the material thickness, and c is the speed of light. From such, RL can be calculated, since RL as a function of the reflection coefficient is the logarithm of the modulus of gamma scaled by a factor of 20, as shown in Equation (18) [95,100,101]. Although the entirety of this model has occasionally generated questions regarding its validity, [102] as one of the consequences of such derivation is the counterintuitive conclusion that sometimes increasing the thickness of the single-layer plane absorber decreases the absorptivity of the material, nevertheless, the model is the most highly utilized method the for determining material response to incident electromagnetic radiation in the microwave region, and appears to have been demonstrated experimentally [68,101,103].

As the value of the reflection coefficient tends towards 1, the value of the RL derived from experiment goes to the negative infinite, as the logarithm of zero is undefined and the limit of the logarithmic function is -∞. This concept is referred to as impedance matching, [67,104] where the six parameters of complex permittivity and permeability, frequency, and thickness, correlate in such a manner that results in a value of the reflection coefficient being equal to that of one. The theoretical material that possesses such characteristic is what is referred to as a perfect absorber [67,92].

Recently, Green et al. demonstrated a mathematical methodology for determining whether or not a material demonstrates perfect absorption [105,106]. By utilizing interpolating functions so to functionalize the permittivity and permeability of a material as a function of frequency, the input impedance can be reduced to a function of frequency and thickness, shown in Equation (19), only constrained by the frequency boundary conditions established from network analysis. With such framework, if decreasing the step size of the calculation by orders of magnitude towards zero around a point of interest demonstrates a linear response in the decrease of RL around said select area of analysis, then as the limit of the logarithm function goes to -∞ as the change in step goes to zero, so too does the RL of the material, satisfying the characteristic of impedance matching [67,105,106].

$$RL(dB) = 20\log_{10}(S_{11}) \tag{14}$$

$$\Gamma = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \tag{15}$$

$$Z_{in} = \eta \cdot \tanh(\gamma \cdot d) \tag{16}$$

$$Z_{in} = Z_0\left[\frac{\mu' - i\cdot\mu''}{\varepsilon' - i\cdot\varepsilon''}\right]^{\frac{1}{2}} \cdot \tanh\left(i\frac{2\pi f\cdot d}{c}[(\varepsilon' - i\cdot\varepsilon'')(\mu' - i\cdot\mu'')]^{\frac{1}{2}}\right) \tag{17}$$

$$RL(db) = 20\log_{10}(|\Gamma|) \tag{18}$$

$$Z_{in}(f,d) = Z_0\left[\frac{\mu'(f) - i\cdot\mu''(f)}{\varepsilon'(f) - i\cdot\varepsilon''(f)}\right]^{\frac{1}{2}} \cdot \tanh\left(i\frac{2\pi f\cdot d}{c}\{[\varepsilon'(f) - i\cdot\varepsilon''(f)][\mu'(f) - i\cdot\mu''(f)]\}^{\frac{1}{2}}\right) \tag{19}$$

The optimization of RL can furthermore be realized through the process of attenuation as the electromagnetic wave propagates through the material medium. Materials which have sufficiently large permittivity and permeability values will also demonstrate a strong attenuating response to incident electromagnetic radiation [75]. However, this specific mechanism of action should be contrasted against the understanding that the resulting increase in the complex parameters will have subsequent effects upon a material's ability to generate the impedance matching conditions, which results in a tendency to reflect the incident radiation off of the material boundary [97,107].

The effective bandwidth is the span of frequencies over which RL is greater than some determined threshold value [62]. Such a property is useful in systems where the frequency parameter that is to be engaged with is not precisely known, or the material interaction requires a light/matter action across a range of operable frequencies. Two of the common proficiency thresholds in the literature are −10.0 dB, equivalent to a 90% absorption threshold, and −20.0 dB, equivalent to a 99% absorption threshold, etc. The −10.0 dB threshold is the more common feature represented in the literature and thus is the proficiency parameter utilized herein. For example, as demonstrated in Fig. 4, if the RL curve for a material thickness of 3.0 mm starts at 2.0 GHz with an RL value of −2.0 dB, and proceeds with an upwards trajectory towards greater RL as a function of increasing frequency. Material response crosses

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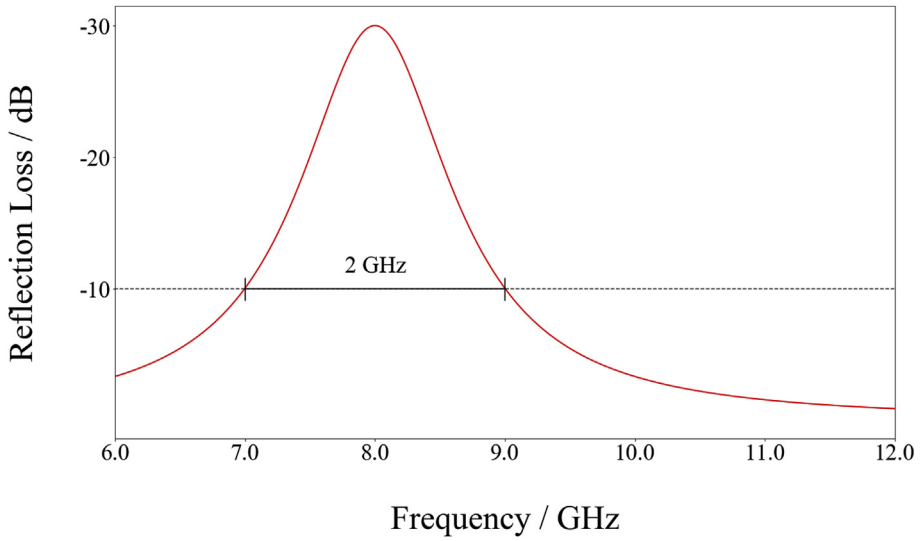


Fig. 4. Example RL curve (red), juxtaposed against the −10 dB proficiency threshold (dashed black), with the proficiency bandwidth (solid black) demarcated.

the −10.0 dB threshold at 7.0 GHz, from where the curve subsequently increases to reach a maximal RL point, and subsequently attenuates towards lesser RL as a function of increasing frequency, crossing the −10.0 dB threshold again at 9 GHz, and ending at −2.0 dB at 18.0 GHz. For this given RL curve the effective bandwidth is said to be 2.0 GHz. This analytical process can be repeated as a function of changing thickness parameters [62]. To note, some manuscripts report an 'effective bandwidth' as the span of frequencies which can be targeted by *a range* of matching thicknesses – this is not the effective bandwidth and readers should pay ample attention to the distinction.

In terms of associating the mechanism for RL to a nanomaterial's propensity for dielectric and magnetic interaction, one potential method for attempting to deduce the underlying causal pathway is to simulate the function response while setting the permeability parameters to the 'trivial' result of $\mu_r = (1 - i{\cdot}0)$ [104]. This 'zero magnetic susceptibility' calculation, where the magnetic susceptibility parameter χ_m is equivalent to zero, removes the consideration for the effects of the lossless and lossy interactions through the magnetic response to the nanomaterial from the calculation. Comparing the resultant simulation values to that of the experimental results allows for the comparison to be made between a hypothetical nanomaterial with zero magnetic susceptibility, and the authentic nanomaterial representation. Thus, if the simulated result is of similar magnitude to the authentic result, it is thus suggested that the propensity for RL is due to a magnetic interaction. Otherwise, the interaction is considered to be that of dielectric origin.

3.3. Common interaction pathways

A few mechanisms for electromagnetic interaction have been explicitly elucidated through either experimentation or permittivity and permeability analysis, and will be discussed briefly herein. Such mechanisms are Debye relaxation for dielectric interaction, and eddy current loss and natural resonance for magnetic interaction [59].

3.3.1. Debye relaxation

Debye relaxation is a process in which incident electric fields distort the normal state of matter in a way which generates polarization through either the distortion of the electron cloud, localized bond distortions, or the alignment of permanent dipoles [52,65,96]. In a polarized state, dipoles move and orientate against a frictional force which induces energy loss as heat. This process of polarization as a function of frequency is modeled by Debye theory. From Debye theory, the representation of permittivity as shown in Equation (20), where, f is the incident frequency, ε_s is the stationary dielectric constant equal to the limit of ε_r as f goes to zero, ε_∞ is the optical dielectric constant equal to the limit of ε_r as f goes to infinity, and τ is the relaxation time, reduces to in the plane of ε'' and ε' the form of a semicircle, commonly known as a Cole-Cole diagram [52,96]. The exponential variable a is an empirical constant introduced to allow for shape distortion, where in typical cases $a = 0$ and the Cole-Cole diagram takes semicircle form [52]. From this diagram, the relaxation time τ can be determined for a given operating frequency via Equation (21), where β is the slope of a line passing through the operating frequency point and ε_s, or the inverse slope of a line passing through the operating frequency point and ε_∞ [52]. τ can be subsequently calculated from β.

One of the consequences of Debye theory is that conductivity can have specific effects on lossless permittivity, as shown in Equation (22) [59]. Such demonstrates that increases in nanomaterial conductivity have significant effects on the permittivity resultants demonstrated by a given nanomaterial, inducing increases in their relative values, both lossless interaction and lossy interaction respectively as conductivity is defined as the product of angular frequency and lossy permittivity [52]. Thus, increasing the conductivity of given nanomaterials can have demonstratable effects on the materials' reflection loss.

3.3.2. Eddy current loss

Eddy current loss is a phenomenon where the varying of a magnetic field in a material medium induces voltages which drive a conduction current and subsequently Joule heating and power loss [71]. The eddy current loss contribution to reflection loss, as related to the electrical conductivity of the nanomaterial and the matching thickness, is represented by Equation (23), where μ_0 is vacuum permeability. If the magnetic loss is due to eddy current loss, a graphical plot of $d(\mu''/(\mu')^2 f)/df$ with respect to f will be zero over the tested frequency range, or, a graph of $(\mu''/(\mu')^2 f)$ with respect to f will remain flat. If such relationship is anything but, then the magnetic losses demonstrated by the nanomaterial are causal linked to some other mechanism of action [63].

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3.3.3. Natural resonance

Magnetic losses due to natural resonance can be determined by comparing the resonance peaks demonstrated via complex permeability with an analysis of anisotropy fields [56,96]. Various forms of the Landau-Lifshitz-Gilbert (LLG) equations (for example as shown in Equation (24) and Equation (25) for spherical magnetic particles, where f is the frequency, f_0 is the spin resonance frequency, α is the damping coefficient, and $A = M_s/H_a$ with M_s being the saturation magnetization and H_a being the anisotropy field) can be utilized to ultimately determine whether the permeability response is due to natural resonance, where good agreement between the theoretical model and the experimental results suggests natural resonance [96]. For more complicated systems, such as materials with c-plane anisotropy, [56] modified LLG equations can be used to elucidate natural resonance as the casual mechanism. Using the spin resonance frequency determined from permeability, the in-planar anisotropy field H_ϕ can be calculated via the Kittel equation $f_0 = \gamma \cdot sqrt(H_\theta \cdot H_\phi)$, assuming H_θ or the out-of-plane anisotropy field is known from further measurement. Furthermore, H_ϕ can be determined from the expression $\mu_i - 1 = M_s/2\pi H_\phi$, where μ_i is the shape factor modified Bruggeman effective medium theory; A correlation between the resultant H_ϕ values demonstrates that a resonance peak in the permeability results is likely due to natural resonance [56].

$$\varepsilon' - i\cdot\varepsilon'' = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{1 + (i\beta)^{1-a}}; \beta = \frac{\varepsilon_s + 2}{\varepsilon_\infty + 2}\cdot 2\pi f\tau \tag{20}$$

$$\varepsilon''(\varepsilon') = \beta(\varepsilon' - \varepsilon_\infty) = -\frac{1}{\beta}(\varepsilon' - \varepsilon_s) \tag{21}$$

$$\varepsilon' = \frac{\sigma}{(2\pi f)^2\cdot\tau\cdot\varepsilon_s} + \varepsilon_\infty \tag{22}$$

$$\mu'' = 2\pi\mu_0(\mu')^2 d^2 f\sigma \tag{23}$$

$$\mu' = 1 + \frac{A\left[1 - (1-\alpha^2)(f/f_0)^2\right]}{\left[1 - (f/f_0)^2\left(1+\alpha^2\right)\right]^2 + 4\alpha^2\left(f/f_0\right)^2} \tag{24}$$

$$\mu'' = \frac{\alpha A(f/f_0)\left[1 + (1+\alpha^2)(f/f_0)^2\right]}{\left(1 - (f/f_0)^2\left(1+\alpha^2\right)\right)^2 + 4\alpha^2\left(f/f_0\right)^2} \tag{25}$$

3.3.4. Size and shape factors

As previously mentioned, most materials which have been developed for the purposes of microwave absorption are of nanoscale morphology. The size and shape or morphology of nanomaterials can directly alter the dielectric/magnetic properties to provide a good match between them for microwave absorption. As the size of particulate matter is minimized towards the nanoscale, electromagnetic interaction tends to increase [55–60]. As the particle size decreases, parameters which are a function of nanoparticle domain, such as the material surface area or the quantity of active chemical sites, [96] are enhanced, and thus the effects which lead to dielectric or magnetic interaction, such as phase-dielectrics, [61] unpaired spin-states, [62] eddy-current losses, [51,63] resonance effects, [57] et al., are subsequently enhanced so to induce microwave absorption. Furthermore, phenomena such as interfacial polarization and multiple scattering can too be enhanced by the decrease in particle size due to magnifying effects of similar rationale [96].

Similar to that of particle size, the shape of a given nanomaterial also has considerable effect on the response it demonstrates to incident electromagnetic radiation. Typical nanoparticles utilized as microwave absorbing materials are of a simple structure, such as sphere, flake, et al. However, controlling shape factors beyond the simple manifestations, through forming more complex shapes such as disks, [96] rings, [175,208] nets, [340] and core-shell nanomaterials [105,106] are thought to help enhance microwave absorbing properties. Interfacial dipole rotations can be created by simply manipulating the shape and morphology of the nanomaterials. Plasmonic effects can be observed in nanomaterials with size and shape tuning to provide additional mechanisms for microwave absorption.

4. Nanomaterials for microwave absorption

The nanomaterials reviewed herein from the current pantheon of literature are divided into a series of subsets for categorization, based on the fundamental baseline structure that further chemical complexities are established. Many of the nanomaterials currently within the chemical lexicon are derivatives of the following systems: carbons, carbides, oxides, sulfides, phosphides, polymers, and metals/alloys. As such, this text elaborates on each nanomaterial classification. A final section of new nanomaterial developments is also included for the more recent nanomaterial domains published upon which fall outside of the conventional domains. Composite nanomaterials are classified by the fundamental material, i.e. a ferrite-doped carbon nanotube material will be predominately categorized as a carbon material. Material mixtures are classified by the highest quantity material in the highest-preforming material ratio reported herein. Regarding nomenclature within this manuscript, as the full nuance of the reported nanomaterials and their deviations would be cumbersome to fully list in the text, the following short-hand notation is adopted for convenience: dopant identities are separated by a colon punctuation mark and are furthermore separated from the fundamental nanomaterial with a '/' demarcation, with the fundamental nanomaterial listed last. In cases where multiple nanomaterials of similar chemical composition are referred to simultaneously, this demarcation is similarly used, with further clarification being provided in the associated tabulated value and in the referenced report. Composites added to a fundamental nanomaterial are tagged at the front of the nanomaterial name with a '|' demarcation. This allows, for example, a literature source reporting the microwave absorption results for a set of $SrFe_{11.2}Mn_{0.2}Cd_{0.2}Zr_{0.4}O_{19}$, $SrFe_{10.6}Mn_{0.4}Cd_{0.4}Zr_{0.8}O_{19}$, $SrFe_{10.0}Mn_{0.6}Cd_{0.6}Zr_{1.2}O_{19}$ and $SrFe_{9.2}Mn_{0.8}Cd_{0.8}Zr_{1.6}O_{19}$ nanomaterials to be collectively referred to as Mn:Cd:Zr/$SrFe_{12}O_{19}$, as the nanomaterials are all essentially a resultant perturbation of a strontium hexaferrite doped with various quantities of manganese, cadmium, and zirconium. Furthermore, if the nanomaterial set for example was integrated with the addition of polyaniline nanomaterial so to form a composite, the subsequent material reference would be PANI|Mn:Cd:Zr/$SrFe_{12}O_{19}$ to indicate the composite nature between the constituents, in this case the constituents being polyaniline added to the perturbed strontium hexaferrites. Atomic identities not separated by a colon are typically respective nanomaterial alloys. The '@' demarcation references the core-shell phenomenon, with the core material preceding, and the shell material proceeding, i.e. Co@C for a carbon-coated cobalt core/shell nanoparticle. All nanomaterials referenced herein are reported with respect to performance in both maximal reflection loss as well as the sum of the proficiency

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bandwidth over their reported testing ranges. Approximate relative permittivity and permeability values associated with the point of maximal reflection loss, as well as the frequency and material thickness, are also tabulated.

4.1. Carbons

Carbon nanostructures are one of the cornerstone solid-state nanomaterials upon which many further material derivations are applied. Intrinsically, their interaction with incident electromagnetic energy is understood to be purely electronic – so much so that in many cases the permeability values used in the literature are established by axiom. Carbon-based nanomaterials as a form of microwave-absorbing material tend to coalesce into distinct classes of carbon nanomaterials, such as carbon nanotubes, carbon fibers, and graphenes. There are exceptions however, and these materials will be discussed as well.

4.1.1. Carbon nanotubes (CNTs)

To isolate the effects of carbon-based microwave absorbing nanomaterials, Qi et al. [66] synthesized highly purified CNTs by chemical vapor deposition using a water-soluble potassium chromate catalyst coupled with an acetylene carbon source. These nanomaterials demonstrated a maximal RL value of −14.9 dB at 10.3 GHz with a 3.0 mm thick absorber, as shown in Fig. 5. The permittivity values at the point of maximal RL were shown to be $\varepsilon_r = (6.5 - i2.3)$ for the purified nanomaterials. Permeability was shown to be $\mu_r = (0.99 + i0.03)$, which is within a few hundredths of a.u. from the trivial result of $\mu_r = (1 - i0)$; the characteristic value of a nanomaterial with no magnetic interaction between it and an incident electromagnetic wave. The proficiency bandwidth of the purified nanotubes was recorded to be over a span of 2.9 GHz, which can also be observed on the color contour plot in Fig. 5, along with its deviations as a function of changing thicknesses. Standard catalyzed CNTs were synthesized as a comparison nanomaterial,[37] where the catalyst was not systematically removed, and it was found that the impurities due to the trace contamination inhibited the nanomaterials' RL performance; the raw CNTs were shown to achieve a −9.9 dB maximal RL at 10.3 GHz, shown in Fig. 5A. Permittivity at the point of maximal RL was shown to be

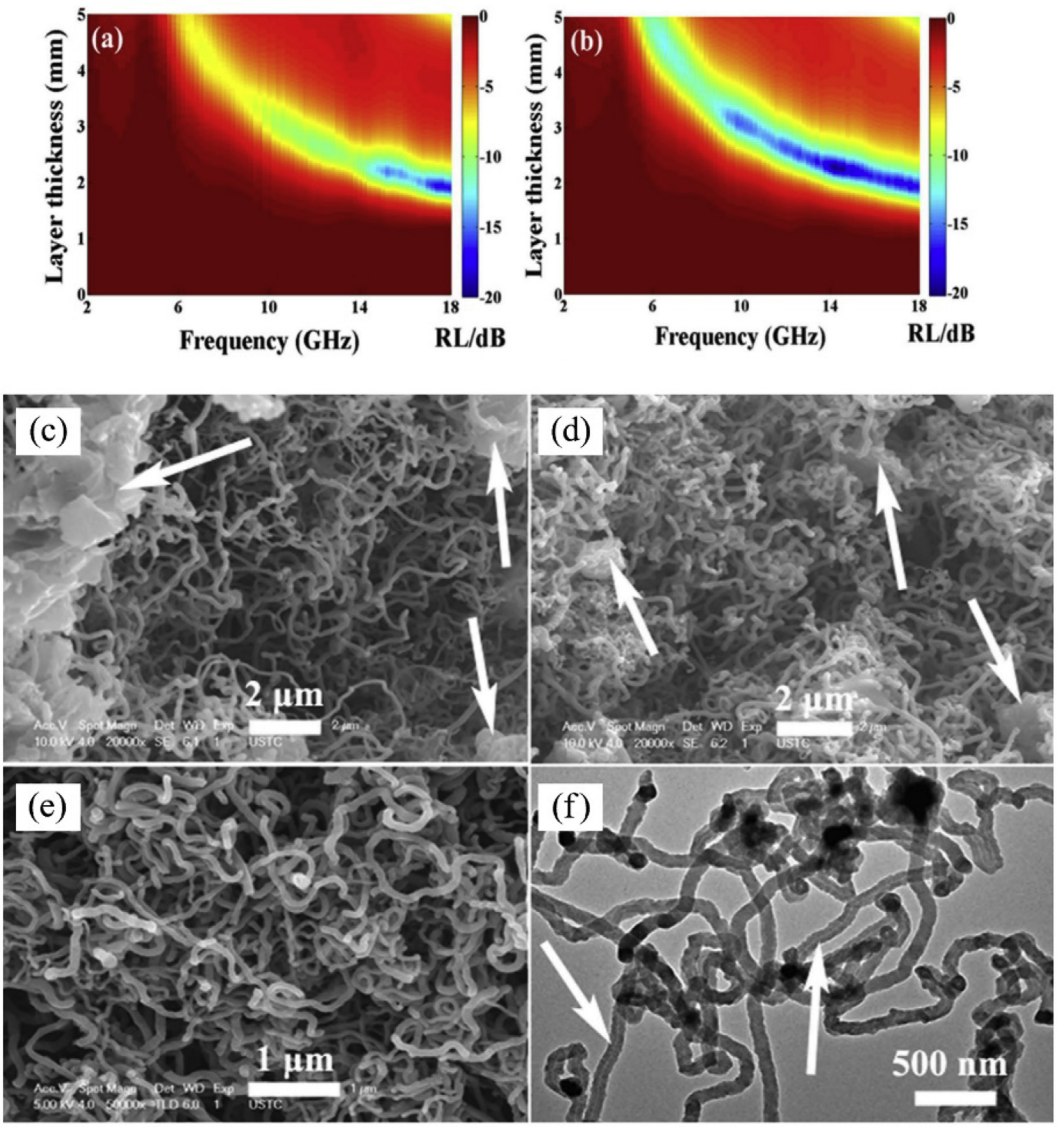


Fig. 5. Microwave absorption characteristics of the (a, c, d) raw and (b, e, f) purified carbon nanotubes. Reproduced with permission from ref. 66. Copyright 2016 Springer Nature.

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$\varepsilon_r = (6.3 - i1.7)$ for the raw nanomaterials, and permeability was shown to be $\mu_r = (0.97 + i0.02)$. The mechanism presented to give reason for the nanomaterial interaction was predominately attributed to dielectric loss, as the derived $\tan\delta_\varepsilon$ values attributed to the nanomaterial were significantly greater than the values associated with $\tan\delta_\mu$. This dielectric loss was attributed to the inherent increase in conductivity of the nanomaterial, as described by debye theory, whereby increasing the conductivity via the systematic removal of impurities induced the increase in permittivity that was ultimately responsible for the observed reflection loss [66].

Carbon nanotubes have been the foundational nanomaterial for multiple sets of dielectric and magnetic nanomaterials, such as Co/CNTs, [108] Fe/CNTs, [109–112] FeCo/CNTs [110] and FeCoNi/CNTs [110], $CoFe_2O_4$/CNTs, [113] and Fe/BN CNTs [114]. Generally, magnetic nanomaterials such as iron, nickel, and cobalt, in agglomerate form as well as in their -oxide derivates, have typically been selected as doping agents in an attempt to induce perturbation in the magnetic interaction via magnetic loss [68,74,115]. For example, Che et al. [109] synthesized a set of carbon nanotubes filled with iron material, which induced an increase in the magnetic interaction with electromagnetic radiation from the nanomaterial, beyond the trivial carbon permeability value of 1. The result of this coupling yielded a reflection loss value of −24.8 dB at 10.9 GHz with a 1.2 mm thick material, derived from the permittivity and permeability values of $\varepsilon_r = (29.7 - i40.7)$ and $\mu_r = (1.94 - i1.91)$. Furthermore, the material demonstrated therein had a proficiency bandwidth of 16.0 GHz. The full results of these materials and similar entities are tabulated in Table 1. Similar to that of carbon nanotubes, multiwalled carbon nanotubes (MWCNTs) have been developed with dopants to induce microwave absorption, through the generation of nanomaterials such as Ag/MWCNTs, [116] Ni/MWCNTs, [117] Fe:Fe_3C/MWCNTs, [118] Er_2O_3/MWCNTs, [119] and Sm_2O_3/MWCNTs [120]. The results of these materials and their proficiency manifestations are shown in Table 1. These nanomaterials utilize similar doping techniques so to change the electromagnetic interaction by integration of magnetic and dielectric constituents into the nanomaterial domain. For example, Xu et al. synthesized a set of MWCNTs by chemical vapor deposition, where after such synthesis the subsequent nanotubes were doped with ferromagnetic Fe/Fe_3C [118]. From such, a reflection loss of −14.1 dB was reported, at 4.6 GHz with a 3.5 mm material thickness. The addition of the metallic particulates demonstrated an increase in the permeability of the material, where the permeability value was reported to be $\mu_r = (1.37 - i0.55)$ at 4.6 GHz. The permittivity of the respected material was $\varepsilon_r = (16.0 - i3.9)$.

4.1.2. Carbon fibers (CFs)

The fundamental magnetic and dielectric interactions of CFs were demonstrated by Chu el al, [121] demonstrating that a solid variant yielded greater RL values compared to that of the hollow CFs. The fibers were generated by use of coaxial electrospinning, where electric potential to induce charge on the surface of a droplet of polymer solution, so to induce the flow of liquid through a spinneret, which ultimately forms long, ultrathin fibers, as shown in Fig. 6 [122]. The hollow fibers' reflection loss values were demonstrated to be −14.8 and −17.0 dB, at 5.0 and 6.4 GHz, for the 1.1 and 210 μm hollow CFs, respectively, and −14.9, −22.9, and −20.1 dB, at 6.9, 7.6, and 9.0 GHz, for the 1.08, 1.43, and 7.40 μm solid CFs, respectively, shown in Fig. 6E. The solid materials' frequency of maximum absorption appears to increase asymptotically as a function of the outer diameter of the CFs. The proficiency bandwidth associated with the solid fiber was consistent over the range of fiber diameters that were reported with proficiency, between 2.0 and 2.3 GHz. The proficiency bandwidth for the hollow CFs decreased as the fiber diameter decreased, from 1.8 GHz for the 210-micron fiber, to 1.1 GHz for the 1.1-micron fiber. Regarding the complex parameters, permeability was established axiomatically as $\mu_r = (1 - i0)$ given the nature of the nanomaterial being predominately carbon, which is known to not interact with the incident electromagnetic wave through magnetic interactions. Permittivity at maximal RL was shown to increase with a decrease of the fiber diameter along the real axis and remained relatively stable along the complex. The complex permittivity at the point of maximal RL for the highest preforming materials was $\varepsilon_r = (15.2 - i6.5)$ for the 210 μm hollow CFs, and $\varepsilon_r = (11.8 - i4.3)$ for the 1.43 μm solid cCFs, respectively. Absorption of the electromagnetic wave was characterized by interaction as a function of networked CFs within the material domain, which as consequence increased the associated complex permittivity, thereby inducing microwave absorption.

CF nanomaterials have furthermore been used in tandem with many doping materials, generating composites such as silica|CFs, [123] CoO_x|CFs [124] and Co|CFs, [124] CF@CoFe, [125] FeCo@CFs,

Table 1
Microwave Absorption Performance of Carbon Nanotube based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε′ at Max	ε" at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ" at Max	$tg\delta_\mu$ at Max	Ref.
raw CNT's	3.0	0	−9.9	10.6	6.4	1.6	0.25	0.98	−0.01	−0.01	[66]
purified CNT's	3.0	2.9	−14.9	10.3	6.5	2.3	0.35	0.99	−0.03	−0.03	[66]
CNT	1.0	3.4	−21.5	11.4	32.4	34.3	1.06	1.0	0.01	0.01	[108]
Co/CNT's	1.0	2.9	−22.9	12.2	22.9	9.9	0.43	1.1	0.08	0.07	[108]
Fe/CNT's (D)	1.2	16.0	−18.2	11.8	28.6	43.1	1.51	2.02	1.6	0.79	[109]
Fe/CNT (E)	1.2	16.0	−24.8	10.9	29.7	40.7	1.37	1.94	1.91	0.98	[109]
FeCo/CNT's	2.0	8.0	−15.5	9.5	n/a	n/a	n/a	n/a	n/a	n/a	[110]
FeCoNi/CNT's	2.0	5.6	−28.2	15.2	n/a	n/a	n/a	n/a	n/a	n/a	[110]
CNT's	3.5	2.9	−22.9	11.4	33.2	34.9	1.05	1.0	0.01	0.01	[111]
Fe/CNT's	3.5	2.9	−31.8	13.2	20.2	25.3	1.25	1.1	0.1	0.09	[111]
Fe/CNT's	3.5	4.2	−22.7	15.6	34.0	55.0	1.62	0.71	0.03	0.04	[112]
$CoFe_2O_4$/CNT's	1.4	7.0	−18.0	9.0	n/a	n/a	n/a	n/a	n/a	n/a	[113]
Fe_3O_4/BN CNT's	2.0	8.1	−42.2	14.5	3.8	0.9	0.24	0.89	0.26	0.29	[114]
Fe_3O_4/BN CNT's	2.5	13.1	−47.9	10.5	3.5	1.9	0.54	1.13	0.23	0.20	[114]
Ag/MWCNT's 10 wt%	1.0	1.8	−19.2	7.8	42.5	38.8	0.91	0.82	0.01	0.01	[116]
Ni/MWNT 5 wt%	4.0	4.4	−23.1	8.0	5.26	2.84	0.54	1.11	0.14	0.13	[117]
Ni/MWNT-10 wt%	4.0	3.4	−17.8	7.2	6.02	3.31	0.55	1.15	0.22	0.19	[117]
Fe/Fe_3C/MWCNT	2.0	2.4	−12.5	9.0	14.9	3.2	0.21	1.0	0.38	0.38	[118]
Fe:Fe_3C|MWCNT	3.5	2.3	−14.1	4.6	16.0	3.9	0.24	1.37	0.55	0.40	[118]
Er_2O_3/MWCNT	2.0	2.3	−27.7	10.0	14.8	4.8	0.32	0.99	−0.01	−0.01	[119]
Sm_2O_3/MWCNT	2.0	1.6	−21.5	9.4	10.0	3.4	0.34	0.89	0.02	0.02	[120]

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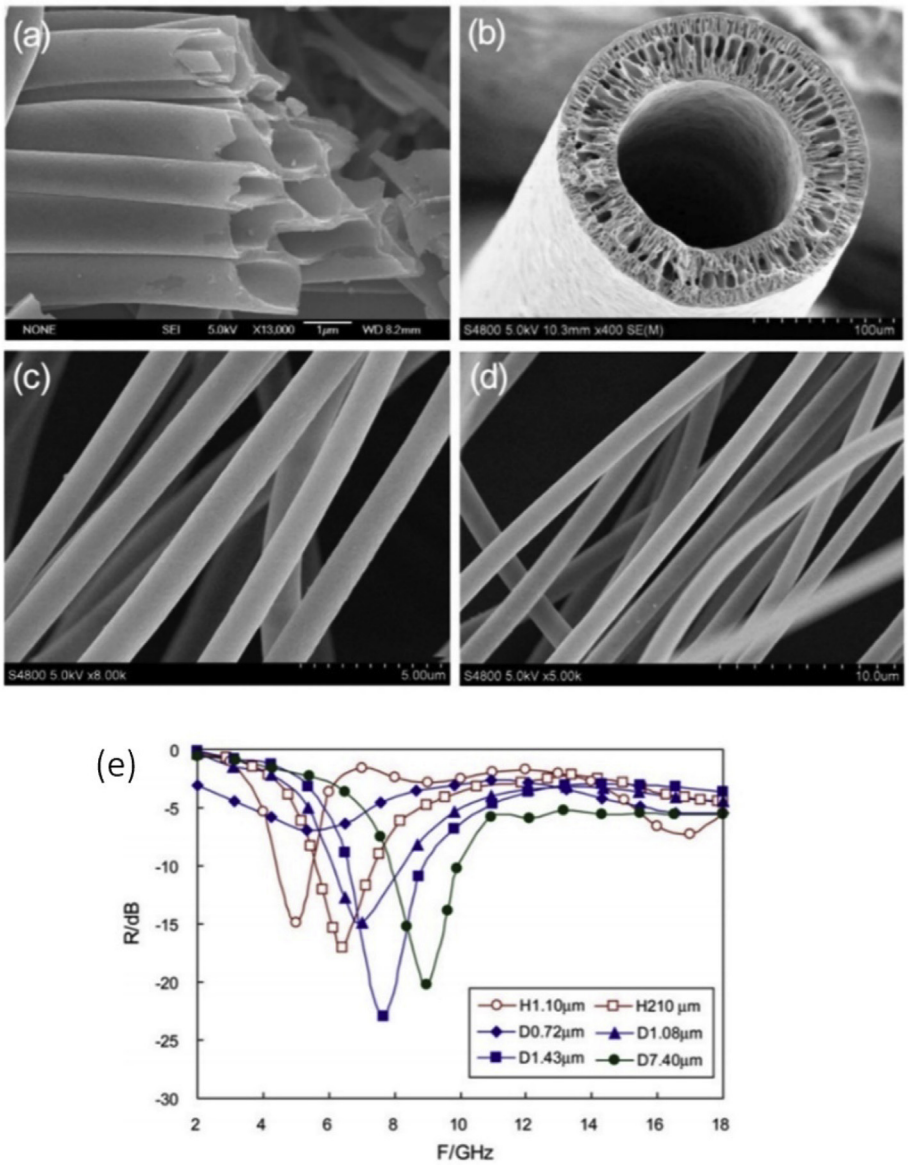


Fig. 6. (a, b) Hollow and (c, d) solid CFs, with (e) the associated RL curves demonstrated via a 3 mm thickness. Reproduced with permission from ref. 122. Copyright 2004 American Chemical Society.

[126] Fe|CFs, [127,128] Fe_3O_4/CFs, [129] Zn/CFs, [130] Zn:ZnO|CFs, [131] CuO/CFs, [132] and CuO:Co/CFs [132]. The introduction of impurities into the CF framework in some cases induced changes in the magnetic interaction to electromagnetic radiation by the nanomaterial, [133] for example via ferromagnetism, [124] though in some instances, such as the Fe_3O_4/metal-coated CF nanomaterial, [129] the tested material only demonstrated deviation in response as a function of the electronic interaction of the material, manifesting in a substantial electronic loss factor compared to a negligible magnetic loss [129].

In other cases, such as the carbonyl-iron doped CFs synthesized by Zhang et al. [127] and shown in Fig. 7, a spike in permeability was directly correlated with a point of maximal RL, with a −41.7 dB RL value being reported for a 2.0 mm thick absorber at 13.4 GHz. The associated permeability value for the reported maximal RL value was $\mu_r = (1.5 - i\ 0.75)$, compared to a similar-magnitude permittivity value of $\varepsilon_r = (4.9 - i1.7)$. The perturbation of the electromagnetic response doesn't appear to be guaranteed to be predominated by dielectric or magnetic interaction with respect to the causal action for RL, though the introduction of impurities and dopants does guarantee at least a change of the aforementioned responses. The tabulation of these nanomaterial responses is given in Table 2.

4.1.3. Graphene

Graphene-based nanomaterials were investigated by Wang et al. [134] for application as a usable microwave absorbing material. However, the nanomaterials intrinsic interaction with gigahertz-range electromagnetic radiation was demonstrated to be lacking, and the nanomaterial on its own was not considered to be a worthwhile microwave absorbing material. It was postulated by Wang et al. that the interaction characteristics through material doping could dramatically change the light/matter interaction so to induce absorption in the microwave region. This has been confirmed experimentally, [135] with the generation of many graphene-derived nanomaterials demonstrating a high-proficiency

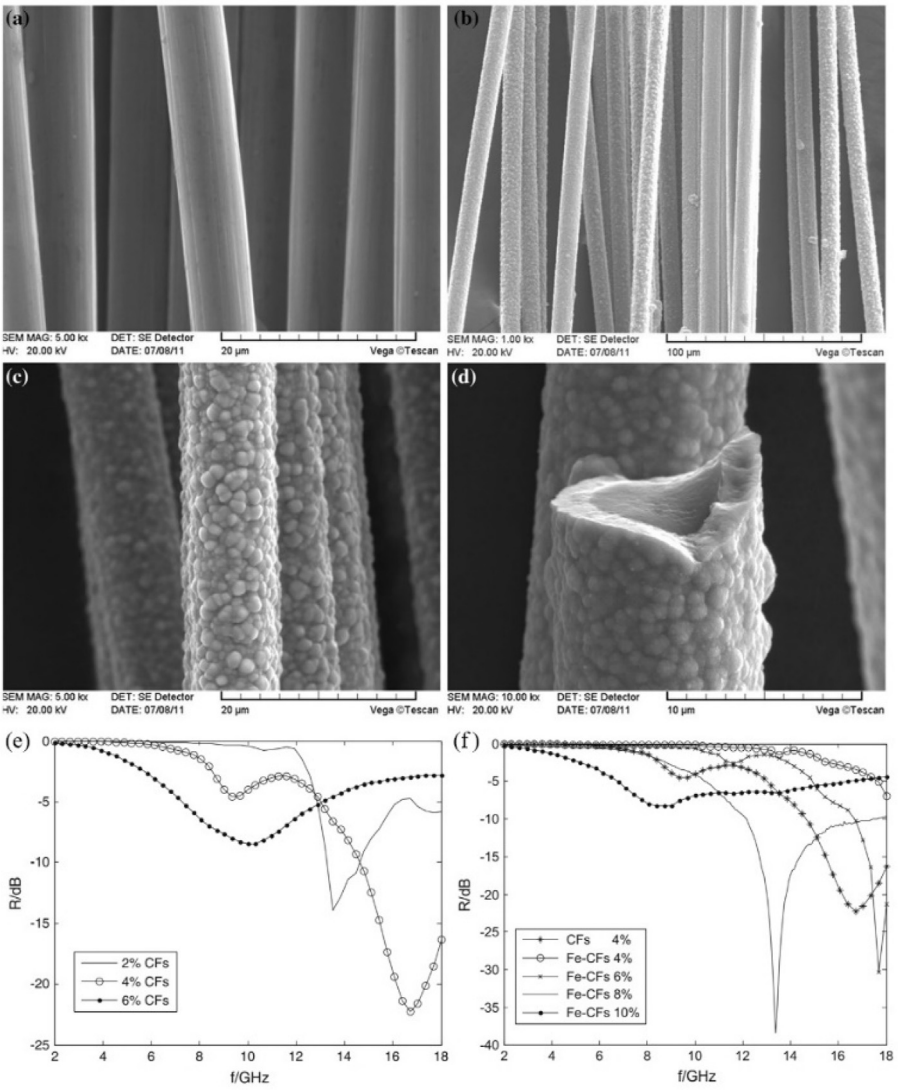


Fig. 7. SEM images of (a) pristine and (b–d) carbonyl iron-doped CFs, with associated RL curves (pristine, e, and doped, f). Reproduced with permission from ref. 127. Copyright 2015 Springer US.

Table 2
Microwave Absorption Performance of Carbon Fiber based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε′ at Max	ε" at Max	tgδ$_\varepsilon$ at Max	μ' at Max	μ" at Max	tgδ$_\mu$ at Max	Ref.
H1.1 μm CFs	3.0	1.1	−14.8	5.0	27.3	8.7	0.32	1	0	0.00	[121]
H210μm CFs	3.0	1.8	−17.0	6.4	15.2	6.5	0.43	1	0	0.00	[121]
D1.43 μm CFs	3.0	2.3	−22.9	7.6	11.7	4.5	0.38	1	0	0.00	[121]
D7.40 μm CFs	3.0	2.0	−20.1	9.0	10.0	4.0	0.40	1	0	0.00	[121]
Silica/CFs	5	0.9	−10.2	9.9	n/a	n/a	n/a	n/a	n/a	n/a	[123]
CoO_x/CF	1.52	3.6	−45.2	13.4	10.8	1.1	0.10	0.5	1.0	2.00	[124]
CoFe/CF-25	1.8	2.0	−37.7	9.0	22.5	4.6	0.20	1.06	0.07	0.07	[125]
FeCo@CFs	1.4	3.8	−44.8	18	8.1	3.3	0.41	1.1	0.05	0.05	[126]
FeCo@CFs	1.6	5.9	−47.5	16.6	8.7	3.3	0.38	1.1	0.06	0.05	[126]
Fe/CFs (8 wt%)	2.0	6.5	−41.7	13.4	4.9	1.7	0.35	1.5	0.75	0.50	[127]
Carbonyl Iron/CF	1.5	8.0	−22.9	9.7	22.0	1.05	0.05	2.05	1.67	0.81	[128]
Fe_3O_4/MCCF-2	3	0.8	−28.9	4.6	29.2	6.0	0.21	1.0	0.04	0.04	[129]
Fe_3O_4/MCCF-2	4	0.6	−30.0	3.4	29.9	6.3	0.21	1.03	0.04	0.04	[129]
Zn/CF's	4.35	0.7	−32.9	3.3	24.2	6.6	0.27	1.1	0.0	0.00	[130]
Zn/CF's	4.0	0.8	−27.2	3.7	24.1	6.7	0.28	1.1	0.0	0.00	[130]
Zn/CF's	3.6	0.9	−24.0	4.0	24.0	6.8	0.28	1.1	0.0	0.00	[130]
Zn:ZnO/CF's	4.5	0.8	−39.4	3.23	24.2	6.56	0.27	1.13	0.0	0.00	[131]
CuO/CFs	1.9	1.3	−24.3	7.2	n/a	n/a	n/a	n/a	n/a	n/a	[132]
CuO/Co/CFs	2.0	4.0	−42.7	10.75	n/a	n/a	n/a	n/a	n/a	n/a	[132]

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in absorbing microwave radiation. Polymer-enhanced materials such as PEO|graphene [136] and PANI|graphene, [137] as well as oxide-based dopants integrated into graphene nanomaterials such as γ-Fe_2O_3/rGO [138] and Fe_3O_4/rGO, [139,140] Fe_3O_4|graphene [141] and other doping materials such as CNT|GN, [142] Silica|rGO, [143] Ni/GN, [144] Ni/rGO, [145] NiO/rGO, [146] $ZnFe_2O_4$/rGO, [147] $NiFe_2O_4$/rGO, [148] C@Ni|graphene, [149] $MnFe_2O_4$/rGO, [150] Co_3O_4/rGO, [151] Fe_3O_4:ZnO/rGO, [152] $CoFe_2O_4$/rGO, [153,154] $CoFe_2O_4$:SnS_2/rGO, [155] and MoS_2/rGO [156] have been used to successfully induce interaction with electromagnetic radiation at the microwave region. As the graphene, similar to other predominately carbon-based microwave absorbers, demonstrates minimal interaction with the magnetic component of an incident electromagnetic wave as shown through the permittivity and permeability results presented in analysis, the quality of the dopant and its intrinsic interaction with the graphene is the driver for interaction between the synthesized nanomaterials and the incident electromagnetic radiation, in both dielectric and magnetic fashion [157–162]. Such a case can be seen by example of the rGO|Fe_3O_4 composite nanomaterials synthesized by Sun et al. [139] as shown in Fig. 8, where the graphene and Fe_3O_4 particles were shown to have compensatory properties due to the introduction of the magnetic Fe_3O_4 particles into the graphene domain. The materials presented therein demonstrated strong RL values, such as −26.4 dB with a 4.0 mm thick absorber at 5.3 GHz [139].

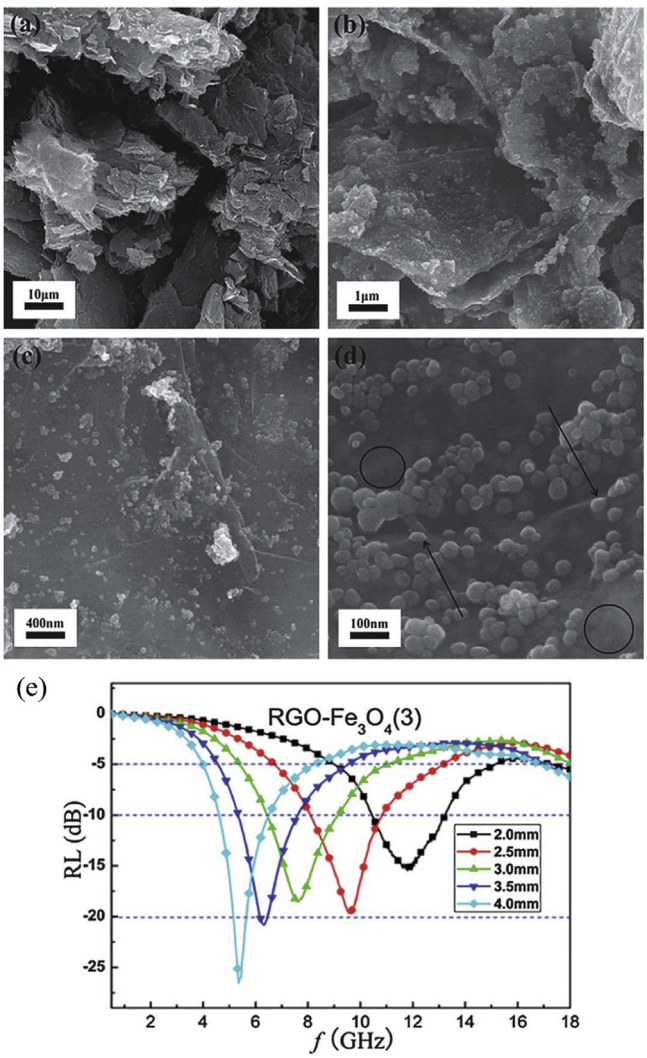


Fig. 8. SEM images of rGO (a) and rGO–Fe_3O_4 (3) composite with (b–d) different magnifications, and (e) corresponding and rGO–Fe_3O_4 (3) RL curves. Reproduced with permission from ref. 139. Copyright 2013 Royal Society of Chemistry.

Other graphene-like nanomaterials such graphene foams [163,164] have been shown to have capacity to interact with gigahertz-range electromagnetic radiation through interaction with the incident electric portion of the EM field. Such interaction is independent of an introduced doping material. For example, Zhang et al. [163] demonstrated graphene foam's propensity for electromagnetic interaction, generating reflection loss values of −28.2 dB with a 2.0 mm thick absorber at 12.3 GHz. Over the 1–18 GHz frequency range the material's proficiency bandwidth was reported to be 8.9 GHz. Furthermore, Zhang et al. demonstrated nanomaterial response at higher frequencies, where the synthesized graphenes were shown to have full bandwidth coverage, for example over the 75–110 GHz frequency range. The full results associated with electromagnetic interactions by graphene-based nanomaterials is given in Table 3.

4.1.4. Amorphous and other carbon materials

Amorphous carbon nanomaterials were reported to also possess intrinsic absorption properties by Liu et al [165]. Liu demonstrated that the nanomaterial had a propensity to interact with the electric portion of gigahertz-range electromagnetic radiation. RL values for various thicknesses of the carbon materials were demonstrated to be −23.6 dB at 17.5 GHz with a 2.0 mm thickness, −18.5 dB at 13.2 GHz with a 2.5 mm thickness, −15.8 dB at 10.4 GHz with a 3.0 mm thickness, −14.1 dB at 8.6 GHz with a 3.5 mm thickness, −13.1 dB at 7.2 GHz with a 4.0 mm thickness, −12.0 dB at 6.2 GHz with a 4.5 mm thickness, and finally −11.4 dB at 5.3 GHz with a 5.0 mm thickness, as shown in Fig. 9 [165]. These results were contrasted against nanomaterials which integrated magnetite into the material domain, of which the effect of magnetic integration demonstrated an enhancement in the materials overall efficiency if absorbing incident microwaves, with RL values for the Fe_3O_4/C going to −33.9 dB at 15.5 GHz, with a 2.5 mm thickness at a 1:1 loading ratio by weight, as shown in Fig. 9. Interesting to note, the nanomaterials characterized by the introduction of magnetite didn't demonstrate much of a magnetic interaction as described by permeability, as the magnetic loss was shown to be negligible [165]. It was instead the decrease in the permittivity values associated with doping which resulted in the increase in material RL, which is characteristic of impedance matching conditions [97]. The comparable reflection loss curves for this study are furthermore tabulated in Table 4.

Other synthesized nanomaterials have been structured upon carbon-based foundational nanomaterials, such as mesoporous carbon, [166] Fe:SiO_2/mesoporous carbon, [167] carbon nanocoils, [168] carbon nanobelts, [169] graphite, [170] PANI|carbon black, [171] Fe_3O_4/C, [165,172] and Ni_2O_3/C [173]. The pure carbon nanomaterials, such as mesoporous carbon, tend to have strong interaction with the electric portion of incident electromagnetic radiation, yielding permittivity values at the point of maximal RL as $\varepsilon_r = (16.1 - i4.7)$, with permeability values associated with the same frequency being very close to unity [166]. Furthermore, the doped carbon nanomaterials tend towards generating strong RL values when magnetic nanomaterials are integrated into to the dielectric material domain, by the perturbation of the nanomaterial's interaction with incident electric and magnetic fields via synergistic effects of the heterogeneous nanomaterials. For example, the results shown by Liu et al. demonstrated that although the addition of Magnetite into an amorphous carbon matrix introduced magnetic nanomaterials into the dielectric domain, the net effect of the addition had little perturbation on the magnetic action of the overall material, though the decrease in the

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Table 3
Microwave Absorption Performance of Graphene based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε′ at Max	ε" at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ" at Max	$tg\delta_\mu$ at Max	Ref.
rGO	2.0	0	−7.1	7.1	42	31	0.74	0.91	0.0	0.00	[134]
PEO|graphene	2.5	4.8	−37.8	16.4	6.8	3.4	0.50	1	0	0.00	[136]
PANI|graphene	3.5	5.3	−36.9	10.3	n/a	n/a	n/a	n/a	n/a	n/a	[137]
γ-Fe_2O_3/rGO	2.5	2.0	−59.7	10.1	9.9	3.7	0.37	0.94	0.01	0.01	[138]
Fe_3O_4|rGO	4.0	2.0	−26.4	5.3	12.3	1.9	0.15	1.14	0.16	0.14	[139]
Fe_3O_4/GN	2.0	2.7	−18	5.2	n/a	n/a	n/a	n/a	n/a	n/a	[140]
Fe3O4|graphene 10 wt%	2.0	3.0	−27.5	18	4.9	2.5	0.51	1.01	0.053	0.05	[141]
Fe3O4|graphene 10 wt%	4.0	3.6	−27.1	8.1	5.8	2.4	0.41	1.00	0.054	0.05	[141]
CNT/GN	3.0	3.3	−44.6	8.7	8.6	3.1	0.36	1.02	0.11	0.11	[142]
Silica|rGO 4.1% wt	3.3	3.1	−36.7	12.3	3.8	2.4	0.63	1	0	0.00	[143]
Silica|rGO 4.1% wt	3.5	3.8	−28.4	11.5	3.8	2.6	0.68	1	0	0.00	[143]
Ni/GN	5.0	1.0	−17.8	3.5	9.5	2.48	0.26	0.98	0.06	0.06	[144]
Ni/rGO	2.0	3.9	−34.3	16.3	6.0	1.5	0.25	1.0	0.0	0.00	[145]
Ni/rGO	2.0	2.5	−42.5	17.8	5.2	1.5	0.29	0.9	0.0	0.00	[145]
NiO/rGO	5.0	8.6	−55.5	10.6	4.3	1.25	0.29	1.08	0.05	0.05	[146]
$ZnFe_2O_4$/rGO	2.5	2.6	−41.1	9.3	11.6	3.7	0.32	0.94	0.07	0.07	[147]
$NiFe_2O_4$/rGO	3.0	3.0	−39.6	9.2	8.0	3.0	0.38	1.0	0.1	0.10	[148]
$NiFe_2O_4$/rGO	1.9	5.2	−36.9	14.5	9.0	1.6	0.18	0.78	0.2	0.26	[148]
C@Ni|graphene	1.6	3.2	−34.2	14.1	11.8	4.2	0.36	0.87	−0.09	−0.10	[149]
$MnFe_2O_4$/rGO	3.0	4.9	−29.0	9.2	6.4	3.2	0.50	1.22	0.1	0.08	[150]
Co_3O_4/rGO	3.3	4.6	−43.7	13.8	4.1	1.0	0.24	0.98	0.02	0.02	[151]
Fe_3O_4:ZnO/rGO	5.0	1.4	−35.0	11.0	n/a	n/a	n/a	n/a	n/a	n/a	[152]
$CoFe_2O_4$/rGO	2.5	4.2	−37.2	11.6	7.1	2.8	0.39	1.0	0.1	0.10	[153]
$CoFe_2O_4$/rGO	2.3	4.1	−47.8	12.4	7.5	3.0	0.40	1.0	0.0	0.00	[153]
$CoFe_2O_4$/rGO	1.6	4.5	−44.1	14.7	9.4	9.3	0.99	0.98	0.5	0.51	[154]
$CoFe_2O_4$:SnS_2/rGO	1.6	12	−54.4	16.5	7.3	2.7	0.37	1.0	0.1	0.10	[155]
MoS_2/RGO	2.3	5.6	−50.9	11.7	8.2	3.3	0.40	n/a	n/a	n/a	[156]
Graphene Foam	2.0	8.9	−28.2	12.3	n/a	n/a	n/a	n/a	n/a	n/a	[163]

material permittivity from such doping action yielded a strong RL [165]. These results are tabulated in Table 4.

4.2. Carbide

Silicon carbide particles were mechanically processed via a high energy planetary ball mill over interval timespans by Kumar et al [55]. The mechanical process generated a series of nanometer-scale materials with average particle sizes that decreased asymptotically as a function of time. The RL of the milled materials does not appear to converge on a functionable trendline, though specific materials did demonstrate respectable magnitudes for RL as a function of interaction of the material with the electric component of the incident electromagnetic wave. The premium RL values reported therein were −50.7, −43.6, and −27.1 dB, at 9.4, 9.8, and 10.5 GHz, for the 3hr-, 2hr-, and 20hr-milled silicon carbide nanoparticles, respectively. Permeability stayed within a ~0.04 and ~0.2 a.u. range from zero magnetic susceptibility for the real and imaginary components, respectively, over the presented 8–12.5 GHz frequency range. The permeability values associated with the given carbides' maximal RL were reported as $\mu_r = (1.098 - i{\cdot}0.073)$, $\mu_r = (1.094 - i{\cdot}0.084)$, and $\mu_r = (1.18 - i\ 0.19)$ for the 3, 2, and 20 h milled materials. The corresponding relative permittivity values were $\varepsilon_r = (16.05 - i{\cdot}4.25)$, $\varepsilon_r = (15.2 - i{\cdot}3.94)$, and $\varepsilon_r = (13.9 - i{\cdot}2.85)$ for the milled silicon carbide nanomaterials. Due to such, the material processing was determined to induce an increase in the dielectric loss of the materials, resulting in the electronic interaction being the causal driver for microwave absorption [55].

Kumar et al. also demonstrated within their report the effects of material doping of their SiC nanoparticles, utilizing aluminum, cobalt, chromium, manganese, nickel, titanium, and zinc impurities for material perturbation [55]. Similar deviations in permittivity and permeability were reported, where relative permittivities of $\varepsilon_r = (16.5 - i{\cdot}4.3)$, $\varepsilon_r = (12.2 - i{\cdot}6.4)$, $\varepsilon_r = (15.1 - i{\cdot}5.2)$, $\varepsilon_r = (15.1 - i{\cdot}4.9)$, $\varepsilon_r = (16.5 - i{\cdot}4.2)$, $\varepsilon_r = (15.6 - i{\cdot}5.3)$, $\varepsilon_r = (15.0 - i{\cdot}5.2)$, and permeabilities of $\mu_r = (1.06 - i{\cdot}0.01)$, $\mu_r = (1.08 + i{\cdot}0.13)$, $\mu_r = (1.11 - i{\cdot}0.01)$, $\mu_r = (1.11 - i{\cdot}0.00)$, $\mu_r = (1.11 - i{\cdot}0.13)$, $\mu_r = (1.09 + i{\cdot}0.02)$, and $\mu_r = (1.09 + i{\cdot}0.02)$, were reported for the aluminum, cobalt, chromium, manganese, nickel, titanium, and zinc composites, respectively. RL values for the given materials were reported as −19.6 dB at 8.3 GHz for the 2.2 mm Al/SiC absorber, −24.3 dB at 11.6 GHz for the 1.8 mm Co/SiC absorber, −37.8 dB at 10.9 GHz for the 1.7 mm Cr/SiC absorber, −43.4 dB at 10.3 GHz for the 1.8 mm Mn/SiC absorber, −24.9 dB at 11.8 GHz for the 1.5 mm Ni/SiC absorber, −40.2 dB at 8.7 GHz for the 2.1 mm Ti/SiC absorber, and finally −43.7 dB at 9.4 GHz for the 2.0 mm Zn/SiC absorber. These material results are shown in Fig. 10, and are tabulated in Table 5.

Silicon carbide as a foundational material have been utilized in the formation of composites such as Ni/SiC, [174] NiO/SiC, [175] Ni:Co:P/SiC, [176] Co/SiC, [177] Fe/SiC, [178] Al/SiC, [179] carbon black|SiC, [180] CNT|Ni/SiC, [181] CF|SiC, [182] and graphene|SiC [183]. Magnetic atomic dopants consistently did not induce additional lossy interactions through permeability, as the permeability values tended to stay at or near the trivial value. The majority of the studies presented indicate that the interaction of the materials tends to be predominately achieved through the dielectric action of the doped SiC materials, and the dopants were successful in altering the material's propensity to interact as a dielectric. These results are tabulated in Table 5.

4.3. Oxides

Oxide based nanomaterials have a wide range of responses to gigahertz-range electromagnetic radiation, due to the functional capacity of the metal center to which the oxygen anion is bonded to, [184–189] as well as the structural manifestation of the material as determined by the crystalline lattice of the material phase [190–193]. As such, these materials have a wide array of potential manifestations which will interact with gigahertz-range electromagnetic radiation, through both electronic and magnetic pathways.

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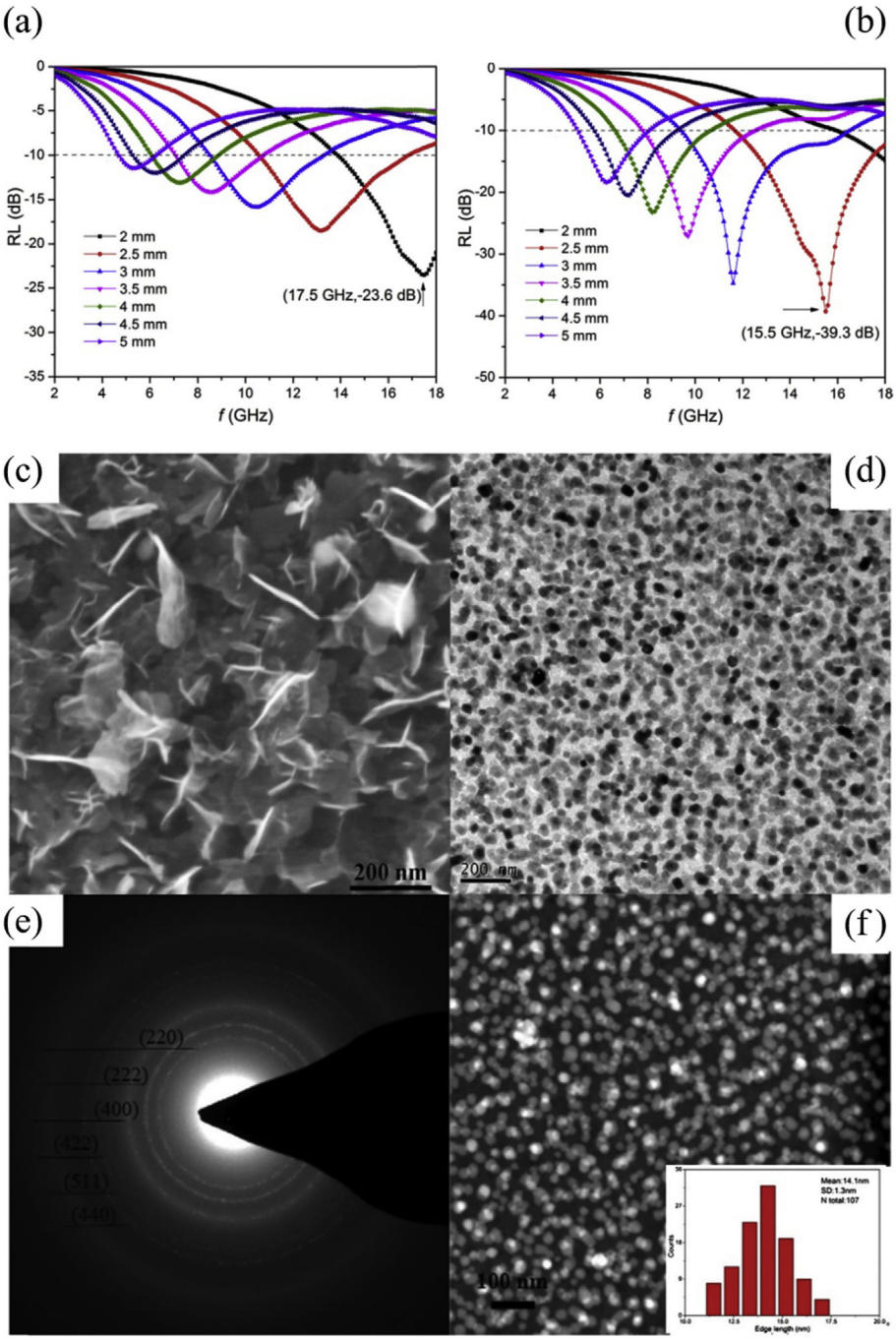


Fig. 9. RL curves for amorphous carbon (a), Fe_3O_4/C (b), and the SEM (c), TEM (d), SAED (e), and HAADF (f) images of the Fe_3O_4/C nanomaterial of 1:1 ratio. Insert is the particle size distribution. Reproduced with permission from ref. 165. Copyright 2015 Elsevier.

4.3.1. Iron oxide

Much of the current research in the field of materials study for microwave absorption can be traced back to the work of Naito et al. [67,92] regarding the use of ferrites as a potential material for electromagnetic wave absorption in the MHz region of the electromagnetic spectrum. The iron species in the oxide materials such as magnetite acts ferromagnetically, where the non-zero spin states of the ferrite due to unpaired electrons induce a propensity to interact with external magnetic fields from sources such as incident electromagnetic radiation [67,74,92]. Modern studies have built upon this understanding and applied the principles such as reflection loss to the gigahertz region of the electromagnetic spectrum.

For example, Ni et al. synthesized both a set of Fe_3O_4 nanocrystals [194] and microspheres [195] so to investigate the magnetic interactions of magnetite with gigahertz range electromagnetic radiation in detail. Both the nanocrystals, as shown in Fig. 11A–D, and the Fe_3O_4 microspheres were synthesized by a hydrothermal method, using ammonium iron (II) sulfate hexahydrate as the metal source, though sodium fluoride was added in the synthesis of the microspheres so to utilize its reducing properties. Both the nanocrystals and the microspheres

Table 4
Microwave absorption performance of amorphous and other carbon-based nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε' at Max	ε'' at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ'' at Max	$tg\delta_\mu$ at Max	Ref.
Amorphous Carbon	2.0	4.1	−23.6	17.5	5.1	3.2	0.63	1.01	−0.019	−0.02	[165]
Amorphous Carbon	2.5	6.2	−18.5	13.2	5.6	3.7	0.66	1.02	−0.002	0.00	[165]
Fe_3O_4/C	2.5	6.3	−39.3	15.5	4.5	2.5	0.56	1.04	−0.019	−0.02	[165]
Fe_3O_4/C	3.0	7.1	−34.7	11.6	4.9	2.8	0.57	1.03	0.002	0.00	[165]
Mesoporous Carbon (600 °C)	2.0	3.1	−24.9	17.8	4.3	2.7	0.63	1.0	−0.1	−0.10	[166]
Mesoporous Carbon (600 °C)	2.2	5.2	−27.1	16.1	4.7	2.9	0.62	1.0	0.09	0.09	[166]
Mesoporous Carbon (600 °C)	2.5	6.7	−21.5	14.3	5.0	3.2	0.64	1.0	−0.07	−0.07	[166]
Fe:SiO_2/mesoporous carbon	2.0	5.0	−20.5	16.6	3.5	1.0	0.29	1.14	0.60	0.53	[167]
Fe:SiO_2/mesoporous carbon	2.0	5.7	−20.7	13.5	7.5	2.2	0.29	0.98	−0.08	−0.08	[167]
Fe:SiO_2/mesoporous carbon	2.0	5.0	−34.4	13.1	8.6	1.9	0.22	0.98	−0.06	−0.06	[167]
Carbon Nanocoils	9.6	4.7	−32.3	12	19.6	12.3	0.63	1.0	0.02	0.02	[168]
Carbon Nanobelts (PF-BK-B)	1.2	2.7	−19.7	17.6	10.4	4.9	0.47	1.1	−0.1	−0.09	[169]
Carbon Nanobelts (PF-BK-B)	1.8	3.1	−19.5	10.9	11.9	5.1	0.43	1.1	0	0.00	[169]
Graphite 4hr milling	2.0	2.0	−12.0	14.4	n/a	n/a	n/a	n/a	n/a	n/a	[170]
Graphite 8hr milling	2.0	3.5	−19.2	14.9	n/a	n/a	n/a	n/a	n/a	n/a	[170]
Graphite 12hr milling	2.0	2.9	−25.3	14.5	n/a	n/a	n/a	n/a	n/a	n/a	[170]
PANI|Carbon Black	2.0	2.9	−40.0	11.3	n/a	n/a	n/a	n/a	n/a	n/a	[171]
Fe_3O_4/C	3.4	1.2	−45.0	6.2	11.4	2.9	0.25	1.07	0.12	0.11	[172]
Ni_2O_3/C	2.0	5.5	−33.5	15.0	7.8	5.2	0.67	0.79	−0.21	−0.27	[173]
Ni_2O_3/C	2.5	3.4	−40.5	10.9	9.0	4.3	0.48	0.86	−0.10	−0.12	[173]

demonstrated high permittivity values, associated with the presence of the Fe^{2+} ions in the material domain, much like that of classic Magnetite [194,195]. Similarly, the permeability values associated with the materials were elevated beyond the trivial value, indicating contribution to electromagnetic absorption through magnetic interaction [194]. For the nanocrystals, maximal RL values were reported at −29.5 dB at 3.9 GHz, −26.9 dB at 5.4 GHz, and −21.1 dB at 8.2 GHz, for the 5.0, 4.0, and 3.0 mm thicknesses respectively at 30 wt% in paraffin, shown in Fig. 11. The permittivity and permeability values associated with these points of maximal RL were reported as $\varepsilon_r = (9.2 - i0.2)$ and $\mu_r = (1.5 - i0.72)$, $\varepsilon_r = (9.3 - i0.3)$ and $\mu_r = (1.2 - i0.52)$, and $\varepsilon_r = (9.5 - i0.4)$ and $\mu_r = (0.9 - i0.38)$, for the 5.0, 4.0 and 3.0 mm nanocrystal materials respectively. For the microspheres, maximal RL values were reported as −44.7 dB at 4.7 GHz, −27.8 dB at 5.4 GHz, and −23.3 dB at 6.9 GHz, for 4.0, 3.5, and 3.0 mm thicknesses at 40% wt in paraffin. The permittivity and permeability values associated with these points of maximal RL were reported as $\varepsilon_r = (11.0 - i1.3)$ and $\mu_r = (1.3 - i\,0.58)$, $\varepsilon_r = (11.0 - i1.4)$ and $\mu_r = (1.23 - i0.54)$, and $\varepsilon_r = (11.1 - i1.6)$ and $\mu_r = (1.06 - i\,0.0.45)$, for the 4.0, 3.5 and 3.0 mm microsphere materials respectively. The results for materials analysis are reported in Table 6A.

More recently, undoped ferrites have been generated as Fe_3O_4 nanowires [196] and nanosheets,[166] and even synthesized from ore cinder to generate the Fe_3O_4 from industrial waste [197]. Furthermore, simple ferrite nanomaterials have been both doped and utilized in composite synthesis for systems such as Ni:B/Fe_3O_4, [198] Cu:Co:Ni:Zn/Fe_2O_4, [199] Ni:Co/Fe_2O_4, [200] Co:Mn/Fe_2O_4, [201] Bi:La/FeO_3, [202] PANI|Co:Zn/Fe_2O_4, [203] PANI|Li:Zn/Fe_3O_4, [204] C|Fe_3O_4, [205–208] ZnO|Fe_3O_4, [209] Fe:Mn/Fe_3O_4, [210] and Fe-phthalocyanine oligomer/Fe_3O_4, [211] in attempt to utilize iron's propensity for inducing a magnetic interaction through magnetic interaction so to drive reflection loss [212]. This effect however is not always guaranteed, as many of the synthesized nanomaterials demonstrate permeability values close to trivial, as shown in Tables 6A and 6B. For example, Liu et al. [207] generated a set of Fe_3O_4@C core/shell nanoparticles and nanosheets, generated by calcinating α-Fe_2O_3 in acetone, and examined their electromagnetic response via permittivity and permeability. The resultant nanomaterial had non-trivial magnetic interaction at the low end of the frequency spectrum, but quickly reduced to the trivial result of near $\mu_r = (1 - i0)$ at the higher end. It was at this higher end of the frequency spectrum where maximal RL was predominantly achieved, with nanomaterials such as 60 nm Fe_3O_4@C nanosheets generating RL values of −41.9 dB at 12.8 GHz with a 2.0 mm material thickness [207]. The results of this research and from materials of similar quality can be found tabulated in Table 6A. The introduction of differing magnetic materials such as cobalt and nickel have been shown to affect the nanomaterial's interaction with the magnetic portion of electromagnetic radiation, and the addition of dielectric nanomaterials such as manganese induces perturbation on the nanomaterial's propensity to interact with the electric portion of the wave [176]. Furthermore, complex formations of iron oxides have also been utilized as absorbing nanomaterials, [115,213–215] where nanomaterials such as Nd/$BiFeO_3$, [216] Ni:Co/$BaTiFe_{10}O_{19}$, [217] Mn:Ni:Co/$BaTiFe_{10}O_{19}$, [218] ZnO|$BaFe_{12}O_{19}$, [219] Ce/$BaFe_{12}O_{19}$, [220] Zn:Ni/$SrFe_{12}O_{19}$, [221] Mn:Cd:Zr/$SrFe_{12}O_{19}$, [222] Nd:Co/$SrFe_{12}O_{19}$, [223] Co:Zn:Al:Ce/$BaFe_{16}O_{27}$, [224] Co:Zn/$BaFe_{16}O_{27}$ [225,226] and Co:Zn:La/$BaFe_{16}O_{27}$, [226] (Sb:SnO_2)|Zn:Co:Gd/$BaFe_{16}O_{27}$, [227] and Ba:La/$Co_2Fe_{36}O_{60}$, [228] demonstrate interactions with EM radiation which are driven through both dielectric and magnetic losses. For example, Feng et al. [225] reported the strong RL of a zinc and cobalt doped barium ferrite, with a maximal RL reported as −28.5 dB with a 2.2 mm thick absorber at 11.6 GHz, and a 9.3 GHz proficiency bandwidth. Feng also demonstrated the analysis of multi-layered absorbers therein, utilizing the aforementioned barium ferrite in tandem with carbonyl iron so to extend the proficiency bandwidth of the material to 13 GHz. The results of the analysis of these nanomaterials are presented in Tables 6A and 6B.

4.3.2. Manganese oxide

Manganese oxide nanomaterials have been demonstrated to interact with incident microwave radiation as a dielectric nanomaterial. Yuping et al. [229] demonstrated that this interaction with gigahertz-range electromagnetic radiation varied as a function of crystallographic structure, comparing manganese dioxide particles of differing synthesis times thus generating differing α, β, and γ crystal forms, shown in Fig. 12. The permittivity of the MnO_2 nanomaterials increased as a function of synthesis time, showing that of the three, the β-MnO_2 demonstrated the highest overall lossless and lossy dielectric action. The opposite was generally true

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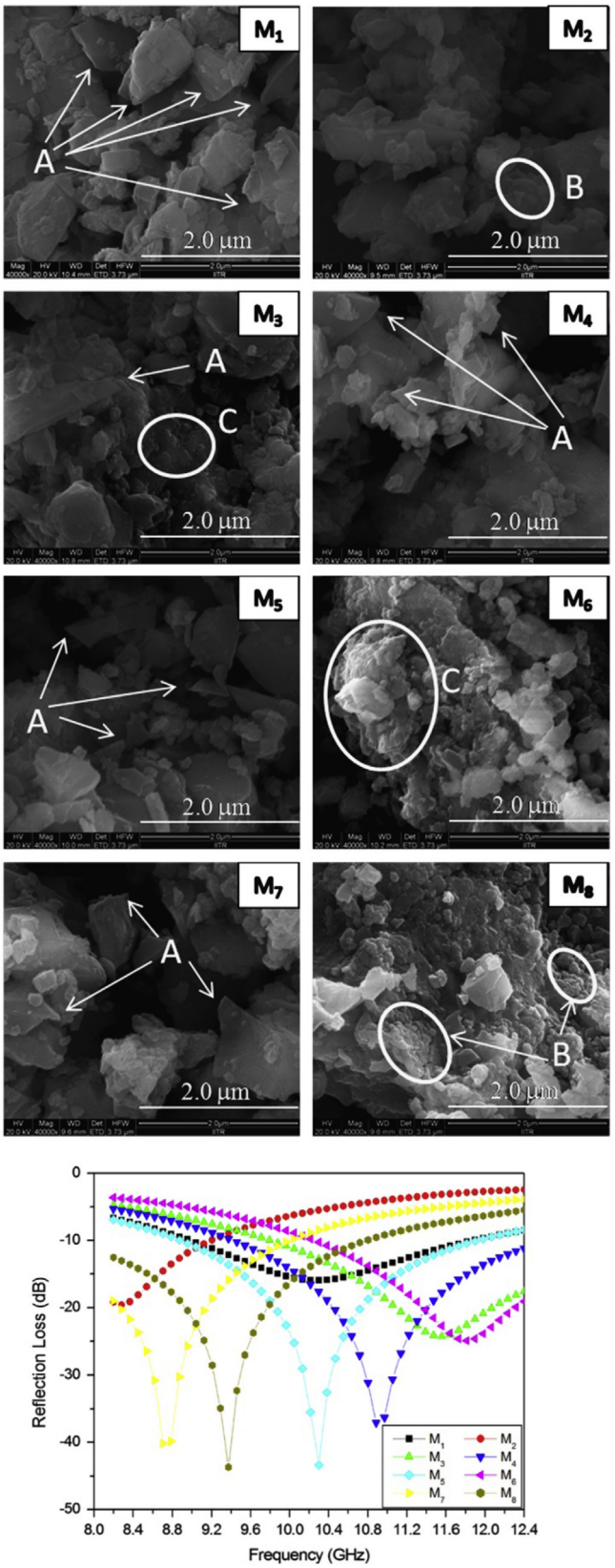


Fig. 10. SEM and RL curves for (M1) pure silicon carbide nanomaterials, (M2) Al/SiC, (M3) Co/SiC, (M4) Cr/SiC, (M5) Mn/SiC, (M6) Ni/SiC, (M7) Ti/SiC, and (M8) Zn/SiC. Reproduced with permission from ref. 55. Copyright 2014 Elsevier.

for permeability, showing that the longer the synthesis time, the lower the permeability values [229]. Calculations for reflection loss at 2 mm thicknesses demonstrated that the agglomerate nanomaterial of α, β, and γ-MnO_2 yielded the highest RL, reaching −35.6 dB at 12.3 GHz. Pure phases of γ-MnO_2 and β-MnO_2 demonstrated RL of −24.2 dB at 17.6 GHz, and −25.2 dB at 10.9 GHz, respectively, shown in Fig. 12. The permittivity and permeability values associated with these points were reported as $\varepsilon_r = (10.8 - i3.7)$ and $\mu_r = (1.0 - i0.02)$, $\varepsilon_r = (4.5 - i1.9)$ and $\mu_r = (1.2 - i0.3)$, and $\varepsilon_r = (14.9 - i\,4.2)$ and $\mu_r = (1.0 - i0.01)$, for the agglomerate, γ-MnO_2, and β-MnO_2 nanomaterials, respectively [229].

Further research on manganese oxide nanomaterials and composites such as urchinlike α-MnO_2, [230] Mn_3O_4, [231] and La:Sr/MnO_3, [232] has also demonstrated strong RL returns in the gigahertz range, where the manganese species therein showed propensity to induce dielectric interaction, and the La:Sr dopants demonstrate an in tandem increased response to both dielectrinc and magnetic interaction [232]. The full results of these analyses is given in Table 7.

4.3.3. Titanium oxide

The most prevalent of the titanium oxide class nanomaterials, titanium dioxide, fundamentally lacks an intrinsic mechanism through which to interact with gigahertz-range electromagnetic radiation [61]. However, nanomaterials which integrate impurities and perturbated crystalline phases into the TiO_2 domain have been shown on specific occasion to induce interaction with microwaves. For example, carbonyl iron|TiO_2 composites constructed by An et al.*ia* [101] have been demonstrated to absorb microwave radiation, with a reported RL value of −62.0 dB at 6.69 GHz with a 2.06 mm thickness. Doped TiO_2 nanomaterials such as Fe/TiO_2 nanowires[204] have also been demonstrated to interact favorably with incident radiation.

Furthermore, hydrogenated TiO_2 nanoparticles [61,234] (as shown in Fig. 13) and nanosheets [235] have also been shown to absorb incident radiation, through interaction with the electric portion of incident light across a two-phase boundary, generated via the distorting process of hydrogenation. This engineered disordering via hydrogenation has also been demonstrated to synergistically coordinate with metal reductases so to generate reduced oxide forms for core-shell formation [105,106]. Finally, mesoporous carbon/TiO_2 composites have been shown to generate desirable RL values at high thicknesses, with the 9.0 mm composite of equal proportions demonstrating a −53.8 dB RL at 12.1 GHz [236]. Complex titanium-derived oxides such as barium titanate have also been integrated with dopants and structural defects so to induce microwave absorption [237,238].

Jing et al. [237] demonstrated the electromagnetic characteristics of flake $BaTiO_3$, where the synthesized nanomaterials were shown to have high permittivity values associated with the nanomaterial. The maximally represented RL value was −29.6 dB at 12.0 GHz with a 4.0 mm thickness. The associated permittivity and permeability values were $\varepsilon_r = (23.4 - i2.7)$ and $\mu_r = (0.94 - i0.00)$ for the material. Furthermore, nanomaterials such as Ni/$BaTiO_3$, [238] and hydrogenated $BaTiO_3$, [239] demonstrate that nanomaterials processing and doping can also be utilized so to drive RL values from complex titanium oxides, through high permittivity activity and increased lossy magnetic interaction. These results are tabulated in Table 8.

4.3.4. Zinc oxide

Zinc oxide has been consistently utilized as a foundational framework in nanomaterials development, even though the intrinsic interaction between ZnO and GHz-range radiation has consistently been shown to be lacking, as the material has no

Table 5
Microwave Absorption Performance of Carbide based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε′ at Max	ε" at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ" at Max	$tg\delta_\mu$ at Max	Ref.
SiC-3hr milling	1.9	2.5	−50.7	9.5	16.05	4.25	0.26	1.098	0.073	0.07	[55]
SiC-2hr milling	1.9	2.5	−43.6	9.8	15.2	3.94	0.26	1.094	0.084	0.08	[55]
SiC-20hr milling	1.8	2.7	−27.1	10.5	13.9	2.85	0.21	1.18	0.19	0.16	[55]
SiC-10hr milling	1.6	2.9	−15.9	10.3	18.1	5.8	0.32	1.15	0.15	0.13	[55]
Al/SiC	2.2	1.0	−19.6	8.3	16.5	4.3	0.26	1.06	0.01	0.01	[55]
Co/SiC	1.8	2.6	−24.3	11.6	12.2	6.4	0.52	1.08	−0.13	−0.12	[55]
Cr/SiC	1.7	3.0	−37.8	10.9	15.1	5.2	0.34	1.11	0.01	0.01	[55]
Mn.SiC	1.8	3.0	−43.4	10.3	15.1	4.9	0.32	1.11	0.00	0.00	[55]
Ni/SiC	1.5	2.1	−24.9	11.8	16.5	4.2	0.25	1.11	0.13	0.12	[55]
Ti/Sic	2.1	1.8	−40.2	8.7	15.6	5.3	0.34	1.09	−0.02	−0.02	[55]
Zn/SiC	2.0	2.6	−43.7	9.4	15.0	5.2	0.35	1.09	−0.02	−0.02	[55]
Ni/SiC 673K anneal	2.1	4.2	−47.6	10.0	19.6	6.7	0.34	n/a	n/a	n/a	[174]
Ni/SiC 573K anneal	2.1	4.0	−45.4	10.4	18.7	4.4	0.24	n/a	n/a	n/a	[174]
NiO/SiC	2.0	4.2	−46.9	10.6	13.3	6.7	0.50	n/a	n/a	n/a	[175]
Ni:Co-P/SiC	2.5	1.9	−32.4	6.4	23.1	5.9	0.26	1.0	0.04	0.04	[176]
Co/SiC	1.7	2.8	−44.7	10.4	18.6	5.5	0.30	n/a	n/a	n/a	[177]
Fe/SiC	2.25	1.9	−46.3	6.4	25.0	5.8	0.23	1.15	0.05	0.04	[178]
Al/SiC	2.4	2	−25.4	8.4	10.2	3.1	0.30	n/a	n/a	n/a	[179]
Carbon Black|SiC	3.0	6.0	−41.5	9.0	n/a	n/a	n/a	n/a	n/a	n/a	[180]
CNT|Ni/SiC	1.9	5.1	−37.6	14.5	6.5	1.75	0.27	1.1	0.1	0.09	[181]
CF|SiC	3.0	1.1	−12.8	3.75	n/a	n/a	n/a	n/a	n/a	n/a	[182]
graphene/SiC	3.0	4.3	−47.3	10.52	7.1	1.5	0.21	n/a	n/a	n/a	[183]

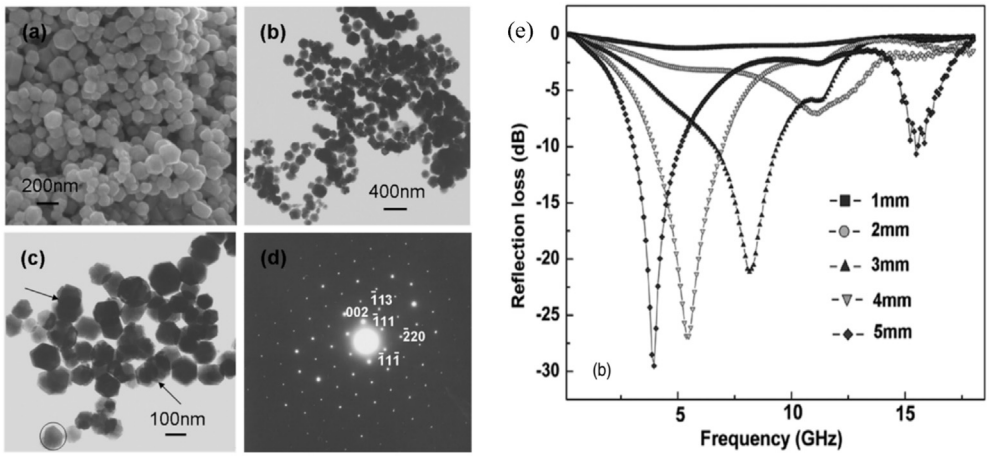


Fig. 11. (a) SEM, (b, c) TEM, (d) SAED, and (e) RL curves (e) for Fe_3O_4 nanoparticles. Reproduced with permission from ref. 194. Copyright 2009 IOP Publishing Ltd.

demonstrated magnetic interaction nor a strong dielectric action so to drive RL [240,241]. However, the introduction of dopants so to induce electromagnetic interaction has been generally successful in the realm of nanomaterials development. Nanomaterials and composites such as PANI|ZnO, [242] Zn:Fe/ZnO, [243] Ni|ZnO, [244] rGO|ZnO [245] and rGO|T-ZnO [246] have been demonstrated to interact with microwaves of specific frequencies. Integration of magnetic entities such as iron and nickel induced perturbation in the nanomaterial response to the magnetic portion of incident electromagnetic waves, and carbon-based integrations such as graphene and various polymers demonstrated an induced change in nanomaterial response to the electric portion of incident electromagnetic waves.

Furthermore, deviations in physical manifestation through synthesis techniques have been utilized, generating nanomaterials such as porous hollow ZnO, [247] dendritic ZnO (shown in Fig. 14), [248] netlike ZnO, [249] as well as hydrogenated ZnO, [250] where the perturbed particles were shown to have both an improved lossless dielectric response, as well as lossy response, to incident radiation compared to that of pristine ZnO nanomaterials, resulting in the increase of the nanomaterial's RL. These results are tabulated in Table 9.

4.3.5. Silicon oxide

The primary silicon oxide nanomaterial, silicon dioxide, possesses a similar lacking in an intrinsic response to incident electromagnetic radiation compared to that of ZnO, barely registering a RL value; for example, −1.2 dB at 12.3 GHz with a 3 mm thickness [241]. It too however has been successfully paired with dopants so to induce interaction with incident radiation. For example, ZnO|SiO_2 composites, [241] even though neither of the materials intrinsically interact with gigahertz-range electromagnetic radiation, when paired together demonstrate elevated RL values, through the composite's lossy and lossless interaction as described by permittivity. For these ZnO|SiO_2 composites, as shown in Fig. 15, RL was reported to be −10.8 dB at 12.8 GHz with a 3.0 mm thickness. SiO_2 has also been doped by means of heterogeneous atomic species so to induce dielectric relaxation in the otherwise inactive material medium [54]. Furthermore, complex silicates such as ceramics have also been utilized in microwave absorption, such as

Table 6A
Microwave Absorption Performance of Iron Oxide based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε' at Max	ε'' at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ'' at Max	$tg\delta_\mu$ at Max	Ref.
Fe_3O_4 30 wt%	5.0	2.1	−29.5	3.9	9.2	0.2	0.02	1.5	0.72	0.48	[194]
Fe_3O_4 30 wt%	4.0	3.1	−26.9	5.4	9.3	0.3	0.03	1.2	0.52	0.43	[194]
Fe_3O_4 30 wt%	3.0	2.7	−21.1	8.2	9.5	0.4	0.04	0.9	0.38	0.42	[194]
Fe_3O_4 40 wt%	3.0	3.1	−23.3	6.9	11.1	1.6	0.14	1.06	0.45	0.42	[195]
Fe_3O_4 40 wt%	3.5	2.6	−27.8	5.5	11.0	1.4	0.13	1.23	0.54	0.44	[195]
Fe_3O_4 40 wt%	4.0	2.1	−44.7	4.7	11.0	1.3	0.12	1.30	0.58	0.45	[195]
Fe_3O_4 nanowires	2.5	2.7	−16.7	8.3	6.2	0.2	0.03	0.98	0.20	0.20	[196]
Fe_3O_4 nanosheets	2.5	0.0	−8.89	10.2	8.1	2.1	0.26	0.90	0.014	0.02	[196]
Fe_3O_4 90%	1.0	4.3	−42.3	14.1	23.6	2.9	0.12	0.3	0.3	1.00	[197]
Fe_3O_4 80%	1.3	3.2	−33.7	15.1	17.7	3.9	0.22	0.5	0.06	0.12	[197]
Fe_3O_4 70%	5.0	1.5	−34.9	3.4	12.1	1.2	0.10	1.2	0.6	0.50	[197]
Ni:B/Fe_3O_4	6.0	2.6	−28.4	5.4	6.6	1.0	0.15	0.8	0.3	0.38	[198]
$(Ni_{0.5}Zn_{0.5})Fe_2O_4$	2.97	5.2	−36.2	11.9	7.5	0.05	0.01	0.5	0.2	0.40	[199]
$(Ni_{0.4}Cu_{0.2}Zn_{0.4})Fe_2O_4$	9.2	1.6	−35.6	1.1	8.1	−0.01	0.00	2.1	3.8	1.81	[199]
$(Ni_{0.4}Co_{0.2}Zn_{0.4})Fe_2O_4$	4.2	3.6	−48.1	4.1	7.05	0.03	0.00	1.56	1.49	0.96	[199]
$Ni_{0.8}Co_{0.2}Fe_2O_4$	2.5	3.2	−35.5	11.5	9.6	−0.13	−0.01	1.0	0.3	0.30	[200]
$Ni_{0.5}Co_{0.5}Fe_2O_4$	2.5	3.4	−30.6	11.9	8.3	−0.25	−0.03	0.7	0.3	0.43	[200]
$MnFe_2O_4$	2	0.3	−42.5	9.4	1.5	0.35	0.23	4.7	0.73	0.16	[201]
$Co_{0.5}Mn_{0.5}Fe_2O_4$	2	1.0	−47.0	10.5	3.8	0.36	0.09	7.1	1.02	0.14	[201]
$CoFe_2O_4$	2	1.0	−39.8	10.7	7.5	0.46	0.06	8.9	1.42	0.16	[201]
$Bi_{0.8}La_{0.2}FeO_3$.	6	1.1	−29.7	11.5	9.8	1.4	0.14	1.0	0.9	0.90	[202]
PANI|$Co_{0.5}Zn_{0.5}Fe_2O_4$	2.0	10.5	−39.8	22.3	19.3	13.1	0.68	1.7	0.3	0.18	[203]
PANI|$Li_{0.35}Zn_{0.3}Fe_{2.35}O_4$	2.0	4.8	−37.5	14.6	8.2	5.0	0.61	1.04	0.1	0.10	[204]
Fe_3O_4@C (17.84 wt%C)	5.0	1.7	−35.8	4.7	9.4	2.9	0.31	1.1	0.2	0.18	[205]
Fe_3O_4@C (23.41 wt%C)	1.5	3.9	−40.2	15.8	10.2	3.7	0.36	1.0	0.0	0.00	[205]
C/Fe_3O_4	1.9	2.5	−52.8	12.1	12.0	1.5	0.13	0.94	0.10	0.11	[206]
C/Fe_3O_4 60 nm nanosheets	2.0	4.7	−41.9	12.8	7.1	1.9	0.27	1.0	0.06	0.06	[207]
C/Fe_3O_4 250 nm nanosheets	4.3	1.9	−44.0	3.92	8.4	0.4	0.05	1.3	0.4	0.31	[207]
C/Fe_3O_4 NR δ(0.5)	6.2	2.9	−55.7	3.4	11.5	2.7	0.23	1.35	0.54	0.40	[208]
C/Fe_3O_4 NR δ(1)	4.5	2.8	−55.1	6.0	10.6	2.5	0.24	0.88	0.26	0.30	[208]
C/Fe_3O_4 NR δ(2)	4.2	2.7	−51.8	6.1	9.3	1.7	0.18	0.93	0.25	0.27	[208]
C/Fe_3O_4 NR δ(4)	1.9	3.8	−55.3	13.8	6.7	1.6	0.24	0.95	−0.06	−0.06	[208]
ZnO/Fe_3O_4	5.0	0.85	−12.92	1.66	3.4	1.23	0.36	2.4	4.0	1.67	[209]
Fe:Mn/Fe_3O_4	1.85	4.2	−27.7	13.9	8.9	2.5	0.28	1.0	0.1	0.10	[210]
Fe-phthalocyanine oligomer/Fe_3O_4	3.0	2.0	−30.8	8.7	8.9	3.0	0.34	1.0	0.06	0.06	[211]
Fe-phthalocyanine oligomer/Fe_3O_4	5.0	1.0	−28.5	5.4	10.7	2.2	0.21	0.7	0.1	0.14	[211]

Table 6B
Microwave Absorption Performance of Iron Oxide based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε' at Max	ε'' at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ'' at Max	$tg\delta_\mu$ at Max	Ref.
Nd/$BiFeO_3$	2.3	1.2	−42.0	8.5	n/a	n/a	n/a	n/a	n/a	n/a	[216]
$CoBaTiFe_{10}O_{19}$	2	5.7	−23.8	10.4	7.6	0.1	0.01	0.7	0.02	0.03	[217]
$Ni_{0.4}Co_{0.6}BaTiFe_{10}O_{19}$	2	4.7	−46.9	13.3	6.6	0.3	0.05	1.2	0.6	0.50	[217]
$Ni_{0.2}Co_{0.8}BaTiFe_{10}O_{19}$	2	8.6	−48.3	16.0	6.5	0.4	0.06	0.6	0.3	0.50	[217]
$(MnNi)_{0.2}Co_{0.6}BaTiFe_{10}O_{19}$	1.8	5.4	−53.0	13.5	6.7	0.4	0.06	0.8	0.7	0.88	[218]
$(MnNi)_{0.25}Co_{0.5}BaTiFe_{10}O_{19}$	1.8	4.9	−68.9	14.1	6.5	0.4	0.06	1.2	0.7	0.58	[218]
ZnO/$BaFe_{12}O_{19}$	6.8	3	−37.3	16.0	3.1	1.3	0.42	0.7	0.03	0.04	[219]
$BaCe_{0.05}Fe_{11.95}O_{19}$	3.5	8.1	−37.4	12.8	2.72	1.38	0.51	1.10	0.38	0.35	[220]
$BaCe_{0.05}Fe_{11.95}O_{19}$	5.0	3.5	−31.5	11.3	1.6	1.4	0.88	1.4	0.77	0.55	[220]
$NiFe_2O_4/ZnFe_2O_4/SrFe_{12}O_{19}$ (900 °C)	2.5	3.1	−27.6	10.2	12.4	5.9	0.48	1.2	1.7	1.42	[221]
$NiFe_2O_4/ZnFe_2O_4/SrFe_{12}O_{19}$ (1200 °C)	2.5	3.6	−29.7	10.2	13.4	13	0.97	1.3	2.0	1.54	[221]
$(Mn_{0.5}Cd_{0.5}Zr)_{1.4}SrFe_{9.2}O_{19}$	1.9	6.0	−51.3	14.5	n/a	n/a	n/a	n/a	n/a	n/a	[222]
$(Mn_{0.5}Cd_{0.5}Zr)_{1.6}SrFe_{8.8}O_{19}$	1.9	3.7	−44.7	15.5	n/a	n/a	n/a	n/a	n/a	n/a	[222]
$Nd_{0.2}Co_{0.2}Sr_{0.8}Fe_{11.8}O_{19}$	1.8	3.3	−19.1	16.7	5.87	0.63	0.11	0.97	0.34	0.35	[223]
$Nd_{0.2}Co_{0.2}Sr_{0.8}Fe_{11.8}O_{19}$	1.9	3.8	−20.8	16.0	5.76	0.64	0.11	0.97	0.36	0.37	[223]
$CoZnAl_{0.2}Ce_{0.2}BaFe_{15.6}O_{27}$	4.52	1.2	−30.4	6.6	3.9	0.08	0.02	1.1	0.2	0.18	[224]
$CoZnAl_{0.2}Ce_{0.2}BaFe_{15.6}O_{27}$	4.37	1.0	−34.5	6.9	3.9	0.09	0.02	1.1	0.2	0.18	[224]
$Zn_{1.5}Co_{0.5}BaFe_{16}O_{27}$	2.2	9.3	−28.5	11.6	7.4	0.1	0.01	0.98	0.67	0.68	[225]
$Zn_{1.5}Co_{0.5}BaFe_{16}O_{27}$	3.9	0	−6.6	9.4	7.4	0.1	0.01	1.63	0.87	0.53	[225]
$Ba_1Co_{0.9}Zn_{1.1}Fe_{16}O_{27}$	2.0	7.7	−33.6	12.0	4.45	0.33	0.07	1.04	0.21	0.20	[226]
$Ba_{0.8}La_{0.2}Co_{0.9}Zn_{1.1}Fe_{16}O_{27}$	2.0	10.9	−39.7	12.7	4.11	0.35	0.09	0.89	0.24	0.27	[226]
$(Sb:SnO_2)$|Zn:Co:Gd/$BaFe_{16}O_{27}$	1.6	3.9	−19.4	14.8	8.3	1.3	0.16	1.01	0.20	0.20	[227]
$(Ba_{0.7}La_{0.2})_4Co_2Fe_{36}O_{60}$	1.6	2.4	−23.2	9.8	19.1	3.0	0.16	1.40	0.63	0.45	[228]
$(Ba_{0.7}La_{0.2})_4Co_2Fe_{36}O_{60}$	1.8	2.1	−27.8	8.5	19.3	3.6	0.19	1.68	0.63	0.38	[228]

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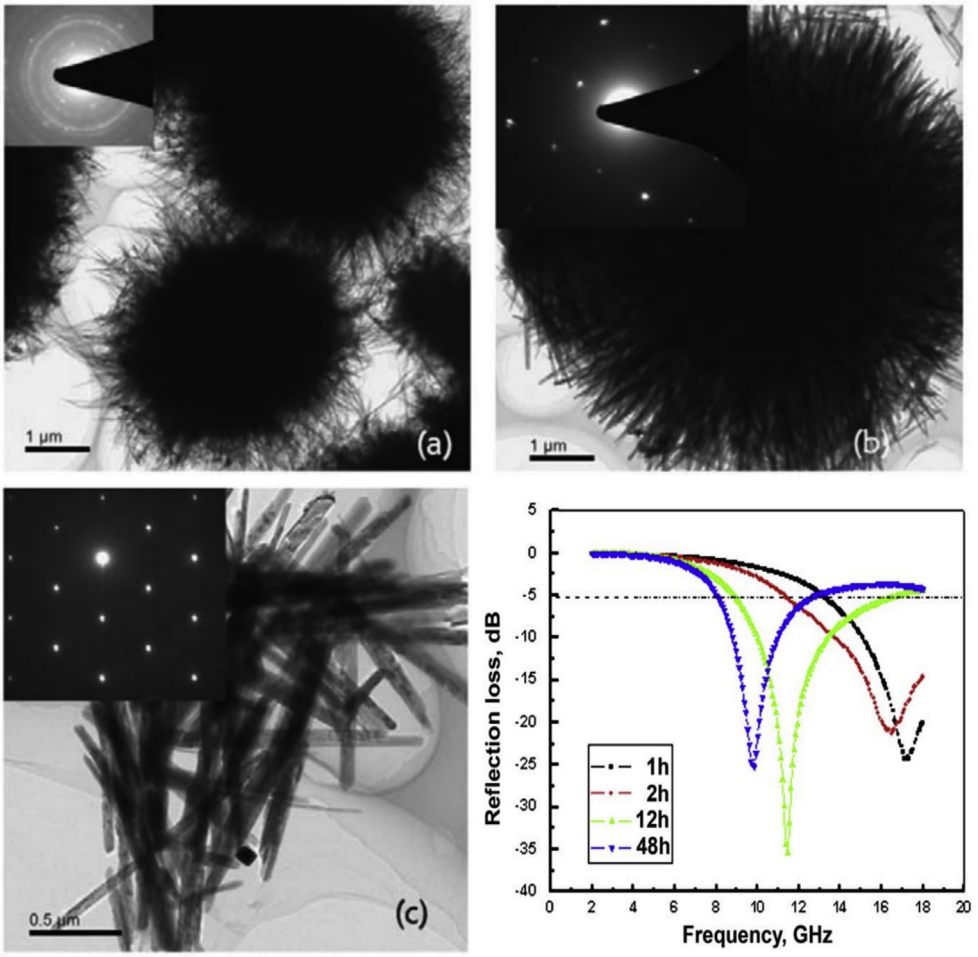


Fig. 12. TEM/SAED and RL curves for (a, 2hr) γ-MnO_2, (b, 12hr) agglomerate-MnO_2 and (c, 48hr) β- MnO_2. Reproduced with permission from ref. 229. Copyright 2013 Elsevier.

Table 7
Microwave absorption performance of manganese oxide based nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε′ at Max	ε" at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ" at Max	$tg\delta_\mu$ at Max	Ref.
β-MnO_2 (48 h Cal.)	2.0	2.0	−25.2	10.9	14.9	4.2	0.28	1.0	0.01	0.01	[229]
aglm-MnO_2 (12 h Cal.)	2.0	3.2	−35.6	12.3	10.8	3.7	0.34	1.0	0.02	0.02	[229]
γ-MnO_2 (1 h Cal.)	2.0	2.6	−24.2	17.6	4.5	1.9	0.42	1.2	0.3	0.25	[229]
α-MnO_2	3.8	0.5	−36.4	2.9	37.5	8.7	0.23	1.3	−0.01	−0.01	[230]
α-MnO_2	1.9	1.8	−41.1	8.7	20.9	4.8	0.23	1.1	0.07	0.06	[230]
Mn_3O_4	7.0	0.7	−24.8	2.5	15.2	4.4	0.29	1.0	0.0	0.00	[231]
Mn_3O_4	6.0	0.8	−27.1	3.0	14.7	4.2	0.29	1.0	0.0	0.00	[231]
$La_{0.6}Sr_{0.4}MnO_3$	1.5	1.9	−22.6	10.6	16.7	0.9	0.05	1.4	0.4	0.29	[232]
$La_{0.6}Sr_{0.4}MnO_3$	2.2	2.4	−40.9	8.2	12.9	0.3	0.02	1.6	0.7	0.44	[232]

SiOC ceramics [251] and Al:ZnO/$ZrSiO_4$ ceramics, [252] through deviation of the nanomaterial's relative permittivity values. These results are reported in Table 10.

4.3.6. Aluminum oxide

Aluminum oxide as an isolated nanomaterial hasn't been demonstrated to intrinsically interact with gigahertz-range electromagnetic radiation, though it too has been utilized as a framework upon which dopants have been introduced so to cause microwave absorption. Nanomaterials such as Fe_3Al/Al_2O_3, [253] as synthesized by Wei et al. demonstrate that coupling aluminum oxide with magnetic particulates demonstrate increased responses to both the electric and magnetic portions of incident electromagnetic radiation in a manner to which RL is induced, by surrounding the dielectric nanomaterial with magnetic particles so to generate further polarization charges on the surface of the dielectric particles, and as a consequence the subsequent interaction introduces to the system an actionable dielectric loss [253]. Similar work was shown to be effective by Zhang et al. [254] shown in Fig. 16, where in reverse order iron nanoparticles were imbedded into an Alumina matrix so to induce reflection loss via strong lossy magnetic interaction as defined by μ", yielding a maximal RL value

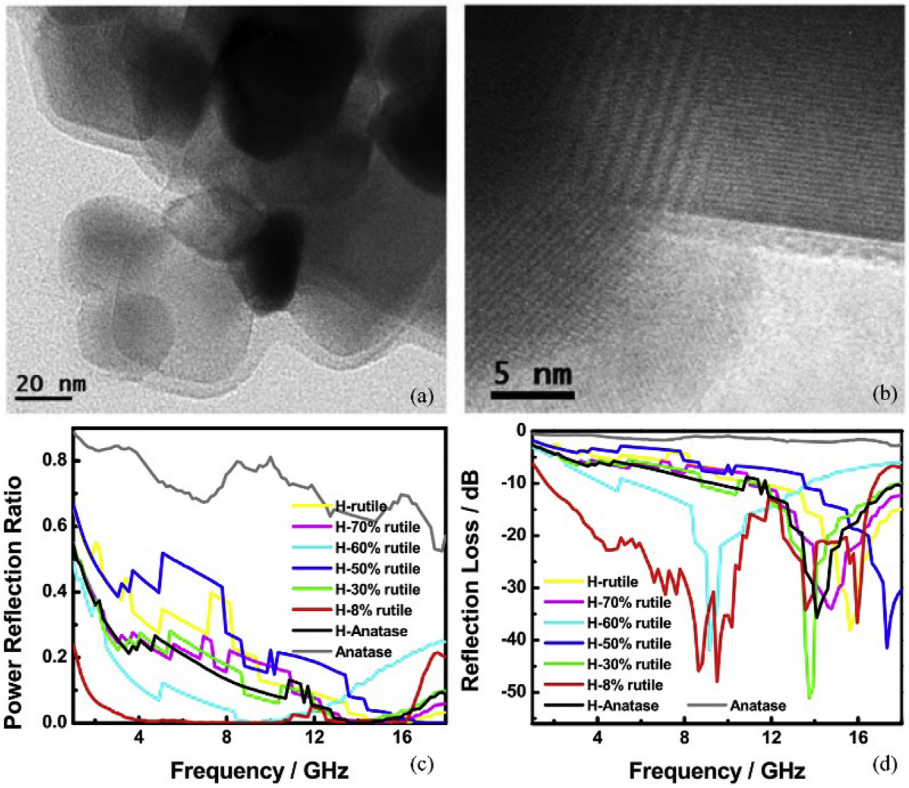


Fig. 13. (a, b) HRTEM images of 60 wt% hydrogenated rutile/anatase composite nanocrystals with their respected (c) power reflection ratio and (d) RL curves. Reproduced with permission from ref. 61. Copyright 2014 Materials Research Society.

Table 8
Microwave Absorption Performance of Titanium Oxide based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε' at Max	ε'' at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ'' at Max	$tg\delta_\mu$ at Max	Ref.
Hydrogenated TiO_2 nanoparticles	2.0	7.4	−36.8	14.0	13.0	5.4	0.42	n/a	n/a	n/a	[61]
Hydrogenated TiO2 8% rutile	2.0	15.0	−48.0	9.6	19.2	6.0	0.31	n/a	n/a	n/a	[61]
Al/H_2 treated TiO_2	1.7	6.2	−21.2	12.8	12.4	5.2	0.33	1.0	0.05	0.05	[105]
Al/H_2 treated TiO_2	3.1	2.1	−58.0	6.61	14.8	5.0	0.33	1.05	0.05	0.05	[105]
Mg/H_2 treated TiO_2	1.7	1.7	−34.7	9.1	24.9	4.6	0.18	1.02	0.04	0.04	[106]
Carbonyl Iron|TiO_2	2.06	2.6	−63.5	6.7	16.5	1.8	0.11	1.7	0.53	0.31	[101]
Fe/TiO_2	3.0	4.5	−45.1	13.9	11.1	0.6	0.05	0.9	1.7	1.89	[233]
Hydrogenated TiO_2 nanosheets	2.0	8.5	−37.9	13.9	n/a	n/a	n/a	n/a	n/a	n/a	[235]
MC|TiO_2	9.0	2.4	−53.8	12.1	4.1	0.4	0.10	1.06	0.01	0.01	[236]
MC|TiO_2	8.0	2.9	−25.9	14.0	4.0	0.4	0.10	1.03	0.01	0.01	[236]
MC|TiO_2	7.0	3.2	−24.8	16.1	3.9	0.4	0.10	1.05	−0.01	−0.01	[236]
$BaTiO_3$	4.0	0.8	−29.6	12.0	23.4	2.7	0.12	0.94	0.0	0.00	[237]
Ni@$BaTiO_3$	1.88	2.6	−41.9	10.7	15.6	1.3	0.08	0.9	0.2	0.22	[238]
Hydrogenated $BaTiO_3$	3.0	3.4	−36.9	11.9	20.4	2.8	0.14	1.0	0.11	0.11	[239]

of −21.4 dB at 13.3 GHz with a 1.4 mm absorber. The results of these nanomaterial sets are summarized in Table 11 for their RL responses and proficiency parameters.

4.4. Sulfides

Though not as exhaustively studied as the oxide nanomaterial class, many nanomaterials studies have focused on sulfide nanomaterials as a means for generating reflection loss. These nanomaterials are thought to be dielectric in nature, as the magnetic portion of electromagnetic response is typically omitted from network analysis or explicitly set to the trivial value of $\mu_r = (1 - i0)$.

4.4.1. Molybdenum sulfide

Intrinsic molybdenum sulfide nanomaterials have been demonstrated to interact with incident gigahertz-range electromagnetic radiation through the development of complex geometries which enhance the light/matter interaction [255–257]. Su et al. synthesized a set of flower-like MoS_2 nanospheres, shown in Fig. 17, and studied how the change in reaction time effected the nanomaterial morphology, and by extension the material complex permittivity [255]. The MoS_2 materials were synthesized hydrothermally with a CH_3CSNH_2 reducing agent/sulfur source. The synthesis was initialized at 16 h at 200 °C, and the reaction time was increased at 4-h increments to yield samples synthesized over 16, 20, 24, and 28-h time intervals. Analysis of the generated

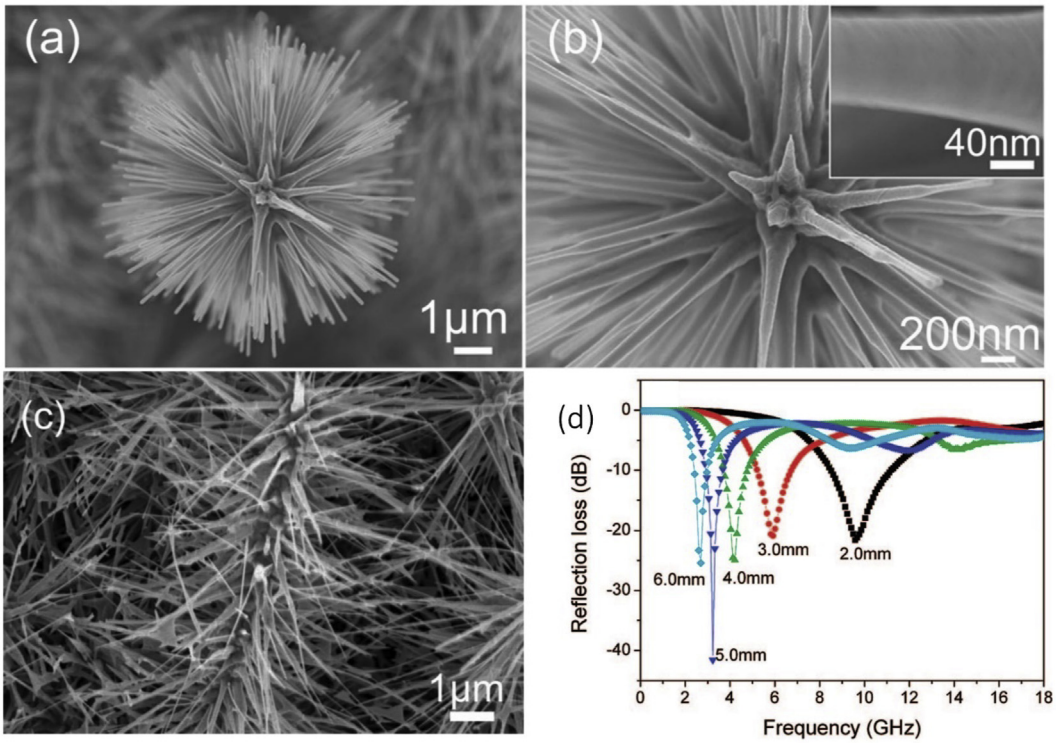


Fig. 14. (a–c) SEM images of dendritic ZnO, with (d) the associated RL curves. Reproduced with permission from ref. 248. Copyright 2008 Royal Society of Chemistry.

Table 9
Microwave Absorption Performance of Zinc Oxide based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε′ at Max	ε" at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ" at Max	$tg\delta_\mu$ at Max	Ref.
ZnO	1.0	0	−3.2	9.1	n/a	n/a	n/a	n/a	n/a	n/a	[240]
PANI/ZnO	3.5	0.9	−41.0	14.0	3.5	2.7	0.77	0.78	0.23	0.29	[242]
Zn:Fe/ZnO	8.0	0.7	20.8	17.9	5.3	0.08	0.02	0.8	0.4	0.5	[243]
Ni@ZnO	1.5	5.3	−27.3	14.6	11.8	4.0	0.34	1.06	0.075	0.07	[244]
Ni@ZnO	2.0	2.7	−48.0	10.4	11.8	4.3	0.36	1.20	0.038	0.03	[244]
rGO/ZnO	2.2	3.3	−45.1	9.7	11.0	4.1	0.37	n/a	n/a	n/a	[245]
rGO/T-ZnO	2.9	6.8	−59.5	14.4	2.4	0.73	0.30	n/a	n/a	n/a	[246]
Hollow ZnO	4	2.2	−36.3	12.8	3.1	0.6	0.19	n/a	n/a	n/a	[247]
Dendritic ZnO	5.0	0.8	−42	3.6	19.3	5.7	0.30	n/a	n/a	n/a	[248]
Netlike ZnO	4.0	2.1	−37.0	6.2	9.2	4.0	0.43	n/a	n/a	n/a	[249]
Hydrogenated ZnO	3.0	14.0	−38.0	15.2	10.8	3.9	0.36	0.98	0.10	0.10	[250]

nanomaterials showed that peak permittivity was achieved at 20 h, resulting in permittivity values ranging between 8.25–8.05 for ε′ and 4.1–3.85 for ε" across the 8–12 GHz frequency testing range. Maximal RL was reported as −52.76 dB, achieved at 8.24 GHz with a reaction time of 28 h and a matching thickness of 3.475 mm, and −34.8 dB, achieved at 11.2 GHz with a reaction time of 16 h and a matching thickness of 2.4 mm.

Molybdenum sulfide nanomaterials have also been integrated with other dielectric and magnetic materials to for form composite fillers which interact favorably with incident radiation. Notably, dielectric amalgams such as rGo|MoS_2, [258–260] C|MoS_2, [261,262] and graphene|MoS_2 [263] nanomaterials have been shown to act as microwave absorbing materials with general success. Furthermore, dielectric/magnetic amalgams such as Ni|MoS_2, [264] MoS_2@FeCo, [265] and MoS_2@$Bi_2Fe_4O_9$ [266] have also been shown to interact with microwaves favorably, generating non-negligible permittivity and permeability values which induce microwave absorption. The results of these nanomaterial sets are summarized in Table 12A for their RL responses and proficiency parameters.

4.4.2. Copper sulfide

The copper sulfide subclass represents a domain of nanomaterials research which hasn't received as much attention from those who work with composites development, as the few studies available for review appear to study the interactions of intrinsic Cu_xS nanomaterials subsets, demonstrated in Table 12B [266–270]. For example, Gu et al. demonstrated that, of varying ion ratios established by the interdiffusion of Cu^{2+} ions into synthesized CuS nanocrystals, intrinsic synthetic CuS covellite nanoparticles demonstrated the greatest overall reflection loss with a reported RL value of −54.84 dB at 12.63 GHz with a 1.95 mm matching thickness, and a permittivity magnitude of $\varepsilon_r = (12.3 - i2.2)$ [267].

The only acquired manuscript which considered integrating CuS nanostructures into a composite system resulted in the only reported instance of a measured RL value below −100 dB, and such

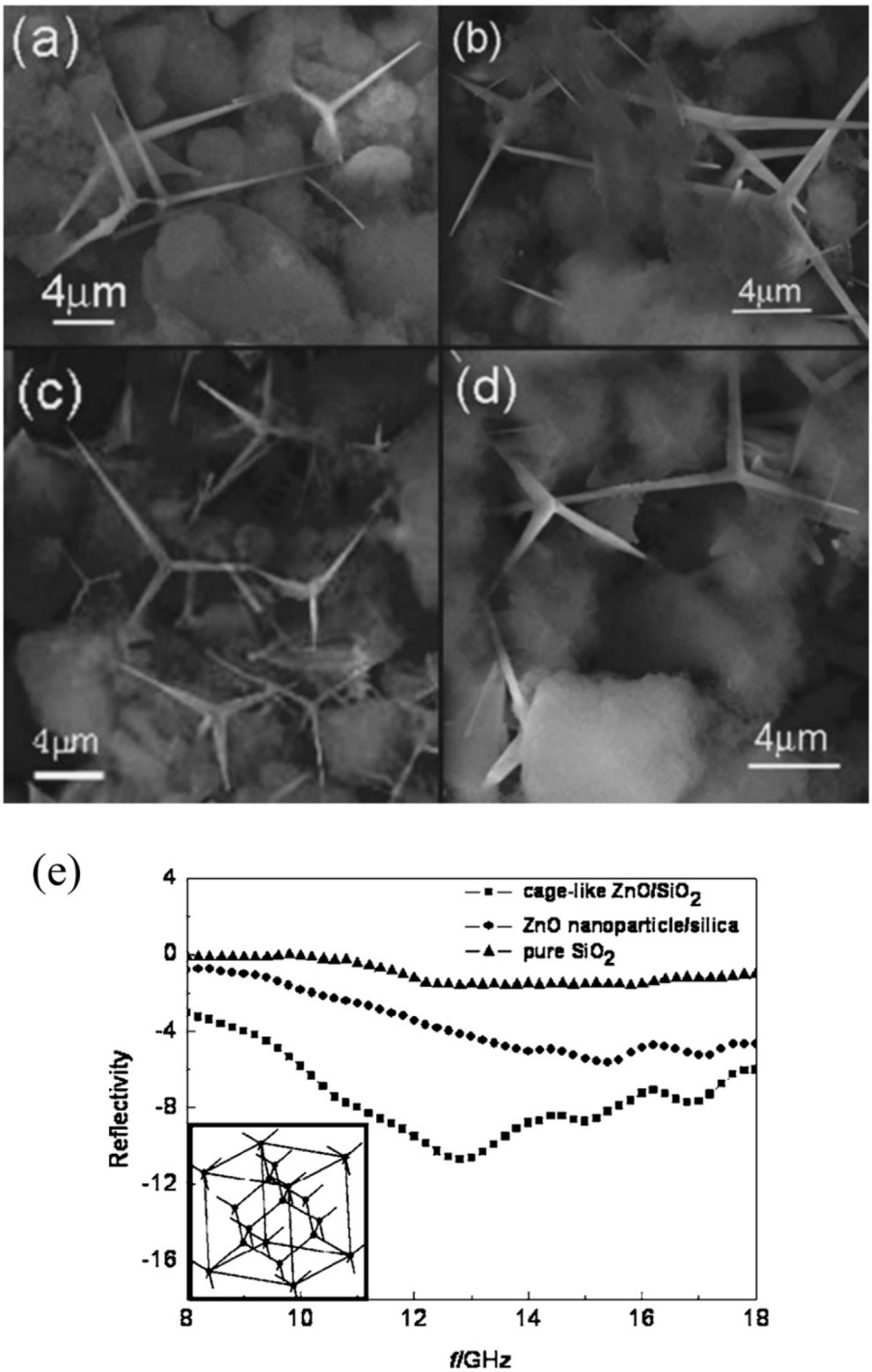


Fig. 15. SEM images of ZnO|SiO_2 nanocomposites (a–d), with associated RL values (e). Reproduced with permission from ref. 241. Copyright 2007 AIP Publishing.

benchmark has yet to be matched in the field [271]. He et al. synthesized the CuS nanostructures using a hydrothermal method, utilizing Copper Nitrate trihydrate and sulfur powder as the CuS precursors. The resulting CuS nanoparticles were mixed with PVDF ultrasonically at room temperature in DMF solvent. The resulting composite nanomaterial was analyzed for complex permittivity and permeability, and the resulting data sets were utilized to determine that the composite demonstrated a maximal RL of −102 dB at 7.7 GHz with a 3.5 mm matching thickness. The permittivity values associated with such measurement were roughly $\varepsilon_r = (7.9 - i3.4)$, with no reporting permeability results [271].

4.4.3. Cobalt sulfide

Cobalt sulfide nanomaterials synthesized with flower-like morphology by Huang et al. have been demonstrated as effective microwave absorbers [272]. The CoS particles were synthesized via a facile solvothermal method, utilizing the cationic detergent, cetyltrimethyl ammonium bromide, as a surfactant to control for morphology. The nanomaterial dispersed in paraffin was able to

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Table 10
Microwave Absorption Performance of Silicon Oxide based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε' at Max	ε'' at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ'' at Max	$tg\delta_\mu$ at Max	Ref.
SiO_2	3.0	0	−1.2	12.3	3.1	0.00	0.00	n/a	n/a	n/a	[241]
ZnO/SiO_2 nanoparticles	0	0	−5.7	15.4	n/a	n/a	n/a	n/a	n/a	n/a	[241]
ZnO/SiO_2 cage structure	1.1	1.1	−10.8	12.8	n/a	n/a	n/a	n/a	n/a	n/a	[241]
SiOC	2.6	2.6	−36.4	11.6	7.3	3.7	0.51	n/a	n/a	n/a	[251]
SiOC	3.2	3.2	−46.5	10.8	7.4	3.8	0.51	n/a	n/a	n/a	[251]
Al-ZnO/$ZrSiO_4$ (2.5%)	3.3	3.3	−28.3	11.2	6.0	2.6	0.43	n/a	n/a	n/a	[252]
Al-ZnO/$ZrSiO_4$ (2.5%)	3.6	3.6	−31.8	9.2	6.2	2.8	0.45	n/a	n/a	n/a	[252]

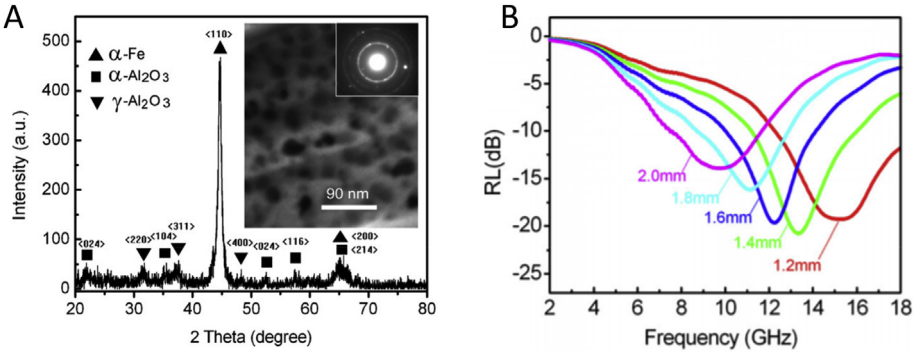


Fig. 16. (A) XRD, (A, insert) TEM, (A, TEM insert) SAED, and (B) RL curves of Fe/Al_2O_3 particles. Reproduced with permission from ref. 254. Copyright 2013 Elsevier.

Table 11
Microwave Absorption Performance of Aluminum Oxide based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε' at Max	ε'' at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ'' at Max	$tg\delta_\mu$ at Max	Ref.
Fe_3Al/Al_2O_3	1.5	4.5	−31.0	16.9	11.2	1.3	0.12	0.77	0.17	0.22	[253]
Fe_3Al/Al_2O_3	2.0	4.9	−25.4	10.2	11.6	1.3	0.11	1.07	0.40	0.37	[253]
Fe_3Al/Al_2O_3	2.2	4.0	−45.4	9.0	11.5	0.9	0.08	1.16	0.44	0.38	[253]
Fe/Al_2O_3	1.4	5.2	−21.3	13.3	7.3	−2.4	−0.33	0.81	0.54	1.28	[254]

generate a maximal RL value of −46.6 dB at a frequency of 15.6 GHz, and with a matching thickness of 2.0 mm. The permittivity values associated with the material were roughly $\varepsilon_r = (7.7 - i1.8)$, with no reporting permeability results. Other CoS particles, as well as CoS particles incorporated into carbon-based nanomaterial composites have demonstrated strong microwave absorbing effects. Materials such as rGO|CoS and MWCNT|CoS composites have consistently generated RL values greater than −50 dB. The results of these studies can be found in Table 12C.

4.4.4. Other sulfide nanomaterials

A multitude of sulfide nanomaterials have been incorporated into core/shell structures so to induce electromagnetic interaction [277–281]. These nanomaterials utilize sulfide cores paired with shells of other dielectric and magnetic materials so to induce domain effects which result in the EM wave absorption. For example, Wang et al. synthesized flower-like In_2S_3 and In_2S_3@CNT core/shell nanostructures by microwave-assisted hydrothermal techniques [277]. The pure In_2S_3 nanomaterials were intrinsically demonstrated to be poor microwave absorbing materials. However, when integrated with CNT structures, the nanomaterial's dielectric response was considerably increased, resulting in a RL of −42.75 dB at a frequency of 11.96 GHz with a 1.55 mm matching thickness. The permittivity values associated with this measurement were roughly $\varepsilon_r = (15.1 - i5.4)$, with no reporting permeability results. Other reported sulfides, such as (ZnS|Ni_3S_2)@Ni, [278] NiS_2@MoS_2, [279] (rGO|NiS_2)@MoS_2, [280] rGO|MnS_2, [281] and graphene|NiS|Ni_3S_2 [282] have also been generated in manners which induce microwave absorption. Their proficiency parameters are reported in Table 12D.

4.5. Phosphides

Phosphides are a newer class of microwave-absorbing materials shown to interact with gigahertz-range electromagnetic radiation. Recently, FeP [88] and Co_2P [89] nanoparticles have been demonstrated to absorb incident microwave radiation through interaction with the electric component of the electromagnetic wave, even though the two nanomaterials involved the integration of atomic species typically characterized by magnetic metal centers. The iron phosphide nanomaterials were synthesized by thermal phosphorization, [283] the resultant particle imaging being shown in Fig. 18. The maximal RL for the FeP was demonstrated was −37.7 dB at 13.6 GHz with a 2.0 mm thickness; the associated permittivity and permeability values were $\varepsilon_r = (17.8 - i3.2)$ and $\mu_r = (0.85 - i0.11)$. The FeP nanomaterial furthermore demonstrated a maximal proficiency bandwidth of 9.7 GHz with a 6.0 mm material thickness. It was suggested that the nanomaterial interactions could be due to polar rotations in the material domain manifesting from the iron-phosphorous bonds [88].

For Co_2P [89] synthesized via thermal phosphorization, [284] the maximal value for RL was given as −39.3 dB at 15.4 GHz with a

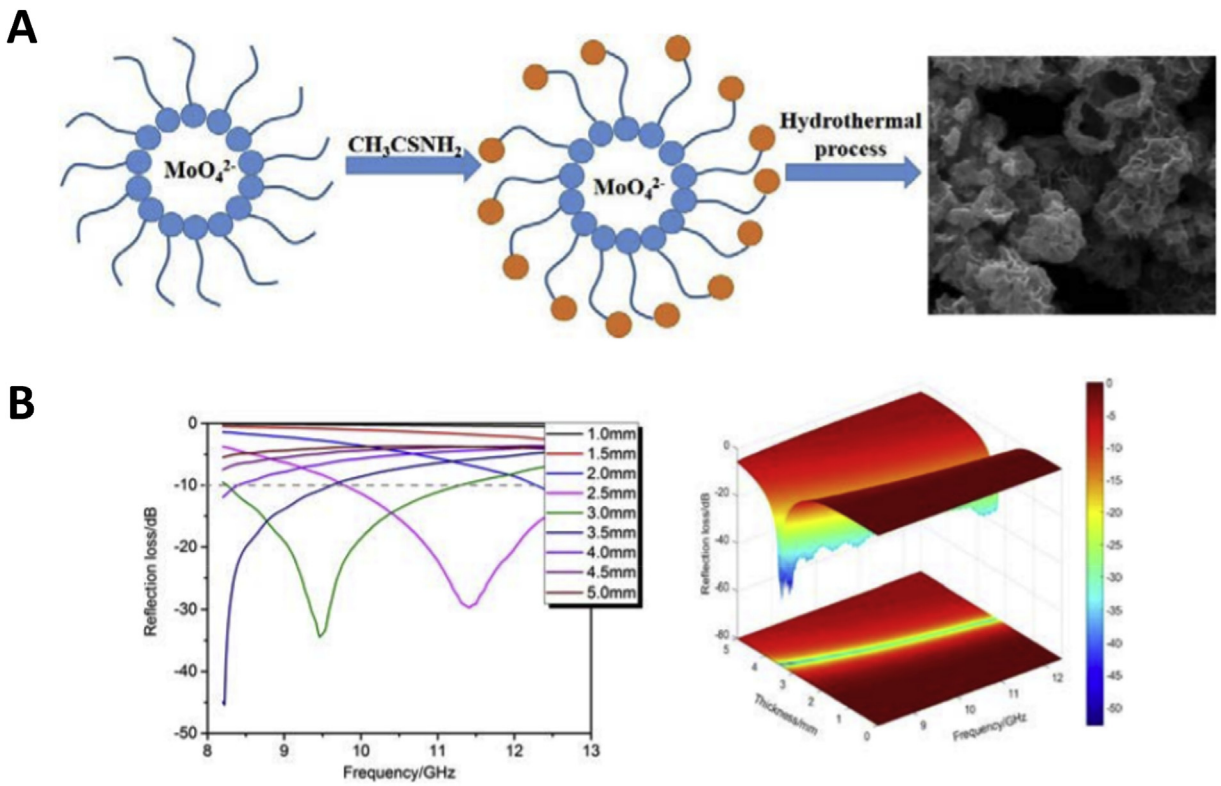


Fig. 17. (A) Schematic of the synthesis process for the MoS_2 nanospheres, and (B) the RL plots of the nanomaterial. Reproduced with permission from ref. 255. Copyright 2018 World Scientific.

Table 12A
Microwave Absorption Performance of Sulfide based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε′ at Max	ε" at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ" at Max	$tg\delta_\mu$ at Max	Ref.
MoS_2	3.5	3.8	−34.8	9.4	7.3	3.47	0.47	n/a	n/a	n/a	[255]
MoS_2	2.5	3.44	−26.11	11.36	8.3	3.3	0.40	n/a	n/a	n/a	[256]
MoS_2	2.4	4.1	−38.4	11.2	8.8	3.48	0.40	n/a	n/a	n/a	[257]
rGO|MoS_2	2.5	5.92	−31.6	13.0	5.5	3.4	0.61	n/a	n/a	n/a	[258]
rGO|MoS_2	2.0	5.72	−33.2	13.0	8.3	3.5	0.42	n/a	n/a	n/a	[259]
rGO|MoS_2	2.6	4.56	−55.0	12.3	6.6	2.2	0.33	n/a	n/a	n/a	[260]
CS|MoS_2	1.4	4.9	−52.6	15.2	6.6	4.0	0.60	n/a	n/a	n/a	[261]
C|MoS_2	1.4	3.32	−44.7	13.0	19.8	7.2	0.36	1.0	0.0	0.0	[262]
graphene|MoS_2	2.5	3.4	−44.8	11.4	9.2	5.8	0.63	1.1	0.0	0.0	[263]
Ni|MoS_2	3.0	2.3	−55.0	5.7	15.4	5.3	0.34	1.18	0.25	0.21	[264]
MoS_2@FeCo	2.0	7.2	−64.64	14.4	7.9	2.1	0.26	0.95	0.24	0.25	[265]
MoS_2@$Bi_2Fe_4O_9$	2.8	5.0	−52.3	12.4	6.2	2.8	0.45	0.78	0.16	0.20	[266]

Table 12B
Microwave Absorption Performance of Sulfide based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε′ at Max	ε" at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ" at Max	$tg\delta_\mu$ at Max	Ref.
CuS	1.95	3.6	−54.84	12.63	12.3	2.2	0.18	n/a	n/a	n/a	[267]
CuS	1.2	4.2	−16.0	14.9	15.8	11.7	0.74	n/a	n/a	n/a	[268]
CuS	2.5	3.8	−29.66	10.9	7.2	4.1	0.56	n/a	n/a	n/a	[269]
CuS	1.8	3.6	−31.5	16.7	7.9	2.6	0.33	n/a	n/a	n/a	[270]
PVDF|CuS	3.5	6.2	−102.0	7.7	7.9	3.4	0.43	n/a	n/a	n/a	[271]

1.1 mm thickness; the associated permittivity and permeability values were given as $\varepsilon_r = (17.8 - i3.6)$ and $\mu_r = (1.1 - i0.1)$. Furthermore, the nanomaterial's dielectric nature was examined by theoretically calculating the material response as if the magnetic interaction was trivial; the resultant demonstrated that specifically at high thicknesses and lower frequencies the magnetic interaction had a non-trivial impact on the material response. As such, it was considered to be both dielectric and magnetic interaction of the

Table 12C
Microwave Absorption Performance of Sulfide based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε' at Max	ε'' at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ'' at Max	$tg\delta_\mu$ at Max	Ref.
CoS	2.0	4.6	−43.6	15.6	7.7	1.8	0.23	1.0	0.0	0.0	[272]
$Co_{1-x}S$	2.5	4.4	−45.8	14.0	5.3	1.6	0.30	0.9	0.3	0.33	[273]
rGO|CoS	4.0	2.1	−54.2	6.7	9.2	1.9	0.21	1.1	0.0	0.0	[274]
rGO|CoS_2	2.2	4.1	−56.9	10.9	9.0	4.2	0.46	1.2	0.1	0.1	[275]
MWCNT|CoS	3.6	2.4	−56.1	6.7	10.2	2.2	0.22	1.0	0.0	0.0	[276]

Table 12D
Microwave Absorption Performance of Sulfide based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε' at Max	ε'' at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ'' at Max	$tg\delta_\mu$ at Max	Ref.
In_2S_3@CNT	1.55	3.8	−42.75	11.96	15.1	5.4	0.36	n/a	n/a	n/a	[277]
$(ZnS\|Ni_3S_2)$@Ni	2.5	1.5	−24.0	16.6	22.3	1.9	0.085	1.3	0.06	0.046	[278]
NiS_2@MoS_2	2.2	4.4	−41.05	12.0	n/a	n/a	n/a	n/a	n/a	n/a	[279]
rGO|MnS_2	2.5	10.6	−52.2	10.72	9.2	4.4	0.48	0.96	0.0	0.0	[280]
$(rGO\|NiS_2)$@MoS_2	4.0	1.4	−29.8	5.3	13.0	4.1	0.32	1.0	0.0	0.0	[281]
graphene|NiS|Ni_3S_2	2.7	2.4	−55.1	6.0	11.5	3.9	0.34	1.1	0.01	0.01	[282]

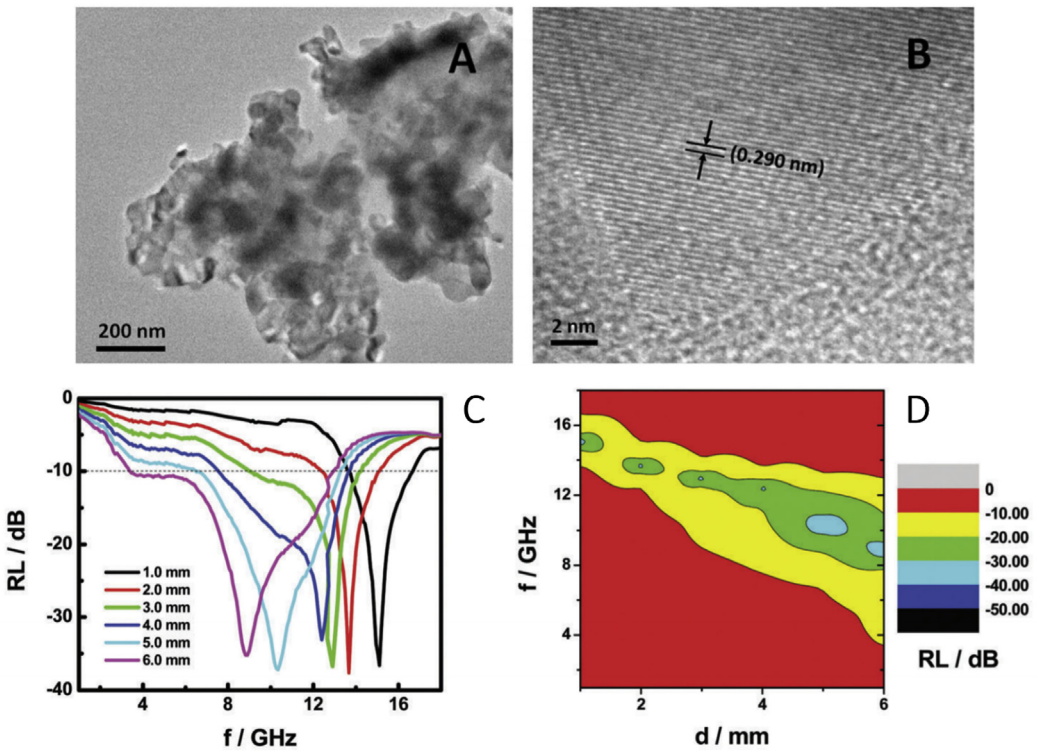


Fig. 18. (A, B) TEM images and (C, D) RL plots of FeP nanoparticles. Reproduced with permission from ref. 88. Copyright 2018 Royal Society of Chemistry.

material that resulted in the strong representation in RL. The results of these studies can be found in Table 13.

4.6. Polymers

Polymer nanomaterials have played an important role in the development of microwave absorbing materials, being utilized for their high dielectric properties as intrinsic absorbers, nanoparticle coatings, and composites. Their application as gigahertz-range electromagnetic absorbers was first developed in the 1990's by Olmedo et al [285]. For example, polypyrrole as a microwave absorbing material was developed and tested for response to incident electromagnetic radiation as a function of the electric portion of the electromagnetic wave. As shown in Fig. 19, over the tested 0.13–20 GHz frequency range investigated, the RL function maximized at 13.6 GHz with a −46.8 dB RL response [285]. The reported permittivity value associated with the optimal response was $\varepsilon_r = (8.0 - j3.5)$. As such, it was concluded that the electrical interaction of the material to the incident wave induced the RL, due to the conductivity of the polymer material which induced a dielectric response [285]. Furthermore, composite polymer nanomaterials such as ZnO|polyester, [240] MnO_2|polydopamine [286] as well and

Table 13
Microwave Absorption Performance of Phosphide based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε' at Max	ε'' at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ'' at Max	$tg\delta_\mu$ at Max	Ref.
FeP	1.0	3.1	−36.5	15.1	18.2	5.4	0.30	0.85	0.11	0.13	[88]
FeP	2.0	2.7	−37.7	13.6	17.8	3.2	0.18	0.88	0.04	0.05	[88]
FeP	3.0	5.4	−36.5	12.9	18.4	2.4	0.13	0.90	0.02	0.02	[88]
Co_2P	1.0	2.7	−23.3	16.7	20.28	4.3	0.21	1.05	0.11	0.10	[89]
Co_2P	1.1	2.0	−38.9	15.5	17.80	3.62	0.20	1.11	0.11	0.10	[89]
Co_2P	2.5	1.2	−30.3	7.0	21.12	3.80	0.18	0.88	0.06	0.07	[89]

other forms of polymer-based nanomaterials such as polyaniline@C, [287] polyprrole@PANI, [288] $BaTiO_3$|polyaniline [289] and $BaFe_{12}O_{19}$|polyaniline, [289–291] $CoFe_2O_4$|$BaFe_{12}O_{19}$|polyaniline, [291] Ni:Zn/Fe_2O_4|polyaniline, [292] MnO_2|polyaniline, [293] $CoFe_2O_4$|graphite|polyaniline, [294] $NiFe_2O_4$|graphene|polyaniline, [295] CNT|polyurethane, [296] and MWCNT|polyethylene terephthalate [297] have been shown to interact with incident electromagnetic radiation in similar dielectric fashions previously discussed, where the integration of magnetic materials and other dielectrics demonstrate deviations in mainly the dielectric response, though strong magnetic responses have been demonstrated as well [291,298]. Polymer materials demonstrate a high degree of flexibility in being integrated with nanomaterials in a manner that consistently perturbs the nanomaterial's response to electromagnetic radiation, predominately through dielectric interaction as shown by the strong permittivity response of the nanomaterial. The results of these nanomaterial sets are tabulated in Table 14.

4.7. Metals/alloys

Many of the pure metal and subsequent alloys which have been used in microwave systems are magnetic-based nanomaterials. Utilization of these magnetic particles such as iron, cobalt, and nickel, as both independent species and in agglomeration with each other, and furthermore with other magnetic and dielectric dopants, has generated sets of nanomaterials which respond differently to incident electric and magnetic fields, depending on the integrated entities [299–304]. For example, market-available carbonyl iron was shown to absorb gigahertz-range electromagnetic radiation by Zhang et al., where the high-loss interaction through the magnetic component of the incident wave, represented by $\mu_r = (1.54 - j1.16)$, which resulted in a −32.5 dB RL at 8.3 GHz with a 2.0 mm thickness [305]. In Zhang's study, it was also shown that the introduction of manganese oxide as a dopant material, consistent with its function as a dielectric material previously discussed, decreased the nanomaterial response to the incident magnetic field, but increased the nanomaterial's response as a dielectric, which resulted in the further increase the MnO_2/CI nanomaterials' reflection loss, from a maximal RL of −32.5 dB being extended to a maximal RLvalue of −39.1 dB with a 30% MnO_2 loading [305]. A select set of these results are shown in Fig. 20. Furthermore, the results from this study are tabulated in Table 15A.

Other metal and alloy-based nanomaterials and composites which have been used to interact with microwaves, [306] including MWCNT|carbonyl iron, [307] graphene-coated iron, [308] SnO_2|carbonyl iron, [309] $BaTiO_3$|carbonyl iron, [310] Fe@C@$BaTiO_3$, [311] C|Fe, [311,312] PVC|CIP, [313] SiO_2|Fe, [314,315] ZnO|Fe, [316] Co@C, [317,318] NiCu, [319] CoNi, [320] CoNi@SiO_2, [320] CoNi@SiO_2@TiO_2, [320] CoNi@Air@TiO_2, [320] CoNi@SiO_2@C, [321] ErHoFe Alloys, [322] CoFe Alloys, [63,323–325] FeSiAl Alloys, [57] Ni powders [326] and Ti_3SiC_2|Ni composites, [326] Al:AlO_x|Ni, [327] Fe_3Ni@C, [328] FeNiMo@C, [329] CoNi, [330] Ni@Co-P, [331] Ni@Basalt, [332] Co@CoO, [333] CoFe@C, [334,335] C|FeNi, [336] PANI|FeNi, [337] and NdFe [338]. For example, Yang et al. [63] synthesized a set of FeCo nanoflakes via a wet chemical method so to interact with incident EM frequencies via magnetic loss. Yang further demonstrated that the nanomaterials' interaction was possibly due to natural resonance. To tease out the mechanism, they graphically showed the value of $\mu''/f\cdot(\mu')^2$ as a function of frequency; if this plot demonstrates a zero derivative the frequency changes, then the magnetic loss can be attributed to eddy-current losses [63]. Upon investigation of the nanomaterial responses however, such was demonstrated not to be the casual mechanism of the interaction. Furthermore, as multidomain nanomaterials are required for exchange resonance, [63] thus the logical conclusion was that the mechanism of action must have been natural resonance.

For further example, carbon-coated iron/-nickel-molybdenum core-shell alloy structures of varying stoichiometric quantities have been shown by Liu et al. [329] to absorb incident microwave

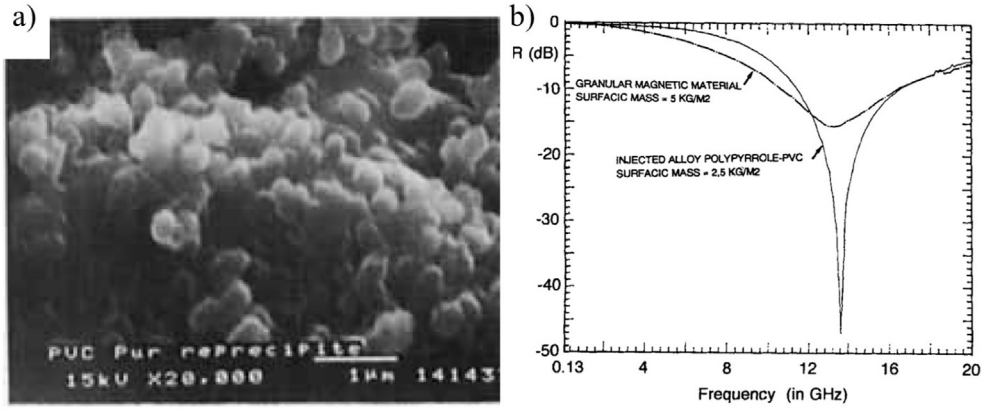


Fig. 19. (a) SEM image and (b) RL plot of polypyrrole. Reproduced with permission from ref. 285. Copyright 1993 Verlag GmbH & Co. KGaA, Weinheim.

Table 14
Microwave Absorption Performance of Polymer based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε' at Max	ε'' at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ'' at Max	$tg\delta_\mu$ at Max	Ref.
ZnO|polyester	1.0	1.7	−12.28	10.75	2.0	0.1	0.05	n/a	n/a	n/a	[240]
ploypyrrole	2.0	4.9	−46.8	13.6	8.0	3.5	0.44	n/a	n/a	n/a	[285]
MnO_2/polydopamine	3.0	3.3	−21.8	9.7	7.3	2.7	0.37	1.0	0.0	0.00	[286]
PANI@C	2.2	5.5	−59.6	15.5	11.0	4.9	0.45	n/a	n/a	n/a	[287]
Polypyrrole@PANI	2.0	4.7	−34.8	13.9	7.9	4.0	0.51	n/a	n/a	n/a	[288]
$BaTiO_3$/PANI	2.0	2.2	−13.8	11.6	n/a	n/a	n/a	n/a	n/a	n/a	[289]
$BaFe_{12}O_{19}$|PANI	2.0	0.5	−13.0	7.8	n/a	n/a	n/a	n/a	n/a	n/a	[289]
$BaFe_{12}O_{19}$|PANI	2.0	0.5	−12.6	7.6	10.9	8.4	0.77	1.4	0.015	0.01	[290]
PANI/BaM	2.9	2.0	−16.7	14.7	8.6	1.8	0.21	4.1	2.1	0.51	[291]
PANI/BaM/CFO	2.9	4.2	−36.4	9.7	5.7	3.7	0.65	1.3	−0.1	−0.08	[291]
$Ni_{0.6}Zn_{0.4}Fe_2O_4$|PANI	2.6	5.0	−41.0	12.8	6.2	6.0	0.97	0.9	−0.05	−0.06	[292]
MnO_2/PANI	2.0	5.4	−21.0	13.6	10.6	5.9	0.56	0.9	−0.04	−0.04	[293]
$CoFe_2O_4$|graphite| polyaniline	0.5	5.9	−19.13	13.3	n/a	n/a	n/a	n/a	n/a	n/a	[294]
$NiFe_2O_4$|graphene| polyaniline	2.5	5.3	−50.5	12.5	6.3	1.9	0.30	0.96	0.02	0.02	[295]
$NiFe_2O_4$|graphene| polyaniline	3.0	3.5	−32.4	10.0	6.6	2.8	0.42	1.1	0.11	0.10	[295]
CNT|Polyurethane	2.0	2.6	−22.0	8.8	n/a	n/a	n/a	n/a	n/a	n/a	[296]
MWCNT|PET	2.0	2.7	−17.0	7.6	n/a	n/a	n/a	n/a	n/a	n/a	[297]

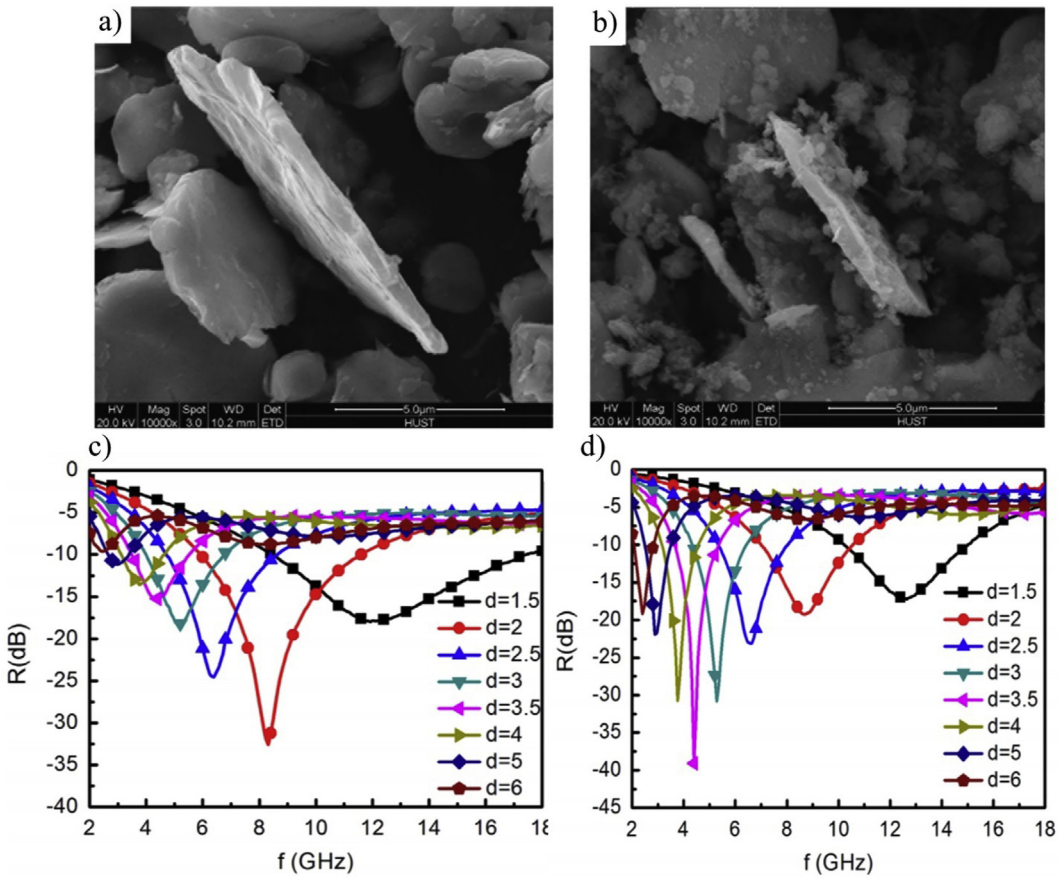


Fig. 20. (a, b) SEM images and (c, d) RL plots of (a, c) carbonyl iron and (b, d) MnO_2/CI 30 wt%. Reproduced with permission from ref. 305. Copyright 2014 Elsevier.

radiation across a diverse set of frequency and thickness profiles, with a reported maximal RL of −64.1 dB at 13.2 GHz with a corresponding thickness of 1.9 mm, as shown in Fig. 21 and tabulated in Table 15A. The material also had a corresponding proficiency bandwidth of 7.9 GHz [329]. Amalgam alloys typically generate resultant RL through dual interaction with both electric and magnetic properties of the nanomaterials, registering relatively strong values for electromagnetic interaction in both domains, depending on the introduced atomic species. For example, the FeSiAl alloys synthesized by Feng et al. [57] demonstrated extremely strong permittivity, as well as relatively strong permeability, resulting in a maximal RL value of −39.7 dB at 1.4 GHz with a 4.0 mm material thickness. Permittivity and permeability were reported to be values of $\varepsilon_r = (25.4 - i1.2)$ and $\mu_r = (3.5 - i1.9)$ at such point of maximal RL. The full results of the metal and alloy nanomaterials previously mentioned are tabulated in Table 15A–Table 15B.

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Table 15A
Microwave Absorption Performance of Metal and Alloy based Nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε' at Max	ε'' at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ'' at Max	$tg\delta_\mu$ at Max	Ref.
Carbonyl Iron	2.0	5.7	−32.5	8.3	9.18	0.17	0.02	1.54	1.16	0.75	[305]
Carbonyl Iron	2.5	3.9	−24.6	6.4	9.1	0.09	0.01	1.83	1.23	0.67	[305]
MnO_2/CI 10 wt%	2.5	3.1	−36.3	5.9	11.9	0.49	0.04	1.73	1.02	0.59	[305]
MnO_2/CI 30 wt%	3.5	1.7	−39.1	4.4	14.0	1.29	0.09	1.54	0.58	0.38	[305]
0.5 wt% MWCNT|spherical CIP	1.5	4.3	−29.5	8.8	13.0	0.26	0.02	1.9	0.85	0.45	[307]
0.5 wt% MWCNT|flaky CIP	1.5	4.0	−24.0	5.7	16.2	0.05	0.00	2.8	1.58	0.56	[307]
1.5 wt% MWCNT|spherical CIP	2.0	3.5	−38.2	6.3	13.5	0.60	0.04	2.1	0.84	0.40	[307]
1.5 wt% MWCNT|flaky CIP	2.0	2.0	−34.4	3.4	13.2	1.37	0.10	4.2	1.71	0.41	[307]
Fe/G	1.5	3.8	−34.3	17.1	7.6	4.3	0.57	1.19	−0.13	−0.11	[308]
Fe/G	3.0	2.6	−45.0	7.1	10.8	3.0	0.28	1.17	0.17	0.15	[308]
SnO_2/CI	7.0	3.5	−37.8	16.1	3.4	0.25	0.07	1.1	0.20	0.18	[309]
SnO_2/CI	8.3	2.2	−57.8	12.2	4.4	0.14	0.03	1.1	0.22	0.20	[309]
Carbonyl Iron	2.0	1.7	−31.4	3.3	27.1	0.7	0.03	3.5	2.0	0.57	[310]
$BaTO_3$/CIP (20/60 wt%)	2.0	2.2	−41.3	4.1	19.8	0.6	0.03	2.7	1.8	0.67	[310]
$BaTO_3$/CIP (40/40 wt%)	2.5	2.0	−38.2	5.1	16.2	0.3	0.02	1.8	0.9	0.50	[310]
Fe@C@$BaTiO_3$	2.24	3.6	−40.1	8.1	12.5	4.4	0.35	1.40	0.108	0.08	[311]
Fe@C	1.7	5.1	−27.2	15.5	6.6	3.0	0.45	1.13	0.052	0.05	[311]
C/Fe	2.0	6.8	−23.0	15.1	6.2	4.3	0.69	1.32	0.0	0.00	[312]
PVC|CIP	2.0	3.9	−20.5	10.5	24.6	2.8	0.11	1.6	1.0	0.63	[313]
SiO_2@Fe	2.0	8.9	−18.3	9.0	n/a	n/a	n/a	n/a	n/a	n/a	[314]
Fe Nanoflakes	2.8	0.3	−10.4	2.4	34.7	14.0	0.40	3.0	1.3	0.43	[315]
SiO_2/Fe nanoflakes	2.8	2.9	−54.9	5.3	11.7	0.58	0.05	1.8	0.98	0.54	[315]
ZnO/Fe	2.0	4.9	−36.8	10.2	8.8	1.0	0.11	1.21	0.094	0.08	[316]
ZnO/Fe	3.0	7.1	−57.1	7.8	9.0	0.8	0.09	1.22	0.25	0.20	[316]
Co@C 50 wt%	2.0	5.5	−23.7	14.2	7.9	3.1	0.39	0.91	0.48	0.53	[317]
Co@C 50 wt%	3.0	3.7	−24.6	8.8	7.9	3.5	0.44	1.0	0.50	0.50	[317]
Co@C	1.65	15.3	−68.7	10.6	13.2	0.8	0.06	1.28	0.48	0.38	[318]
NiCu	1.5	2.8	−31.1	14.2	12.2	4.4	0.36	1.13	−0.02	−0.02	[319]
CoNi	2.1	6.6	−26.0	8.5	9.5	2.2	0.23	2.0	0.6	0.30	[320]
SiO_2@CoNi	2.1	7.1	−27.1	9.5	8.1	2.2	0.27	2.0	0.6	0.30	[320]
TiO_2@SiO_2@CoNi	2.1	10.6	−57.9	10.0	6.4	2.8	0.44	1.9	0.6	0.32	[320]
TiO_2@Air@CoNi	2.1	8.5	−41.2	9.5	8.0	3.2	0.40	2.0	0.6	0.30	[320]
CoNi@SiO_2@C	2.2	5.6	−46.0	10.8	6.3	1.35	0.21	1.06	0.1	0.09	[321]
ErHoFe	2.0	1.84	−35.72	7.68	22.2	4.2	0.19	1.12	0.15	0.13	[322]
CoFe	2.1	5.6	−22.7	14.8	6.8	1.5	0.22	0.9	0.2	0.22	[323]
FeCo	2.0	3.3	−30.8	11.4	4.4	0.5	0.11	1.1	0.1	0.09	[324]
FeCo	1.7	3.2	−27.2	12.9	4.4	0.5	0.11	1.1	0.1	0.09	[324]
Co_3Fe_7	1.55	6.8	−53.6	14.2	9.2	1.4	0.15	1.6	0.5	0.31	[325]
FeCo nanoflakes	1.8	3.3	−43.0	8.1	17.1	3.2	0.19	1.6	0.47	0.29	[63]

4.8. Carbon nitrides

Graphitic carbon nitride (g-C_3N_4) nanosheets were recently reported as a nanomaterial set which induces reflection loss from gigahertz-range electromagnetic radiation at high value material thicknesses. Green et al. [104] generated the g-C_3N_4 nanosheets through thermal condensation, heating melamine to 550 °C for 4 h in atmosphere so to produce the nanomaterial as shown in Fig. 22. Such nanomaterial reached a maximal RL of −36.1 dB at 15.11 GHz with a 19.5 mm thickness, due to tandem magnetic and dielectric action upon the incident electromagnetic wave, as shown in Fig. 22C and D. The permittivity and permeability associated with the point of maximal RL was $\varepsilon_r = (3.16 - i{\cdot}0.10)$ and $\mu_r = (1.00 - i{\cdot}0.08)$. Along with theoretical calculations for zero susceptibility, g-C_3N_4 was used in attempt to elucidate the response of a theoretical non-dielectric, setting the lossy dielectric parameter to zero and taking the average lossless interaction in calculation over the frequency/thickness domain [104]. The calculated theoretical results were used to suggest tandem dielectric and magnetic action of the nanomaterial. These results are tabulated in Table 16.

4.9. Metal-organic frameworks

Recently, an iron-based metal-organic framework was shown by Green et al. [62] to induce RL through dielectric antireflection action. The Fe-MOF, shown in Fig. 23, was synthesized via the utilization of iron metal centers and both 1,4-benzenedicarboxylate and 1,4-diazabicyclo[2.2.2]octane as coordinating ligands as described in synthesis by Chun et al. [339]. The anti-reflection model proposed therein noted that the thickness of local maximal RL followed the relation of $d = (2m-1)\frac{\lambda}{4}$ where m = 1, 2, 3 … positive integers, with 'd' being the material thickness, lambda being the propagating wavelength, and m being the set of positive integer values greater than one which represented the absorption bands traversing the RL profile. Furthermore, the peak wavelength was characterized by $\lambda = \frac{c}{n \cdot f}$ with f being the incident frequency, n being the refractive index, and c being the speed of light. From such, it was shown that frequency of maximal reflection loss and the material thickness were interrelated by the function $d(\lambda) = \frac{(2m-1)\cdot\lambda}{4}$, and conversely $f(d) = \frac{(2m-1)}{4} \cdot \frac{c}{n \cdot d}$. For the Fe-MOF, the point of maximal RL was −39.3 dB at 8.82 GHz with a 2.7 mm thickness. The permittivity and permeability values associated with the point of maximal RL were reported as $\varepsilon_r = (10.27 - i{\cdot}3.47)$ and $\mu_r = (1.01 - i{\cdot}0.05)$, and are also shown in Table 16. Given these associated response values to incident electric and magnetic fields, it was suggested that the RL was due to dielectric action of the framework, possibly manifesting via microwave conductivity originating from polar rotations across the framework interface [62].

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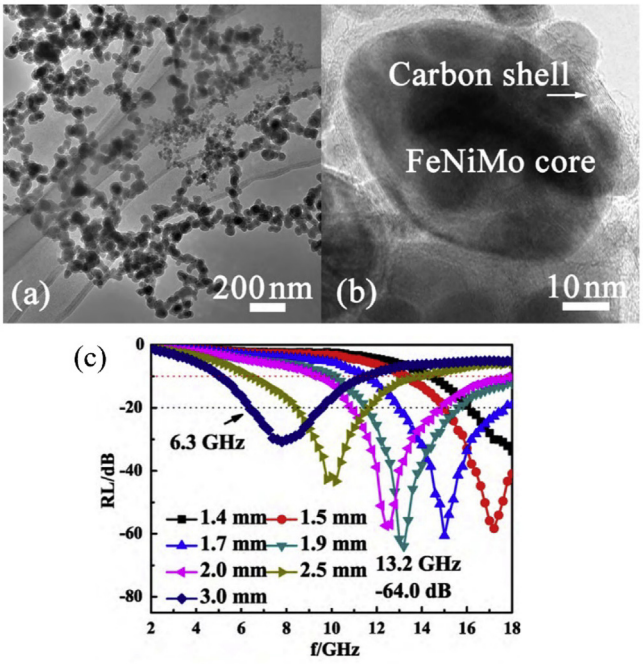


Fig. 21. (a) TEM, (b) HRTEM, and (c) RL plots for C@$Fe_{11}Ni_{79}Mo_{10}$. Reproduced with permission from ref. 329. Copyright 2013 Elsevier.

5. Summary and conclusions

The current pantheon of scientific literature includes a diverse set of nanomaterials which respond to incident electromagnetic radiation in a variety of different fashions. Varying the nanomaterial identity, crystal structure, particle and material doping identities and scales, and composite ratio – all these domains of tuning can have aggregate effects on a nanomaterials propensity to interact with incident electromagnetic radiation in a way that is deemed desirable from the perspective of maximizing reflection loss.

The nature of nanomaterial interaction with electromagnetic radiation is ultimately quantified by the permittivity and permeability of a given bulk material domain. These values are complex numbers, relating to the energy storage and dissipation of the nanomaterial, typically relative to that of free space. There are many ways in which researchers can utilize these nanomaterial properties to quantify various intrinsic interactions with light. Fundamental parameters, such as the refractive index, the extinction coefficient, the attenuation and phase constants, and impedance can be directly derived. Furthermore, the RL of a nanomaterial can be calculated knowing permittivity and permeability as a function of frequency and material thickness. Maximizing this reflection loss value with respect to both a point of maximal RL as well as the effective bandwidth are two of the main goals for researchers in this field.

Normally, to obtain a good microwave absorption performance, there needs to be a good match between the dielectric/magnetic constants with the thickness of the absorber and the wavelength of the microwave irradiation so that a large RL can be realized with the effective cancellation of microwave irradiation reflected from the front and back sides of the microwave coating featuring a quarter-wavelength characteristics with the thickness of the coating [340]. The dielectric/magnetic constants are normally related to the chemical composition and the crystal phase of the compounds, therefore, microwave absorbers made of materials with different constitutions would have different dielectric/magnetic values. Nanomaterials which are used in the development for microwave absorbing materials typically have an intrinsic propensity to interact with gigahertz-range electromagnetic radiation, which can be a function of the nanomaterial identity and composition as dictated by synthesis parameters, crystal structure, particle shape, particle size, et al. Some nanomaterials have intrinsic permittivity and permeability values which demonstrate strong reflection loss. Other nanomaterials can be perturbed so to induce systematic changes in nanomaterial response as demonstrated within ε_r and μ_r. Chemical perturbations such as materials doping can generally, though not axiomatically, lend to increases in the nanomaterial interaction, through respective dielectric and magnetic responses attributed to the dopant identity. Such actionable perturbation is as commonly demonstrated in the dielectric domain as it is within the domain of the magnetic. Structural perturbations, such as core-shell nanoparticles and particles of complex morphologies have also been reported as good microwave absorbing materials. Composite nanomaterials tend towards results which are typically defined by a change in dielectric interaction of the nanomaterial as expressed by permittivity, particularly when integrated in part with materials such as conducting polymers. The size and shape of the nanomaterials can not only alter the dielectric/magnetic properties of the materials, but also create big interfacial dipoles whose rotations might echo with the incident microwave energy, leading to a resonance absorption. Finally, in distinct cases, non-interactive

Table 15B
Microwave absorption performance for metal and alloy nanomaterials.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε′ at Max	ε" at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ" at Max	$tg\delta_\mu$ at Max	Ref.
FeSiAl	4.0	0.9	−39.7	1.4	25.4	1.2	0.05	3.5	1.9	0.54	[57]
Nickel Powder	2.0	1.6	−22.4	9.7	10.8	0.3	0.03	0.99	0.45	0.45	[326]
Ni/Ti_3SiC_2 30/30 wt%	2.2	3.8	−41.2	9.7	17.4	1.3	0.07	0.81	0.18	0.22	[326]
Al:AlO_x/Ni	2.4	2.7	−41.7	9.2	9.8	4.8	0.49	0.1	0.8	8.00	[327]
$FeNi_3$@C NW	2.0	5.2	−43.5	9.9	8.6	3.6	0.42	1.8	0.29	0.16	[328]
$Fe_{11}Ni_{79}Mo_{10}$@C	1.7	6.5	−60.5	15.0	4.8	1.0	0.21	1.21	0.10	0.08	[329]
$Fe_{11}Ni_{79}Mo_{10}$@C	1.9	7.9	−64.1	13.2	5.6	0.9	0.16	1.19	0.14	0.12	[329]
flowerlike CoNi	2.0	5.5	−21.8	8.0	9.1	3.6	0.40	2.9	0.6	0.21	[330]
Co-P@Ni	3.0	1.2	−36.8	8.1	6.4	0.2	0.03	1.0	0.2	0.20	[331]
Basalt@Ni	3.0	1.5	−39.9	8.8	14	8.7	0.62	1.1	0.1	0.09	[332]
Co@CoO	1.5	4.6	−30.4	16.1	n/a	n/a	n/a	n/a	n/a	n/a	[333]
C@CoFe	4.0	2.8	−44.0	8.0	6.3	2.9	0.46	0.95	0.02	0.02	[334]
oriented C@CoFe	2.1	1.7	−48.2	6.4	1.7	1.5	0.88	0.6	0.4	0.67	[335]
C/FeNi	2.0	5.6	−26.9	16.0	6.2	2.5	0.40	0.97	0.12	0.12	[336]
PANI|FeNi	2.5	2.4	−43.0	8.3	9.48	2.0	0.21	0.99	0.28	0.28	[337]
NdFe	3.0	1.3	−55.9	3.6	21.0	3.0	0.14	2.2	0.75	0.34	[338]

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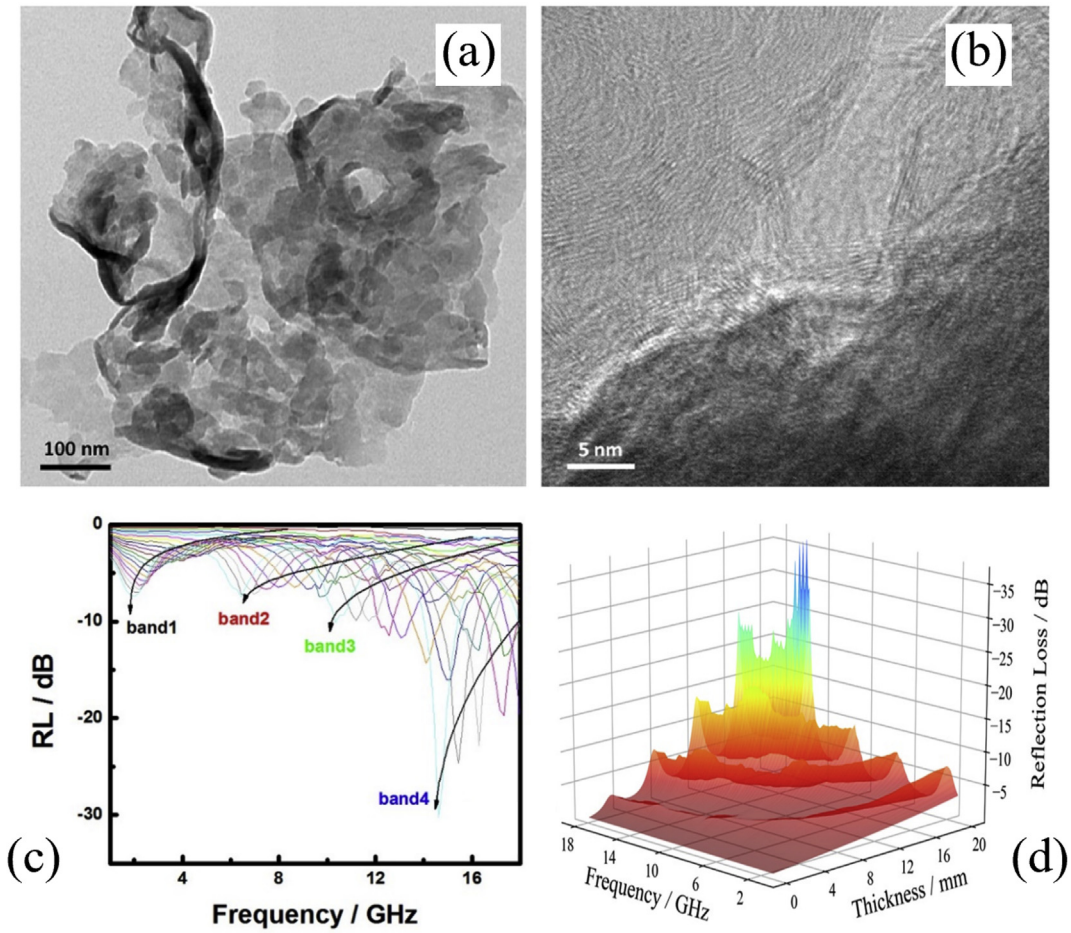


Fig. 22. (a) TEM, (b) HRTEM, and (c, d) RL plots for g-C_3N_4. Reproduced with permission from ref. 104. Copyright 2018 Elsevier.

Table 16
Microwave absorption performance in the recent nanomaterial advances.

Nanomaterial	Performance:										
	d (mm)	$\Sigma\Delta f_{-10}$ (GHz)	Max RL (dB)	Freq (GHz)	ε' at Max	ε'' at Max	$tg\delta_\varepsilon$ at Max	μ' at Max	μ'' at Max	$tg\delta_\mu$ at Max	Ref.
g-C_3N_4	19.5	1.4	−36.1	15.11	3.16	0.10	0.03	1.00	0.08	0.08	[104]
g-C_3N_4	20.0	1.4	−30.2	14.60	3.19	0.10	0.03	1.00	0.08	0.08	[104]
Fe-MOF	2.5	3.1	−34.6	9.59	9.92	3.64	0.37	1.02	0.05	0.05	[62]
Fe-MOF	2.6	2.9	−38.9	9.16	10.12	3.57	0.35	1.01	0.05	0.05	[62]
Fe-MOF	2.7	2.8	−39.3	8.82	10.27	3.47	0.34	1.01	0.05	0.05	[62]

nanomaterials in tandem have been shown to induce microwave interaction.

The major classes of nanomaterials which have been shown to interact with microwaves are carbons, carbides, oxides, phosphides, polymers, and alloys. Research continues to develop new amalgams, composites, and structures of these nanomaterials so to generate new responses to incident electromagnetic radiation. Furthermore, new nanomaterial systems, such metal organic frameworks and carbon nitride, continue to be expanded upon. These and other research works allow for the develop of new niches in the field of microwave absorbing materials, upon which development in parallel to the lines of logic demonstrated in the major nanomaterial classifications can be used, so to expand the totality of our understanding in regard to how nanomaterials interact with microwaves. Finally, as new nanomaterial domains are investigated and reported on for their intrinsic responses, the scientific community will continue to articulate, enhance, and build upon the development of new future nanomaterials which absorb microwave radiation.

As a final note, many of the nanomaterials analyzed and discussed in the literature have gone to great effort highlighting their maximal RL values as the parameter of greatest importance. However, it is coming more apparent with the recent demonstration of regression analyses that such a paradigm should be reconsidered. Regression analysis has demonstrated that in certain systems where it has been applied, for a relatively large RL (appx. < -30 dB), increased precision in the tuning of the frequency and thickness parameters around a point of maximal RL induces a growth in the resulting RL value that is linear on a log scale [105,106]. As such, maximal RL may be more so an artifact of calculation, rather than a parameter which should be acquired via the fine tuning of nanomaterial parameters as expressed through the permittivity and permeability. This observation is further reinforced by the fact that some nanomaterials which demonstrate

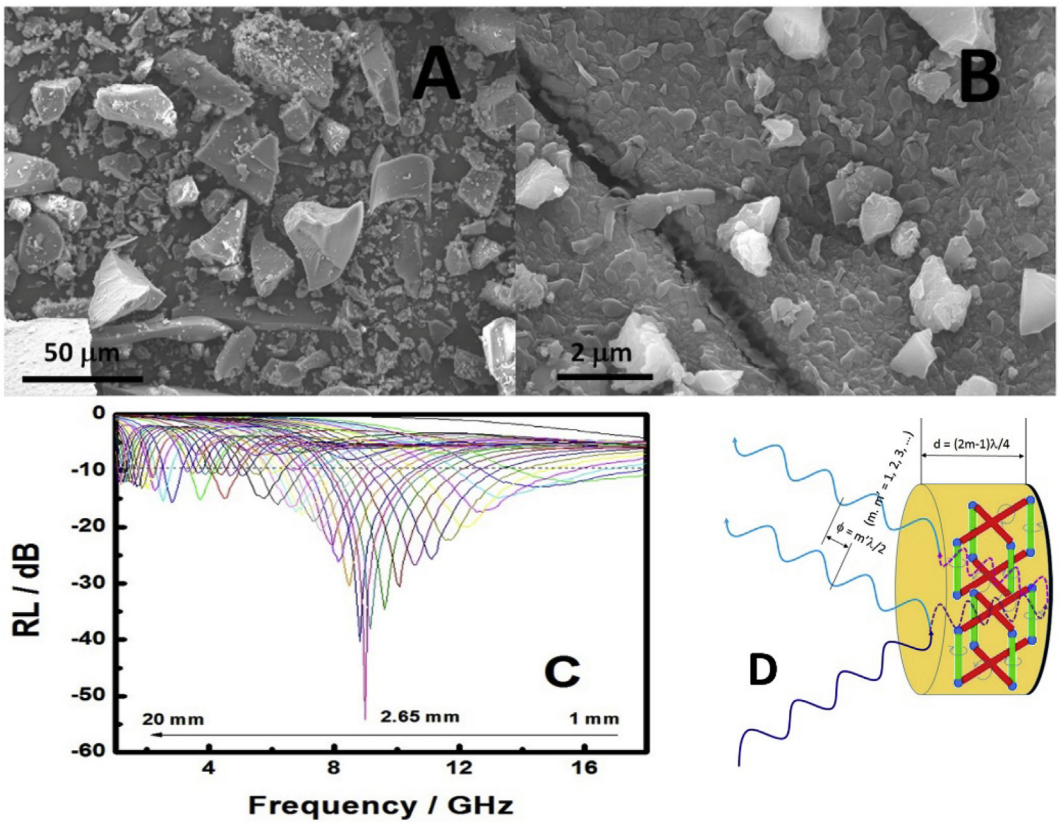


Fig. 23. (a, b) SEM images, (c) RL plots, and (d) schematic for antireflection action for Fe-MOF Reproduced with permission from ref. 62. Copyright 2018 Elsevier.

astronomical RL values don't further demonstrate oddly novel permittivity and permeability resultants, when compared to other nanomaterial resultants from across the field. For example, the −102 dB resultant demonstrated by PVDF|CuS [271] and defined by the input parameters $f = 7.7$, $d = 3.5$, and $\varepsilon_r = (7.9 - i3.4)$ isn't much more differing in input parameters than nanomaterials such as, $NiFe_2O_4$|rGO [148] defined by the input parameters $f = 7.7$, $d = 3.5$, and $\varepsilon_r = (8.1 - i3.2)$, or Ni_2O_3|C [173] defined by the input parameters $f = 7.5$, $d = 3.5$, $\varepsilon_r = (8.4 - i3.5)$. Yet, the microwave absorption reported by the PVDF|CuS is, by definition, demonstrating multiple orders of magnitude greater performance than either the $NiFe_2O_4$|rGO or the Ni_2O_3|C, which demonstrated RL values of −30.0 and −21.3 dB respectively. It is more likely that the frequency·thickness parameter explicitly calculated for the PVDF|CuS nanomaterial via conventional norms of the field simply ran slightly, yet sufficiently, closer to perfect absorption conditions, resulting in the astronomical RL value, despite the nanomaterial not demonstrating a significantly differing electromagnetic interaction as quantified by the permittivity and permeability of the material at the frequency·thickness of interest.

Given such, it seems logical that the field consider the primacy of not the RL, but the effective bandwidth, as the parameter which should be maximized; whereby the presence of RL of an arbitrarily large value simply serves as a conditional which permits for effective bandwidth analysis. The utility of the sufficient absorption of microwave radiation across a greater range of frequencies given the above discussed seems greater than permitting the reflection of 1 photon per 100,000 photons, as compared to 1 photon per 10,000. There are select manuscripts which have demonstrated results which have yielded strong effective bandwidths, [61,109,110,114,203,226,318] but those reports are fewer and farther between, as a maximized effective bandwidth requires the tuning of a function of parameters, and not simply the set that goes into a single RL point. Nevertheless, it seems that the future of research into microwave absorbing materials will inevitably pivot towards such a direction, towards these greater challenges more difficult to attain. It is up to the researchers in this field to assume the challenge presented, and to continue to develop new and exciting nanomaterials which demonstrate such results.

Author contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Conflict of interest

Authors declare that there are no conflicts of interest.

List of abbreviations

aglm	agglomerate
BaM	Magnetic Barium Hexaferrite
BNCNT	Boron/Nitrogen-doped CNT
CF	Carbon Fiber
CFO	$CoFe_2O_4$
CHNF	Carbon Hybrid Nanofibers
CIP	Carbonyl Iron Particles
CI	Carbonyl Iron
CNT	Carbon Nanotube
EM	Electromagnetic
G	graphene
g-	graphitic

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GN	graphene network
HRTEM	High-Resolution Transmission Electron Microscopy
LLG	Landau-Lifshitz-Gilbert
MC	Mesoporous Carbon
MCCF	Magnetite Coated Carbon Fibers
MOF	Metal Organic Framework
MWNT	Multi-Walled Nanotube
NIST	National Institute of Standards and Technology
NR	nanorings
NW	Nanowires
PANI	polyaniline
PEO	poly(ethylene oxide)
PET	Polyethylene terephthalate
PVC	Polyvinyl chloride
PVDF	Polyvinylidene fluoride
rGO	reduced Graphene Oxide
RL	Reflection Loss
SEM	Scanning Electron Microscopy
SAED	Selected Area Electron Diffraction
TEM	Transmission Electron Microscopy
T-ZnO	Tetrapod-like ZnO
XRD	X-ray Diffraction

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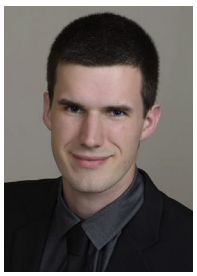
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Michael Green is a graduate student under the supervision of Dr. Xiaobo Chen at the University of Missouri–Kansas City, Department of Chemistry. He received his Bachelors of Science in chemistry with a minor in mathematics from the University of Idaho in 2016, and his Masters of Science in chemistry from the University of Missouri–Kansas City in 2019. His research interests include the development, characterization, modeling, and application of nanomaterials in light/matter interactions, focusing on photolysis, photocatalysis, and microwave absorption, as well as short-range matter/matter interactions with a focus in physical adsorption.



Dr. Xiaobo Chen is an Associate Professor at the University of Missouri – Kansas City, Department of Chemistry. His research interests include nanomaterials, catalysis, electrochemistry, light-materials interactions and their applications in renewable energy, environment protection, and information protection through microwave absorption. His renowned work includes the discovery of black TiO_2 with Professor Samuel S. Mao at the University of California, Berkeley and the new application of black TiO_2 nanomaterials along with other nanomaterials in microwave absorption application. Dr. Chen has published so far 150 peer-reviewed articles with about 41,000 citations.

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